Germanic Lexeme Reports: Lexical Derivation Volume

Alpha 01 assembly scaffold. This document assembles the current lexeme-report corpus in manifest order without revising entry-level prose, citations, locators, or transducer logic.

Introduction

The lexical catalogue is organized by derivation class rather than as a single undifferentiated list. This makes the interpretive burden of each entry explicit. Regular derivations establish the baseline relation between selected input and Old English target under the current cascade. The variant, analogy, reconstructed-comparator, and exception classes then show where that baseline is not sufficient on its own.

This lexical catalogue is a word-centered volume. It shows how individual selected inputs develop to individual Old English targets and how each entry is classified. A later sound-change volume or report should remain separate and rule-centered, covering chronology, interactions among rules, and broader system-level exception handling.

Data and sources

This alpha assembles the current model-entry corpus against the live manifest, compact derivation-trace report, and project bibliography. The lexical data layer is the current aligned Germanic dataset as represented by the model-entry metadata and the current compact trace source; comparative dictionaries, Old English dictionaries, and historical grammars remain in the entry prose exactly as already cited there.

Broad citations are carried forward honestly where the citation-layer audit already judged them mechanically acceptable for assembly. This alpha therefore tests book structure and technical integration rather than attempting a final source-polish pass.

Transducer and derivation method

Each lexical entry retains the pilot structure: a generated derivation summary, a boxed derivation trace split into Earlier Germanic changes and Old English changes, and the current model-entry prose. The summary distinguishes citation reconstruction, selected input, transducer outcome, and selected target where those differ, and the boxed trace remains a compact PDF-oriented rendering of the current compact trace data.

Derivation classes

The lexical catalogue is ordered by the seven live `DERIVATION_CLASS` values in the manifest. Counts in this alpha are taken directly from `manifest_all_by_class.tsv`:

- `regular`: **70**
- `attested_variant`: **4**
- `early_analogy`: **35**
- `late_analogy`: **28**
- `reconstructed_oe`: **3**
- `known_unmodelled`: **2**
- `unexplained_unmodelled`: **5**

Part I. Regular derivations

Regular derivations are entries where the selected transducer input and the Old English target stand in a straightforward relation under the current cascade. These entries form the baseline against which the analogy and exception classes are interpreted.

adder — OE nǣdre

Derivation: **nḗdrōn* > *nǣdre* (regular).

Derivation trace Proto input: **nḗdrōn*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Long Vowel Shortening	**nǣdræ*
[no change]		OE Unstressed AE Merger	**nǣdre*
Northwest Germanic			
NWGmc N Stem N Loss	**nḗdrǭ*		
NWGmc Long E Lowering	**nǣdrǭ*		

Outcome: *nǣdre*

Reconstruction and comparative evidence Kroonen distinguishes the masculine snake word **nadra-* from a feminine ablauting formation **nēdrōn-*, and gives Old English *nǣdre, næddre* under the latter (Kroonen 2013, 426). Orel likewise points from the masculine entry to a feminine **nēdrōn* ~ **nadrōn* type (Orel 2003, 325).

The selected input therefore is not a reshaped convenience form. It is the comparative reconstruction that specifically underlies the Old English noun.

Old English evidence The Old English word is securely represented by *nǣdre*, with *næddre* as a secondary variant. Clark Hall cross-references *næddre* to *nædre*, and Fulk treats *næddre* as the later geminated form beside the older base (Clark Hall 1960, 225; Fulk 2018, 149).

Development to Old English From **nḗdrōn*, the stressed long mid vowel develops to Old English *ǣ*, and the weak feminine ending remains as final *-e*, giving *nǣdre*. The doubled consonant of *næddre* is secondary and does not alter the inherited base form.

bake — OE bacan

Derivation: **bákaną* > *bacan* (regular).

Derivation trace Proto input: **bákaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**bækaną*
[no change]	OE A Restoration	**bakaną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**bakan*
[no change]	OE Secondary Nasalization	**bakąn*
	OE Weak Tail Reduction	**bakan*

Outcome: *bacan*

Reconstruction and comparative evidence Orel reconstructs the verb as **bakanan* and cites Old English *bacan* beside Old High German *backan, bahhan* (Orel 2003). Campbell gives *bacan* as one of the standard examples of Old English A-restoration before a single consonant, and Ringe and Taylor state the same development from **bakan* to Old English *bacan* (Campbell 1959, 61; Ringe and Taylor 2014).

Old English evidence Bosworth-Toller and Clark Hall both record *bacan* as the ordinary Old English verb 'to bake' (Bosworth and Toller 1898, 72; Clark Hall 1960). The target in this entry is therefore the attested infinitive headword itself, not a selected oblique or finite paradigm cell.

Development to Old English From **bákaną*, Anglo-Frisian brightening first gives **bækaną*. A-restoration then returns the stem vowel to *a* before single *k* plus the back-vocalic infinitive suffix, and later apocope and weak-tail reduction yield *bacan* (Campbell 1959, 61; Ringe and Taylor 2014). The development is therefore straightforward: **bákaną* > *bacan*.

beech — OE bōc

Derivation: **bōkō* > *bōc* (regular).

Derivation trace Proto input: **bōkō*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**bōk*
[no change]			
Northwest Germanic			
NWGmc Final Long O Raising	**bōku*		

Outcome: *bōc*

Reconstruction and comparative evidence Kroonen gives the beech noun as **bōk(j)ō-* and cites Old English *boc, bēce* among its reflexes (Kroonen 2013). The selected input **bōkō* is the nominative-singular shape of that family, which is the relevant comparison form here.

Old English evidence Kroonen's Old English evidence already separates the paradigm material: *boc* as the nominative form and *bēce* as an oblique form (Kroonen 2013). The relevant comparator is therefore *bōc*; *bēċe* remains related paradigm evidence rather than the form chosen for this comparison.

Development to Old English With nominative input **bōkō*, the development is compact. Northwest Germanic final long *ō* raises to *u*, and later high-vowel apocope leaves *bōc*. The regular comparison is therefore **bōkō* > *bōc*.

begin — OE beġinnan

Derivation: **bigínnaną* > *beġinnan* (regular).

Derivation trace Proto input: **bigínnaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**bigínnan*
[no change]	OE Secondary Nasalization	**bigínnąn*
	OE Velar Palatalization	**biʤínnąn*
Northwest Germanic	OE Prefix I Reduction	**bĕʤínnąn*
[no change]	OE Weak Tail Reduction	**bĕʤínnan*

Outcome: *beġinnan*

Reconstruction and comparative evidence The verb is modeled here as inherited **bigínnaną*. Ringe and Taylor state that intervocalic **g* is palatalized between front vowels in Old English (Ringe and Taylor 2014), and Campbell lists *ginnan* among familiar examples of palatal *g* in this verb family (Campbell 1959, 174).

Old English evidence Bosworth-Toller and Clark Hall lemmatize the verb as *be-ginnan* / *beginnan* (Bosworth and Toller 1898, 84; Clark Hall 1960). Those plain-*g* dictionary spellings support the same verb that appears here in normalized form as *beġinnan*.

Development note The prefix deserves separate notice. Ringe and Taylor explicitly cite *bi-* > *be-* as an Old English unstressed-prefix development (Ringe and Taylor 2014).

Development to Old English From **bigínnaną*, heavy-syllable nasal apocope yields **bigínnan*. Intervocalic **g* between front vowels then palatalizes to *ġ*, and the unstressed prefix reduces *bi-* to *be-*, giving *beġinnan*.

bier — OE bǣr

Derivation: **bḗrō* > *bǣr* (regular).

Derivation trace Proto input: **bḗrō*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**bǣr*
[no change]			
Northwest Germanic			
NWGmc Final Long O Raising	**bḗru*		
NWGmc Long E Lowering	**bǣ́ru*		

Outcome: *bǣr*

Reconstruction and comparative evidence Kroonen reconstructs the noun as **bērō-* f. 'bier' and cites Old English *bar, bær* among the reflexes (Kroonen 2013, 717). The selected input **bḗrō* is the same lexeme in the accent notation used here.

Old English evidence Clark Hall and Bosworth-Toller lemmatize the noun as *bær*, and Kroonen also records *bar* beside it (Clark Hall 1960; Bosworth and Toller 1898, 73; Kroonen 2013, 717). The target *bǣr* is therefore a normalized long-vowel spelling of the same noun.

Source note Lexicographic spellings vary between *bær* and *bar*. The normalized target *bǣr* simply marks the same long vowel explicitly (Clark Hall 1960; Bosworth and Toller 1898, 73; Kroonen 2013, 717).

Development to Old English From **bḗrō*, Northwest Germanic final long *ō* raises to *u*, long *ē* lowers to *ǣ*, and high-vowel apocope yields *bǣr*. The resulting noun matches the normalized Old English target.

birth — OE byrd

Derivation: **búrdiz* > *byrd* (regular).

Derivation trace Proto input: **búrdiz*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	**byrdi*
[no change]		OE High Vowel Apocope	**byrd*
Northwest Germanic			
PGmc Final Z Deletion	**búrdi*		

Outcome: *byrd*

Reconstruction and comparative evidence Kroonen cites the noun under stem-level **burdi-* and gives Old English *(ge-)byrd* among the reflexes (Kroonen 2013). The selected input **búrdiz* is the nominative-style form that stands behind that stem label.

Old English evidence Clark Hall and Bosworth-Toller both attest simplex *byrd* as an Old English noun meaning ‘birth’ (Clark Hall 1960; Bosworth and Toller 1898, 125). The prefixed form *gebyrd* is also well established in the tradition: Kroonen lists *(ge-)byrd*, Bosworth-Toller has a separate *ge-byrd* entry, and Campbell cites *gebyrd* and *gebyrdu* in his grammatical discussion (Kroonen 2013; Bosworth and Toller 1898, 125; Campbell 1959).

Form note The relevant comparator here is the simplex noun *byrd*. The prefixed forms remain related attested material within the same lexical family, and Hogg’s discussion of deverbal feminines provides the broader derivational setting (Hogg 1992).

Development to Old English From **búrdiz*, loss of final *z* gives **búrdi*. I-umlaut fronts *u* to *y*, and high-vowel apocope then yields *byrd*. The result is the ordinary simplex Old English noun.

bone — OE bān

Derivation: **báiną* > *bān* (regular).

Derivation trace Proto input: **báiną*

Earlier Germanic changes		**Old English changes**	
West Germanic		OE Heavy Syllable Nasal Apocope	**bān*
PWGmc Ai Monophthongization	**bāną*		
Northwest Germanic			
[no change]			

Outcome: *bān*

Reconstruction and comparative evidence Kroonen cites the noun as **baina-*, and Orel gives the same lexeme under **bainan* (Kroonen 2013; Orel 2003). Both are comparative headword conventions for the same neuter noun whose Old English reflex is *bān*.

Old English evidence Clark Hall and Bosworth-Toller record *bān* as the ordinary Old English noun (Clark Hall 1960; Bosworth and Toller 1898). Bright’s glossary also distinguishes citation-form *bān* from oblique *bāne*, which keeps the nominative-accusative singular separate from the rest of the paradigm (Bright 1971).

Source note The comparative headwords **baina-* and **bainan* provide lexeme background. The relevant comparison form here is the nominative-accusative singular **báiną*.

Development to Old English West Germanic monophthongization turns stressed **ai* into *ā*, giving **bāną*; heavy-syllable nasal apocope then yields *bān*. The resulting form matches the attested Old English citation noun.

both — OE bū

Derivation: **bō* > *bū* (regular).

Derivation trace Proto input: **bō*

Earlier Germanic changes		Old English changes
West Germanic		[no change]
[no change]		
Northwest Germanic		
NWGmc Stressed Monosyllable O Raising	**bū*	

Outcome: *bū*

Reconstruction and comparative evidence Kroonen treats the Germanic numeral under **ba-* and gives the inherited paradigm **bai, *bans, *bōz/*bōns, *bō*, with Old English *bēġen*, *bā*, and neuter *bū* (Kroonen 2013, 47). For the present entry, the relevant inherited form is the unextended neuter dual **bō*.

The older explanation of *bēġen* derives it from **bō-jen-*, and Orel still gives *OE bezen (< *bō-jenō)* beside *ON báðir*, *OFris bēthe*, *OS be-thia*, and *OHG bēde, beide* (Orel 2003, 65). Fulk reports that explanation cautiously and notes Seebold's preference for a **bō-þ-* analysis instead (Fulk 2018, sect. 10.1). That debate matters for *bēġen* and for the extended forms behind Modern English *both*, German *beide*, and Dutch *beide*; it does not displace the inherited neuter **bō* > *bū* treated here.

Old English evidence The Old English dual paradigm is well established. Brunner gives masculine *bēġen*, feminine *bā*, and neuter *bū* beside *bā*, with compounds such as *bā twā*, *bū tū*, and *bām twām* (Sievers 1965, sect. 324 Anm. 2). Campbell and Fulk present the same basic pattern: masculine *bēġen*, feminine *bā*, neuter *bā*, *bū*, genitive *bēġra, bēġ(e)a*, and dative *bǣm* (Campbell 1959, sect. 683; Fulk 2018, sect. 10.1).

bū is therefore an attested neuter dual form, not a reconstruction. It is the cleanest target for this entry because *bēġen* belongs to the historically more contested **bō-jen-* / analogical zone, while *bā* remains a partner form within the dual paradigm rather than the most straightforward monosyllabic comparison.

Development to Old English **bō* is a stressed monosyllabic form. Campbell cites *cū*, *hū*, *tū*, and *bū* as examples of final accented *ō* > *ū* in the West Germanic stage leading to Old English (Campbell 1959, sect. 122). Brunner states the same development more directly: *Auslautendes ō erscheint als û in bū … cu … hū, tū* (Sievers 1965, sect. 69).

The development is therefore straightforward: **bō* > *bū*.

Form comparison The comparison below is manual. It separates the inherited OE target from the other forms that belong to the same broader lexical history.

Form	Source / stage	Status	Relevance to this entry
**bō* > *bū*	PGmc neuter dual > OE neuter dual	selected regular comparison	main line of the entry
bēġen	OE masculine dual	attested, but historically contested and at least partly analogical in Kroonen	real OE evidence, not the selected target
bā	OE feminine dual; also neuter variant	attested partner form	part of the OE paradigm, but not the chosen monosyllabic comparator

Form	Source / stage	Status	Relevance to this entry
báðir, beide, both	Norse, continental West Germanic, Modern English extended forms	related but different formation	useful background, not the direct continuation of OE *bū*

bow — OE bīeġan

Derivation: **báugijaną* > *bīeġan* (regular).

Derivation trace Proto input: **báugijaną*

Earlier Germanic changes

West Germanic

[no change]

Northwest Germanic

[no change]

Old English changes

OE Au Fronting	**báeugijaną*
OE Diphthong Leveling	**bēagijaną*
OE Heavy Syllable Nasal Apocope	**bēagijan*
OE Secondary Nasalization	**bēagijąn*
Sievers Law Syncope	**bēagjąn*
OE Velar Palatalization	**bēaʤjąn*
OE I Umlaut	**bīeʤjąn*
OE Weak Tail Reduction	**bīeʤjan*
OE J Loss After Heavy	**bīeʤan*

Outcome: *bīeġan*

Reconstruction and comparative evidence Kroonen reconstructs the weak verb as **baugjan-* 'to (make) bend' and cites Old English *biegan* among its reflexes (Kroonen 2013). Ringe and Taylor give the northwest Germanic and West Saxon development more fully as *PNWGmce *baugijana* > **béagjan* > *WS OE biegan* (Ringe and Taylor 2014). The entry therefore concerns the weak causative member of the bend-family, alongside the related strong verb and noun.

Old English evidence Clark Hall lemmatizes *biegan*, and Bosworth-Toller records *bigan* with examples such as *Ic bēge mīne cneówa* and *Se ord bīgde upp tō þām hiltum* (Clark Hall 1960; Bosworth and Toller 1898, 102). The form *bīeġan* used here is a normalized spelling of that attested Old English weak verb.

Development to Old English From **báugijaną*, the stem reaches pre-Old-English **bēagjan*, after which palatalization of **gj* and i-umlaut yield West Saxon *biegan*; Campbell lists *biegan* among the regular *ie* outcomes of **éa* under i-umlaut (Ringe and Taylor 2014; Campbell 1959, 80). The development is therefore straightforward: **báugijaną* > *bīeġan*.

breeches — OE brēċ

Derivation: **brōkiz* > *brēċ* (regular).

Derivation trace Proto input: **brōkiz*

Earlier Germanic changes

West Germanic

[no change]

Northwest Germanic

PGmc Final Z Deletion	**brōki*

Old English changes

OE Velar Palatalization	**brōʧi*
OE I Umlaut	**brēʧi*
OE High Vowel Apocope	**brēʧ*

Outcome: *brēċ*

Reconstruction and comparative evidence Kroonen cites the noun under **brōk-*, with Old English *brōc* and plural 'breeches' among its reflexes (Kroonen 2013). Ringe and Taylor give the plural development directly as *PNWGmce *brokiz* > **breeci* > *OE bréc* (Ringe and Taylor 2014). The deeper verbal base belongs to the noun's etymological background, while the selected input here is the plural noun form **brōkiz*.

Old English evidence Bright notes *brōc* with plural *brēc*, and Clark Hall gives *brēc fp. breeches* while also listing *broc* as a feminine noun probably represented chiefly in the plural (Bright 1971; Clark Hall 1960, 64). The spelling *brēċ* used here makes the long vowel and palatal consonant explicit; the Old English evidence itself is the attested plural *brēc*.

Development to Old English After loss of final -*z*, the stem ends in -*ki*, so the velar palatalizes and *ō* undergoes i-umlaut to *ē*; final high-vowel apocope then yields *brēċ* (Ringe and Taylor 2014). The development is therefore regular: **brōkiz* > *brēċ*.

calf — OE ċealf

Derivation: **kálbaz* > *ċealf* (regular).

Derivation trace Proto input: **kálbaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**kálb*
[no change]		Anglo Frisian Brightening	**kælb*
Northwest Germanic		OE Breaking	**kealb*
PGmc Final Z Deletion	**kálba*	PGmc B Allophony	**kealβ*
		OE Velar Palatalization	**ʧealβ*

Outcome: *ċealf*

Reconstruction and comparative evidence Kroonen treats the noun under **kalbiz-* and notes an older s-stem **kalbaz, pl. *kalbizō*, while Orel cites **kalbaz* as the citation form and Ringe and Taylor derive West Saxon *Cealf* from **kalbaz, *kalbiz-* (Kroonen 2013; Orel 2003, 248; Ringe and Taylor 2014, 220). The selected input here is the singular **kálbaz*, since the entry concerns the citation-form noun.

Old English evidence Clark Hall gives *cealf I. (œ, e) nm. (nap. cealfru)*, and Bosworth-Toller likewise records *Caelf / Cealf* beside plural forms such as *calfur* and *cealfru* (Clark Hall 1960; Bosworth and Toller 1898, 131). Campbell and Fulk show the same singular-plus--*r*- plural pattern (Campbell 1959; Fulk 2018, 193). The spelling *ċealf* used here makes the palatalized initial explicit; the ordinary attested dictionary headword is *cealf*.

Development to Old English After loss of final -*z* and bare -*a*, Anglo-Frisian brightening gives **kælb*, and breaking before *l* plus consonant yields **kealb*. Ringe and Taylor's account of the lexeme and their rule for initial *k* in front-vocalic environments support the West Saxon palatalized onset represented here as *ċ*-, so **kálbaz* develops regularly to *ċealf* (Ringe and Taylor 2014, 220).

corn — OE corn

Derivation: **kúrną* > *corn* (regular).

Derivation trace Proto input: **kúrną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**kórn*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**kórną*		

Outcome: *corn*

Reconstruction and comparative evidence Kroonen cites the noun as **kurna-*, and Orel gives the citation form **kurnan*, both with Old English *corn* among the reflexes (Kroonen 2013; Orel 2003, 264). The singular form **kúrną* is the nominative-accusative singular appropriate to the citation noun.

Old English evidence Clark Hall gives *corn n. 'corn,' grain*, Bright's glossary lists *corn, n.* with genitive singular *cornes*, and Bosworth-Toller treats *corn* as an ordinary noun headword (Clark Hall 1960; Bright 1971, 347; Bosworth and Toller 1898, 144). The target is therefore an attested citation form, while forms such as *cornes* simply provide paradigm background.

Development to Old English With northwest Germanic lowering, **kúrną* becomes **kórną*, and later loss of final nasal after a heavy syllable yields **kórn*, whence *corn*. The oblique form **kurnăn* belongs to comparative background rather than to the derivational input of this entry.

deed — OE dǣd

Derivation: **dḗdiz* > *dǣd* (regular).

Derivation trace Proto input: **dḗdiz*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**dǣd*
[no change]			
Northwest Germanic			
PGmc Final Z Deletion	**dḗdi*		
NWGmc Long E Lowering	**dǣdi*		

Outcome: *dǣd*

Reconstruction and comparative evidence Orel reconstructs the noun as **dēdiz*, and Ringe and Taylor derive the same inherited i-stem from Proto-Germanic **dédiz* through northwest Germanic **dadiz* (Orel 2003; Ringe and Taylor 2014). The stress-marked form **dḗdiz* represents that same inherited noun in a notation that keeps the stressed long vowel explicit.

Old English evidence Campbell states that Primitive Germanic *ē* appears as West Saxon *ǣ* but in other Old English dialects mostly as *ē*, and Brunner gives the contrast explicitly as West Saxon *dǣd* beside non-West-Saxon *dēd* (Campbell 1959; Sievers 1965). Clark Hall likewise lists *dǣd* and cross-refers Anglian *dēd* to it (Clark Hall 1960). West Saxon *dǣd* is therefore the relevant Old English form here, with Anglian *dēd* as a dialectal doublet.

Development to Old English From inherited **dēdiz*, loss of final -*z* and the West Saxon lowering of stressed long *ē* yield *dǣd*; Anglian *dēd* preserves the non-West-Saxon outcome (Campbell 1959; Sievers 1965). The development treated here is therefore the regular West Saxon line.

door — OE dor

Derivation: **dúrą* > *dor* (regular).

Derivation trace Proto input: **dúrą*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**dór*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**dórą*		

Outcome: *dor*

Reconstruction and comparative evidence Kroonen reconstructs a neuter **dura-* 'gate, (single) door' and cites Old English *dor* among its reflexes. In the same entry he separates Old English *duru*, Old Frisian *dore*, and Old High German *tura* as reflexes of **durō-* instead (Kroonen 2013).

Old English evidence Clark Hall records *dor* as a neuter noun and separately records feminine *duru* with its own inflection (Clark Hall 1960). Ringe and Taylor likewise treat *duru* as an early Old English u-stem, originally a root noun shifted into that class (Ringe and Taylor 2014). The selected target here is therefore the attested neuter *dor*, while *duru* remains a parallel Old English reflex from another stem history.

Development to Old English From **dúrą*, Northwest Germanic u-lowering gives **dórą*, and heavy-syllable nasal apocope then yields *dor*. The regular development treated in this entry is therefore **dúrą* > *dor*; the feminine *duru* belongs to the separate line identified by Kroonen and Ringe-Taylor (Kroonen 2013; Ringe and Taylor 2014).

fare — OE faran

Derivation: **fáraną* > *faran* (regular).

Derivation trace Proto input: **fáraną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**færaną*
[no change]	OE A Restoration	**faraną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**faran*
[no change]	OE Secondary Nasalization	**farąn*
	OE Weak Tail Reduction	**faran*

Outcome: *faran*

Reconstruction and comparative evidence Kroonen gives the inherited strong verb as **faran-*, and Orel gives the same lexeme as **faranan*, both with Old English *faran* among the reflexes (Kroonen 2013; Orel 2003, 132). Campbell also uses *faran* as a standard example of Old English A-restoration (Campbell 1959, 61).

Old English evidence Clark Hall lemmatizes the strong verb as *faran* and separately records weak *færan* 'to frighten'; *fære*, *færst*, and *færð* belong to present-tense forms of *faran* rather than to the infinitive itself (Clark Hall 1960). Bosworth-Toller preserves the same distinction (Bosworth and Toller 1898, 108). The selected target is therefore the attested citation infinitive *faran*.

Development to Old English From **fáranq*, Anglo-Frisian brightening first gives **færanq*, but A-restoration before single *r* returns **faranq*; later apocope and weak-tail reduction yield *faran* (Campbell 1959, 61). Fulk's contrast with participial *faren-* < **faræn-* < **faran-* shows why fronting elsewhere in the paradigm does not alter the infinitive headword (Fulk 2018).

fell — OE fell

Derivation: **féllą* > *fell* (regular).

Derivation trace Proto input: **féllą*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**féll*
[no change]			
Northwest Germanic			
[no change]			

Outcome: *fell*

Reconstruction and comparative evidence Kroonen reconstructs the noun as **fella-* 'membrane, skin, hide' and cites Old English *fell* beside Dutch *vel* and German *Fell* (Kroonen 2013). The selected input **féllą* is the derivable singular form of that same inherited noun.

Old English evidence Clark Hall records *fell* as the noun 'fell, skin, hide', and Bright's glossary likewise gives *fell* with inflected forms such as accusative singular *fel* and dative plural *fellum* (Clark Hall 1960; Bright 1971). The target is therefore the attested noun *fell*, not the verb *fellan* or the preterite *feoll*.

Development to Old English With **féllą*, no special earlier reshaping is needed: heavy-syllable nasal apocope yields **féll*, surfacing as *fell*. The regular development treated here is therefore **féllą* > *fell*.

fern — OE fearn

Derivation: **fárnaz* > *fearn* (regular).

Derivation trace Proto input: **fárnaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**fárn*
[no change]		Anglo Frisian Brightening	**færn*
		OE Breaking	**fearn*
Northwest Germanic			
PGmc Final Z Deletion	**fárna*		

Outcome: *fearn*

Reconstruction and comparative evidence Kroonen cites the noun as masculine **farna-* and gives Old English *fearn, fern*, while Orel gives the same lexeme as neuter **farnan* with Old English *fearn* (Kroonen 2013; Orel 2003, 133). Those are comparative headword conventions rather than competing Old English outcomes; the modeled input here is the nominative-style **fárnaz*.

Old English evidence Clark Hall gives *fearn* as an Old English noun, and Bosworth-Toller records *fearn* with inflected forms such as *fearnes*, *fearna*, and *fearne* (Clark Hall 1960; Bosworth and Toller 1898, 219). Kroonen's additional *fern* remains useful comparative background, but the best-supported citation target in the local lexical sources is *fearn* (Kroonen 2013).

Development to Old English From **fárnaz*, loss of final *-z* and final *-a* gives **fárn*; Anglo-Frisian brightening then yields **færn*, and breaking before *r* plus consonant gives *fearn* (Campbell 1959; Ringe and Taylor 2014). The development treated here is therefore the regular *rC*-breaking line.

field — OE feld

Derivation: **félθuz* > *feld* (regular).

Derivation trace Proto input: **félθuz*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**féld*
PWGmc L Th Voicing	**félduz*		
Northwest Germanic			
PGmc Final Z Deletion	**féldu*		

Outcome: *feld*

Reconstruction and comparative evidence Ringe and Taylor treat Old English *feld* as one of the cases where earlier **felþu-* ~ **feldu-* may reflect either inherited **þ* ~ **d* alternation or the regular West Germanic development **lþ* > *ld* (Ringe and Taylor 2014, 170). The broader proto label **félθuz* can remain as comparative background, while that narrower historical ambiguity does not affect the regular classification.

Old English evidence Clark Hall records *feld* with oblique forms such as *felda* and *felde*, and Campbell notes early place-name spellings in *-felth* beside the later standard form (Clark Hall 1960, 114; Campbell 1959, 169). The selected target is therefore the attested citation noun *feld*, with the older *-felth* spellings as historical support rather than as rival targets.

Development to Old English In the modeled pathway, medial **lþ* becomes *ld*, final *-z* is lost, and high-vowel apocope then yields *feld*. Whether the voiced dental ultimately reflects inherited alternation or the regular **lþ* > *ld* development, both accounts converge on the same Old English form (Ringe and Taylor 2014, 170).

fly — OE flēogan

Derivation: **fléuganą* > *flēogan* (regular).

Derivation trace Proto input: **fléuganą*

Earlier Germanic changes	Old English changes	
West Germanic	OE Diphthong Leveling	**flēoganą*
[no change]	OE Heavy Syllable Nasal Apocope	**flēogan*
Northwest Germanic	OE Secondary Nasalization	**flēogąn*
[no change]	OE Weak Tail Reduction	**flēogan*

Outcome: *flēogan*

Reconstruction and comparative evidence Ringe and Taylor derive the verb as **fleugana > OE fléogan* and elsewhere contrast West Saxon *fléogan* with Anglian *flégan*, alongside related forms *fléoge / flége* (Ringe and Taylor 2014). The selected input **fléuganą* represents that inherited strong verb in the notation used here.

Old English evidence Clark Hall and Bosworth-Toller record *flēogan* as the ordinary Old English strong verb, and Bright gives the familiar paradigm *flēag, flugon, flogen* with present *fleogeð* (Clark Hall 1960; Bosworth and Toller 1898; Bright 1971). The target in this entry is therefore the attested verbal infinitive itself.

Form note Ringe and Taylor also list related *fléoge / flége* and Anglian *flégan*, which belong to the same family but do not replace the infinitive *flēogan* treated here (Ringe and Taylor 2014).

Development to Old English From **fléuganą*, Old English diphthong leveling gives **flēoganą*; heavy-syllable nasal apocope and weak-tail reduction then yield *flēogan* (Ringe and Taylor 2014). The development is therefore regular: **fléuganą > flēogan*.

forlorn — OE lēosan

Derivation: **léusaną* > *lēosan* (regular).

Derivation trace Proto input: **léusaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Diphthong Leveling	**lēosaną*
[no change]	OE Heavy Syllable Nasal Apocope	**lēosan*
Northwest Germanic	OE Secondary Nasalization	**lēosąn*
[no change]	OE Weak Tail Reduction	**lēosan*

Outcome: *lēosan*

Reconstruction and comparative evidence Kroonen reconstructs the verb under **leusan-* and cites prefixed daughters such as Gothic *fra-liusan* and Old English *for-lēosan*; Orel likewise gives Old English *for-leósan* (Kroonen 2013; Orel 2003). The inherited verbal base is therefore clear, though the daughter set often appears with the prefix.

Old English evidence The direct Old English evidence behind English *forlorn* lies in the prefixed verb *forlēosan* and especially in the participle *forloren*, recorded by Ringe and Taylor and in the dictionaries (Ringe and Taylor 2014; Clark Hall 1960; Bosworth and Toller 1898). The simplex infinitive *lēosan* represents the verbal base itself.

Form note As a base-form comparison, the simplex infinitive is *lēosan*, while the English adjective continues the prefixed Old English family *forlēosan / forloren* (Ringe and Taylor 2014).

Development to Old English From **léusaną*, Old English diphthong leveling gives **lēosaną*, and later nasal apocope and weak-tail reduction yield *lēosan* (Ringe and Taylor 2014). The prefixed forms follow the same verbal base with added *for-*.

gang — OE gang

Derivation: **gángaz* > *gang* (regular).

Derivation trace Proto input: **gángaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**gáng*
[no change]			
Northwest Germanic			
PGmc Final Z Deletion	**gánga*		

Outcome: *gang*

Reconstruction and comparative evidence Orel reconstructs the noun as **gangaz* and cites Old English *gang* beside Old Norse *gangr*, Old Frisian *gang / gong*, Old Saxon *gang*, and Old High German *gang* (Orel 2003). The selected input **gángaz* is the same lexeme in the accent notation used here.

Old English evidence Clark Hall and Bosworth-Toller both record *gang* as the noun 'going, journey, way', and Bright's glossary gives *gong (gang), m., path, course* (Clark Hall 1960; Bosworth and Toller 1898, 159; Bright 1971, 392). The target is therefore the attested noun headword itself.

Form note This entry concerns the noun *gang*, not the separate verb *gangan* (Clark Hall 1960; Bosworth and Toller 1898, 159).

Development to Old English From **gángaz*, loss of final *-z* gives **gánga*, and later loss of final bare *-a* yields *gang*. The development is therefore regular: **gángaz* > *gang*.

give — OE ġiefan

Derivation: **gébaną* > *ġiefan* (regular).

Derivation trace Proto input: **gébaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**géban*
[no change]	OE Secondary Nasalization	**gébąn*
Northwest Germanic	PGmc B Allophony	**géβąn*
[no change]	OE Velar Palatalization	**ʤéβąn*
	OE Ws Palatal Diphthongization	**ʤíeβąn*
	OE Weak Tail Reduction	**ʤíeβan*

Outcome: *ġiefan*

Reconstruction and comparative evidence Kroonen reconstructs the strong verb as **geban-* and cites Old English *giefan* among its reflexes (Kroonen 2013). Ringe and Taylor contrast West Saxon *giefan* with Mercian *for-geofan* and Northumbrian *geafa*, showing that the inherited verb takes different later dialectal shapes (Ringe and Taylor 2014).

Old English evidence Campbell gives *gefan (W-S giefan)* among examples of initial palatalization, and Clark Hall records the verb under plain *giefan* with forms such as *geaf* and *giefen* (Campbell 1959; Clark Hall 1960). The spelling *ġiefan* used here makes the palatal initial explicit.

Dialect note West Saxon *ie* here reflects palatal diphthongization after initial palatalization; non-West-Saxon forms such as *geafa* or *for-geofan* continue the same verb without the West Saxon vocalism (Ringe and Taylor 2014).

Development to Old English From **gébaną*, initial *g* palatalizes before *e*; West Saxon palatal diphthongization then yields *ie*, and later tail reduction gives *giefan* (Campbell 1959; Ringe and Taylor 2014). The result is therefore the regular West Saxon infinitive.

gold — OE gold

Derivation: **gúlθą* > *gold* (regular).

Derivation trace Proto input: **gúlθą*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**góld*
PWGmc L Th Voicing	**gúldą*		
Northwest Germanic			
NWGmc U Lowering	**góldą*		

Outcome: *gold*

Reconstruction and comparative evidence Ringe and Taylor cite the noun as **gulþa- / *gulda-*, and Kroonen gives the same pair (Ringe and Taylor 2014, 42; Kroonen 2013). The selected input **gúlθą* preserves the older consonantal form while leaving open whether the medial stop reflects inherited alternation or regular West Germanic development.

Old English evidence Bosworth-Toller and Clark Hall both record *gold* as the ordinary Old English neuter noun (Bosworth and Toller 1898, 121; Clark Hall 1960, 152). The target is therefore the attested citation form itself.

Development note Ringe and Taylor note that the medial stop can be understood either as alternation **gulþa- / *gulda-* or as the ordinary West Germanic change **lþ* > *ld*; both routes lead to the same Old English consonantism (Ringe and Taylor 2014, 42).

Development to Old English From **gúlθą*, the regular consonant development gives **gúldą*; Northwest Germanic / Old English lowering then yields **góldą*, and apocope gives *gold* (Campbell 1959; Ringe and Taylor 2014, 42).

grave — OE grafan

Derivation: **grábaną* > *grafan* (regular).

Derivation trace Proto input: **grábaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**grœbaną*
[no change]	OE A Restoration	**grabaną*
	OE Heavy Syllable Nasal Apocope	**graban*
Northwest Germanic	OE Secondary Nasalization	**grabąn*
[no change]	PGmc B Allophony	**graβąn*
	OE Weak Tail Reduction	**graβan*

Outcome: *grafan*

Reconstruction and comparative evidence Campbell gives *grafan* among the standard examples of Old English a-restoration before a single consonant and a following back vowel, and Ringe and Taylor describe the same development for Class VI infinitives (Campbell 1959, 61; Ringe and Taylor 2014).

Old English evidence Clark Hall records *grafan* as the verb ‘to dig, grave’ and separately records noun *græf* ‘grave, trench’ (Clark Hall 1960). The target here is the attested infinitive headword of the verb.

Development to Old English From **grábaną*, Anglo-Frisian brightening first gives a fronted stem vowel. A-restoration then returns *a* before single *b* plus the back-vocalic infinitive ending, and later apocope and weak-tail reduction yield *grafan* (Campbell 1959, 61; Ringe and Taylor 2014).

Form note Noun *græf* and verbal forms such as *græfð* or past participial *græfen* belong to other lexical or paradigm positions and do not replace the infinitive *grafan* as the target here (Clark Hall 1960).

guest — OE ġiest

Derivation: **gástiz* > *ġiest* (regular).

Derivation trace Proto input: **gástiz*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**gæsti*
[no change]		OE Velar Palatalization	**ʤæsti*
Northwest Germanic		OE I Umlaut	**ʤesti*
PGmc Final Z Deletion	**gásti*	OE Ws Palatal Diphthongization	**ʤiesti*
		OE High Vowel Apocope	**ʤiest*

Outcome: *ġiest*

Reconstruction and comparative evidence Campbell and Ringe-Taylor treat the noun as an ordinary i-stem whose West Saxon development shows palatal diphthongization, while non-West-Saxon evidence preserves forms of the *gest* type (Campbell 1959; Ringe and Taylor 2014).

Old English evidence Bosworth-Toller and Clark Hall record the word under forms such as *gist*, *gest*, *giest*, and *gyst* (Bosworth and Toller 1898; Clark Hall 1960). The selected target *ġiest* is the normalized West Saxon form within that attested family.

Development to Old English From **gástiz*, Anglo-Frisian brightening gives a *gæst-* stage, and i-mutation affects the front vowel before the lost high-vocalic ending. In West Saxon the initial palatal environment then produces *ie*, so the regular outcome is *ġiest* (Campbell 1959; Ringe and Taylor 2014).

Dialect note West Saxon *ġiest* is the selected target here. Anglian *gest* and related spellings remain real Old English comparators rather than corrections to that choice (Ringe and Taylor 2014; Bosworth and Toller 1898).

hair — OE hǣr

Derivation: **xḗrą* > *hǣr* (regular).

Derivation trace Proto input: **xḗrą*

Earlier Germanic changes		Old English changes	
West Germanic		OE Velar Fricative Palatalization	**çǣrą*
[no change]		OE Heavy Syllable Nasal Apocope	**çǣr*
Northwest Germanic			
NWGmc Long E Lowering	**xǣrą*		

Outcome: *hǣr*

Reconstruction and comparative evidence Kroonen cites the ordinary Proto-Germanic hair word as **hēra-* (Kroonen 2013). The selected input **xḗrą* represents that same long-ē stem in the present derivation.

Old English evidence Clark Hall and Bosworth-Toller record *hær* / *hǣr* as the ordinary Old English noun 'hair' (Clark Hall 1960, 158; Bosworth and Toller 1898, 510). The target is therefore the attested headword itself.

Development to Old English From **xḗrą*, Northwest Germanic lowering gives a long front vowel, and later loss of the final nasal leaves the Old English form *hǣr*. The development treated here is straightforward and does not require any special paradigm choice.

Form note Older references to **xazwăz* belong to a different lexeme, and the separate *haddr / heordan / hād-* material does not displace the ordinary simplex *hǣr* treated here (Kroonen 2013; Clark Hall 1960, 158).

harvest — OE hierfest

Derivation: **xárbistuz* > *hierfest* (regular).

Derivation trace Proto input: **xárbistuz*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**xærbistu*
[no change]		OE Breaking	**xearbistu*
Northwest Germanic		OE Velar Fricative Palatalization	**çearbistu*
PGmc Final Z Deletion	**xárbistu*	PGmc B Allophony	**çearβistu*
		OE I Umlaut	**çierβistu*
		OE High Vowel Apocope	**çierβist*
		OE Med Unstressed I Lowering1	**çierβest*

Outcome: *hierfest*

Reconstruction and comparative evidence Bammesberger and Ringe-Taylor treat **harbist-* as the inherited base and explain that the regular native West Saxon development would be of the *hierfest / hyrfest* type (Bammesberger 1997; Ringe and Taylor 2014).

Old English evidence Bosworth-Toller and Clark Hall record *hærfest*, with *herfest* as a variant in the lexical tradition (Bosworth and Toller 1898; Clark Hall 1960). Those attested forms remain the main dictionary evidence for the noun.

Development to Old English From **xárbistuz*, Anglo-Frisian brightening, breaking, and i-mutation produce a *hierbist-* stage, and later lowering of unstressed medial *i* to *e* gives *hierfest*. That is the regular West Saxon development treated here (Ringe and Taylor 2014; Campbell 1959).

Source note The selected target *hierfest* represents the regular native West Saxon outcome discussed by Bammesberger and Ringe-Taylor. The attested Old English lexical tradition, however, is chiefly *hærfest / herfest*, commonly treated as non-West-Saxon or Anglian material in West Saxon transmission (Bammesberger 1997; Ringe and Taylor 2014).

hedge — OE heġġ

Derivation: **xágjaz* > *heġġ* (regular).

Derivation trace Proto input: **xágjaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**xággj*
PWGmc J Gemination	**xággjaz*	Anglo Frisian Brightening	**xæggj*
Northwest Germanic		OE Velar Fricative Palatalization	**çæggj*
PGmc Final Z Deletion	**xággja*	OE Velar Palatalization	**çæʤʤj*
		OE I Umlaut	**çeʤʤj*
		OE J Loss After Heavy	**çeʤʤ*

Outcome: *heġġ*

Reconstruction and comparative evidence The selected input models a palatal **-gj-** noun whose Old English development includes gemination, palatalization, and i-mutation. The current derivation therefore reaches a palatal-geminate outcome of the *heġġ* type (Campbell 1959).

Old English evidence Bosworth-Toller and Clark Hall record the noun under standard spellings *hecg* / *heċġ* (Bosworth and Toller 1898; Clark Hall 1960). The lexical item itself is therefore well attested even though the selected form is normalized.

Development to Old English From **xágjaz*, West Germanic j-gemination first yields a geminate stop, and later Old English palatalization and loss of final *j* produce *heġġ*. The development is treated as regular rather than exceptional.

Form note Standard dictionary spelling is *heċġ* or *hecg*. Normalized *heġġ* is the selected target here, while the ordinary lexicographic forms remain the main Old English citation evidence (Bosworth and Toller 1898; Clark Hall 1960).

helm — OE helm

Derivation: **xélmaz* > *helm* (regular).

Derivation trace Proto input: **xélmaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**xélm*
[no change]		OE Velar Fricative Palatalization	**çélm*
Northwest Germanic			
PGmc Final Z Deletion	**xélma*		

Outcome: *helm*

Reconstruction and comparative evidence Kroonen cites the helmet noun as **helma-* and separately distinguishes a different lexeme **helman-* ‘rudder’ (Kroonen 2013). The selected input **xélmaz* is the nominative-style form used for the helmet noun itself.

Old English evidence Clark Hall and Bosworth-Toller record *helm* as the ordinary Old English noun for ‘helmet’, while *helma* belongs to a separate rudder lexeme (Clark Hall 1960; Bosworth and Toller 1898, 542).

Development to Old English From **xélmaz*, loss of final *z* and later loss of the short final vowel yield *helm*. The development is therefore a straightforward citation-form match.

Form note Comparative **helma-* is headword notation for the helmet cognate set. It should not be confused with Old English *helma*, which is a different noun meaning ‘helm, rudder’ (Kroonen 2013; Clark Hall 1960).

help — OE helpan

Derivation: **xélpaną* > *helpan* (regular).

Derivation trace Proto input: **xélpaną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Velar Fricative Palatalization	**çélpaną*
[no change]		OE Heavy Syllable Nasal Apocope	**çélpan*
Northwest Germanic		OE Secondary Nasalization	**çélpąn*
[no change]		OE Weak Tail Reduction	**çélpan*

Outcome: *helpan*

Reconstruction and comparative evidence The entry treats the strong verb itself rather than the separate noun *help*. Bright’s principal parts *helpan, healp, hulpon, holpen* show the ordinary Old English strong-verb family continued by this input (Bright 1971).

Old English evidence Clark Hall and Bosworth-Toller record *helpan* as the verbal headword ‘to help’ (Clark Hall 1960; Bosworth and Toller 1898, 542). The target is therefore the attested infinitive citation form.

Development to Old English From **xélpaną*, no special repair is needed beyond the ordinary reduction of the infinitive ending. The derivation therefore reaches *helpan* directly.

Form note Noun *help* belongs to a separate lexical line and should not replace verbal *helpan* as the target here (Clark Hall 1960; Bosworth and Toller 1898, 542).

hind — OE hind

Derivation: **xéndjō* > *hind* (regular).

Derivation trace Proto input: **xéndjō*

Earlier Germanic changes		Old English changes	
West Germanic		OE Velar Fricative Palatalization	**çéndju*
[no change]		OE I Umlaut	**çindju*
Northwest Germanic		OE High Vowel Apocope	**çindj*
NWGmc Final Long O Raising	**xéndju*	OE J Loss After Heavy	**çind*

Outcome: *hind*

Reconstruction and comparative evidence Kroonen cites the animal name as **hindō-* f. 'hind' (Kroonen 2013). The selected input **xéndjō* represents that same noun in the present derivation.

Old English evidence Clark Hall and Bosworth-Toller record *hind* as the noun 'hind, female deer' (Clark Hall 1960; Bosworth and Toller 1898, 554). The target is therefore the attested lexical item itself.

Development to Old English From **xéndjō*, i-mutation produces the front-vocalic Old English stem, and later apocope plus loss of final *j* yield *hind*. The outcome is therefore a regular citation-form derivation.

Form note *hindan* 'from behind, behind' is a different Old English lexeme and does not belong to the noun history of *hind* (Clark Hall 1960; Bosworth and Toller 1898, 554).

hold — OE healdan

Derivation: **xáldaną* > *healdan* (regular).

Derivation trace Proto input: **xáldaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**xældaną*
[no change]	OE Breaking	**xealdaną*
Northwest Germanic	OE Velar Fricative Palatalization	**çealdaną*
[no change]	OE Heavy Syllable Nasal Apocope	**çealdan*
	OE Secondary Nasalization	**çealdąn*
	OE Weak Tail Reduction	**çealdan*

Outcome: *healdan*

Reconstruction and comparative evidence Campbell and Ringe-Taylor treat the verb as a regular **a + lC* breaking case, with West Saxon *healdan* opposed to Anglian and Mercian *haldan* (Campbell 1959; Ringe and Taylor 2014).

Old English evidence Bright gives the ordinary strong-verb citation form and principal parts *healdan, heold, heoldon, healden* (Bright 1971). The target is therefore the attested infinitive headword itself.

Development to Old English From **xáldaną*, Anglo-Frisian brightening first yields a fronted vowel, and West Saxon breaking then produces *ea* before *ld*. Later reduction of the infinitive ending gives *healdan* (Campbell 1959; Ringe and Taylor 2014).

Dialect note West Saxon *healdan* is the selected target here. Anglian and Mercian *haldan* are genuine non-West-Saxon doublets rather than corrections to that choice (Campbell 1959; Ringe and Taylor 2014).

horn — OE horn

Derivation: **xúrną* > *horn* (regular).

Derivation trace Proto input: _*xúrną_

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	_*xórn_
[no change]			
Northwest Germanic			
NWGmc U Lowering	_*xórną_		

Outcome: _horn_

Reconstruction and comparative evidence Kroonen and Orel cite lemma-style Proto-Germanic headwords of the _*hurna-_ / _*xurnan_ type for this noun (Kroonen 2013; Orel 2003, 234). The selected input _*xúrną_ is the nominative-style form used in the derivation here.

Old English evidence Clark Hall, Bosworth-Toller, and Bright all record _horn_ as the ordinary Old English noun (Clark Hall 1960; Bosworth and Toller 1898, 108; Bright 1971). The target is therefore the attested citation form.

Development to Old English From _*xúrną_, Northwest Germanic u-lowering gives _*xórną_, and later loss of the final nasal leaves _horn_. The development treated here is fully regular.

Form note The note's oblique _*xurnăn_ belongs to comparative stem background only. It does not replace the selected input _*xúrną_ as the derivational form used here (Kroonen 2013; Orel 2003, 234).

lead — OE lǣdan

Derivation: _*láidijaną_ > _lǣdan_ (regular).

Derivation trace Proto input: _*láidijaną_

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	_*lādijan_
PWGmc Ai Monophthongization	_*lādijaną_	OE Secondary Nasalization	_*lādijąn_
Northwest Germanic		Sievers Law Syncope	_*lādjąn_
[no change]		OE I Umlaut	_*lǣdjąn_
		OE Weak Tail Reduction	_*lǣdjan_
		OE J Loss After Heavy	_*lǣdan_

Outcome: _lǣdan_

Reconstruction and comparative evidence Ringe and Taylor derive Old English _lǣdan_ from Proto-Germanic _*laidijaną_, and Kroonen likewise cites a weak verb of the _*laidjan-_ type for 'lead' (Ringe and Taylor 2014; Kroonen 2013, 363).

Old English evidence Clark Hall and Bosworth-Toller both record _lædan_ / _lǣdan_ as the ordinary Old English verb 'to lead, guide, conduct' (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English From _*láidijaną_, monophthongization of _*ai_ first gives a _*lād-_ stage. Later syncope, i-mutation, weak-tail reduction, and loss of _j_ after a heavy stem yield _lǣdan_, so the development represented here is fully regular (Ringe and Taylor 2014).

learn — OE liornian

Derivation: _*líznōjaną_ > _liornian_ (regular).

Derivation trace Proto input: **líznōjaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Breaking	**líornōjaną*
[no change]	OE Heavy Syllable Nasal Apocope	**líornōjan*
Northwest Germanic	OE Secondary Nasalization	**líornōjąn*
[no change]	OE I Umlaut	**líornējąn*
	OE Unstressed Long Vowel Shortening	**líornejąn*
	OE Weak Tail Reduction	**líornejan*
	OE Intervocalic J Vocalization	**líorneian*
	OE Unstressed EI Contraction	**líornian*

Outcome: *liornian*

Reconstruction and comparative evidence Ringe and Taylor place the verb in a class-II weak family of the **liznō-* type (Ringe and Taylor 2014), and Kroonen keeps the same comparative base for the Germanic lexeme (Kroonen 2013). The selected input therefore requires no change of stem class or paradigm cell.

The non-obvious issue is dialectal. Campbell records Northumbrian *liornian* beside *leornian*, and he states that the *eo* of *leornian* is secondary, while Northumbrian preserves forms with *io*, where original *eo* and *io* remain distinct (Campbell 1959, sect. 123 n. 2).

Old English evidence The form modeled here is *liornian*, the Northumbrian member of the Old English family. Dictionary practice more often privileges *leornian* as the ordinary headword (Clark Hall 1960; Bright 1971), but Campbell's dialect evidence shows that *liornian* is a genuine Old English form (Campbell 1959, sect. 123 n. 2).

This entry therefore remains compact. The point is to state clearly that the selected target belongs to the Northumbrian side of the OE evidence rather than to the leveled *leornian* headword tradition.

Development to Old English From **líznōjaną*, the expected Old English developments include rhotacism of *z*, followed by breaking before *r* plus consonant, and the ordinary reduction of the weak verbal ending. The result is *liornian*, preserving the *io* spelling that Campbell associates with Northumbrian where original *eo* and *io* remain distinct (Campbell 1959, sect. 123 n. 2).

Leornian reflects the later *eo* development that Campbell treats as secondary [Campbell (1959), §123 n. 2; §296]. The selected Northumbrian target is therefore the regular comparison form for the *i*-grade member of the family.

Form comparison The comparison below is manual. It distinguishes the regular Northumbrian form modeled here from the better-known West Saxon headword.

Form	Status	Relevance to this entry
**líznōjaną* -> *liornian*	computed regular output; attested Northumbrian comparison form	selected comparison
leornian	attested later *eo* form and dictionary headword	useful control, but not the target of this entry

lid — OE hlid

Derivation: **xlídą* > *hlid* (regular).

Derivation trace Proto input: **xlídą*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**xlíd*
[no change]		
Northwest Germanic		
[no change]		

Outcome: *hlid*

Reconstruction and comparative evidence Orel cites a neuter lexeme of the **xliđ-* type with Old English *hlid*, and Lloyd includes OE *hlid* beside ON *hliþó* and OHG *(h)lit* among forms that retain *i* (Orel 2003; Lloyd 1966).

Old English evidence Clark Hall and Bosworth-Toller record *hlid* as the noun 'lid, cover, door, gate' (Clark Hall 1960; Bosworth and Toller 1898, 563).

Development to Old English The selected input already represents the later Germanic *hliđ-* stage used for the derivation here. From **xlídą*, heavy-syllable apocope yields *hlid*, and the form belongs to the retained-*i* set noted by Lloyd rather than to the lowered *e* type (Lloyd 1966).

Form note An earlier etymological stage **liþuz* belongs to comparative background only. The form represented here is the later **xlídą* > *hlid* line that matches the attested Old English noun (Orel 2003; Lloyd 1966).

light — OE līehtan

Derivation: **léuxtijaną* > *līehtan* (regular).

Derivation trace Proto input: **léuxtijaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Diphthong Leveling	**lēoxtijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**lēoxtijan*
Northwest Germanic	OE Secondary Nasalization	**lēoxtiją n*
[no change]	Sievers Law Syncope	**lēoxtją n*
	OE I Umlaut	**līextją n*
	OE Weak Tail Reduction	**līextjan*
	OE J Loss After Heavy	**līextan*

Outcome: *līehtan*

Reconstruction and comparative evidence Fulk gives Proto-Germanic **liuxtijanan* with Old English *līehtan* 'illuminate', and Ringe and Taylor likewise derive West Saxon *liehtan* from the same weak-verb formation (Fulk 2018; Ringe and Taylor 2014).

Old English evidence Clark Hall and Bosworth-Toller preserve the verb family under spellings such as *liehtan*, *lihtan*, and *līhtan*, distinct from the related noun *lēoht* and adjective *leoht/liht* (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English From **léuxtijaną*, the regular verbal line preserves **xt*, passes through a West Saxon *liehtan* stage, and is represented here by normalized *līehtan*. The word treated in this entry is therefore the verb 'to light, illuminate', not the related noun from **leuxtą* (Fulk 2018; Ringe and Taylor 2014).

Dialect note Ringe and Taylor and Campbell distinguish West Saxon *liehtan* from Anglian *lihtan*, while later West Saxon also shows *lyhtan* (Ringe and Taylor 2014; Campbell 1959).

linden — OE lind

Derivation: **líndō* > *lind* (regular).

Derivation trace Proto input: **líndō*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**línd*
[no change]			
Northwest Germanic			
NWGmc Final Long O Raising	**líndu*		

Outcome: *lind*

Reconstruction and comparative evidence Kroonen cites **lindō-* 'lime tree' and gives Old English *lind* as the relevant reflex (Kroonen 2013).

Old English evidence Clark Hall and Bosworth-Toller both record *lind* as the noun 'lime-tree, linden' (Clark Hall 1960; Bosworth and Toller 1898, 630).

Development to Old English From **líndō*, Northwest Germanic final **ō* raising first gives a **líndu* stage, and later high-vowel apocope yields *lind*. The development is therefore straightforward and regular.

Form note The Old English noun represented here is *lind*. Clark Hall also has a separate adjectival *linden* 'made of linden-wood', but that is not the noun counterpart for this entry (Clark Hall 1960).

milk — OE meoloc

Derivation: **mélukz* > *meoloc* (regular).

Derivation trace Proto input: **mélukz*

Earlier Germanic changes		Old English changes	
West Germanic		OE Med Unstressed U Lowering	**mélok*
[no change]		OE Back Mutation	**méolok*
Northwest Germanic			
PGmc Final Z Deletion	**méluk*		

Outcome: *meoloc*

Reconstruction and comparative evidence Kroonen and Orel reconstruct the noun as **meluk- / *melukz*, and the nominative-style input used here is **mélukz* (Kroonen 2013; Orel 2003, 306).

Old English evidence Old English preserves a mixed dossier for this noun. Ringe and Taylor describe West Saxon *meolc* < *meoluc* < **meluk*, Campbell likewise discusses *meoluc* and *meoloc*, and Anglian shows *milc* (Ringe and Taylor 2014; Campbell 1959).

Development to Old English The unsyncopated line from **mélukz* loses final **z*, lowers unstressed *u* to *o*, and with back mutation yields *meoloc*. That fuller unsyncopated outcome is the form represented here (Ringe and Taylor 2014).

Form comparison Syncopated *meolc* and Anglian *milc* belong to the competing leveled tradition associated with oblique forms, whereas *meoloc / meoluc* preserves the fuller nominal shape (Campbell 1959; Ringe and Taylor 2014).

mother — OE mōder

Derivation: **mōdēr* > *mōder* (regular).

Derivation trace Proto input: **mōdēr*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Long Vowel Shortening	**mōdær*
[no change]		OE Unstressed AE Merger	**mōder*
Northwest Germanic			
NWGmc Long E Lowering	**mōdǣr*		

Outcome: *mōder*

Reconstruction and comparative evidence Kroonen and Orel cite the Proto-Germanic r-stem kinship noun as **mōder- / *mōdēr* (Kroonen 2013; Orel 2003).

Old English evidence The transmitted Old English headword tradition is *mōdor / modor*, with oblique *mēder* in the paradigm. Clark Hall, Campbell, and Ringe and Taylor all preserve that contrast (Clark Hall 1960; Campbell 1959; Ringe and Taylor 2014).

Development to Old English From **mōdēr*, the regular suffixal development yields *mōder*. That regular nominative reflex is the form represented here, while the more familiar citation form *mōdor* reflects later levelling within the r-stem paradigm (Campbell 1959; Ringe and Taylor 2014).

Form comparison The note therefore concerns inherited vocalism rather than a different lexeme: *mōder* is the regularized nominative represented here, but dictionaries usually print *mōdor / modor*, and the oblique evidence survives in *mēder* (Clark Hall 1960; Ringe and Taylor 2014).

net — OE nett

Derivation: **nátją* > *nett* (regular).

Derivation trace Proto input: **nátją*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**nættją*
PWGmc J Gemination	**náttją*	OE Heavy Syllable Nasal Apocope	**nættj*
Northwest Germanic		OE I Umlaut	**nettj*
[no change]		OE J Loss After Heavy	**nett*

Outcome: *nett*

Reconstruction and comparative evidence Orel gives **natjan* with Old English *nett*, and Fulk's account of West Germanic gemination before *j* explains the geminate outcome after a short vowel (Orel 2003; Fulk 2018).

Old English evidence Clark Hall and Bosworth-Toller record *nett* as the noun, and Campbell notes that final geminates are often graphically simplified in Old English spelling (Clark Hall 1960; Bosworth and Toller 1898, 29; Campbell 1959).

Development to Old English From **nátją*, West Germanic j-gemination first gives **náttją*. Later brightening, loss of the weak ending, and loss of final *j* after a heavy stem yield *nett*, so the development represented here is regular (Fulk 2018).

Form note Spellings in *net* can therefore be graphic simplifications, but the lexical target supported by the dictionary evidence is *nett* (Campbell 1959; Orel 2003).

nightmare — OE mare

Derivation: **márōn* > *mare* (regular).

Derivation trace Proto input: **márōn*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**mærǭ*
[no change]		OE A Restoration	**marǭ*
Northwest Germanic		OE Unstressed Long Vowel Shortening	**marœ*
NWGmc N Stem N Loss	**márǭ*	OE Unstressed AE Merger	**mare*

Outcome: *mare*

Reconstruction and comparative evidence Ringe and Taylor treat the lexeme as Proto-Germanic / Proto-Northwest-Germanic **marōn-*, with Old English *mare, maran*, and variant *mere*; Orel preserves the same comparative lemma though with a different Old English headword tradition (Ringe and Taylor 2014; Orel 2003).

Old English evidence Clark Hall records *mare* 'nightmare, monster' and also preserves related variant forms *mera / mere* (Clark Hall 1960, 213).

Development to Old English The selected simplex input **márōn* regularly gives *mare* after brightening, A-restoration before the n-stem ending, and later reduction of the final vowel. The word represented here is the attested simplex noun, not an attested compound (Ringe and Taylor 2014).

Form note The concept corresponds to an unattested compound **nihtmare*, but the Old English lexical evidence is for simplex *mare*, with oblique *maran* and variant *mere / mera* (Ringe and Taylor 2014; Clark Hall 1960, 213).

coat — OE rocc

Derivation: **rúkkaz* > *rocc* (regular).

Derivation trace Proto input: **rúkkaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**rókk*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**rókkaz*		
PGmc Final Z Deletion	**rókka*		

Outcome: *rocc*

Reconstruction and comparative evidence Orel cites a masculine **rukkaz* for the garment word, while Kroonen gives **hrukka-*. Both treat this as the garment lexeme and not as the separate stone word (Orel 2003; Kroonen 2013, 290).

Old English evidence Clark Hall and Bosworth-Toller record *rocc* as an over-garment or tunic and preserve compounds such as *bisceoprocc* and *breóstrocc* (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English With **rúkkaz* as the selected input, Northwest Germanic u-lowering and later loss of final *-a* yield *rocc* as a regular outcome.

Source note This entry concerns the garment noun only. The stone word seen in *stānrocc* belongs to a different lexical history (Clark Hall 1960; Bosworth and Toller 1898).

sheep — OE sċēap

Derivation: **skḗpą* > *sċēap* (regular).

Derivation trace Proto input: **skḗpą*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**skǣp*
[no change]		OE Sk Palatalization	**ʃǣp*
Northwest Germanic		OE Ws Palatal Diphthongization	**ʃēap*
NWGmc Long E Lowering	**skǣpą*		

Outcome: *sċēap*

Reconstruction and comparative evidence Ringe and Taylor cite a later West Germanic **skap* > *WS scéap*, while Orel preserves a Proto-Germanic noun of the **skēp-* type for the same lexeme (Ringe and Taylor 2014; Orel 2003).

Old English evidence Clark Hall records *scēap* with spelling variation, and Campbell likewise lists West Saxon *scéap* among the palatal-diphthongized forms (Clark Hall 1960; Campbell 1959).

Development to Old English From **skḗpą*, Northwest Germanic lowering gives **skǣpą*; after apocope and palatalization the West Saxon branch diphthongizes to *sċēap*. The development represented here is therefore fully regular.

Dialect note Ringe and Taylor contrast West Saxon *scéap* with Mercian and Kentish *scép*, and Campbell also notes Northumbrian *scip*. The form represented here is the West Saxon headword (Ringe and Taylor 2014; Campbell 1959).

shilling — OE sċilling

Derivation: **skíllingaz* > *sċilling* (regular).

Derivation trace Proto input: **skíllingaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**skílling*
[no change]		OE Sk Palatalization	**ʃílling*
Northwest Germanic		OE Med Unstressed I Lowering1	**ʃílleng*
PGmc Final Z Deletion	**skíllinga*	OE Med Unstressed I Lowering	**ʃílling*

Outcome: *sċilling*

Reconstruction and comparative evidence Kroonen treats the cognate set under **skellinga- ~ *skillinga-* and connects it with **skeld-linga-*, while Orel likewise gives the coin word with OE *scilling* among the reflexes (Kroonen 2013; Orel 2003). The selected input **skíllingaz* is the nominative-style form used here to represent that inherited **-ing-* derivative.

Old English evidence Clark Hall records *scilling*, and Campbell cites *scilling* among nouns whose derivational *-ing* keeps *i* in unstressed syllables (Clark Hall 1960; Campbell 1959). The target represented here is the ordinary OE citation form, normalized as *sċilling*.

Development to Old English From **skíllingaz*, loss of final *-az* yields **skílling*. Old English palatalization of initial *sk* before front vocalism then gives *sċilling*. The note matters because derivational *-ing-* keeps *i*, so the regular outcome is *sċilling*, not **sċilleng* (Campbell 1959; Hogg 1992).

Form note Kroonen's **skellinga- ~ *skillinga-* and his internal analysis **skeld-linga-* belong to the etymological background of the cognate set. The selected input **skíllingaz* is the specific form used for the derivation represented here (Kroonen 2013).

show — OE sċēawian

Derivation: **skáwōjaną* > *sċēawian* (regular).

Derivation trace Proto input: **skáwōjaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Aw Long Diphthong	**skḗawōjaną*
[no change]	OE Heavy Syllable Nasal Apocope	**skḗawōjan*
Northwest Germanic	OE Secondary Nasalization	**skḗawōjąn*
[no change]	OE Sk Palatalization	**ʃḗawōjąn*
	OE I Umlaut	**ʃḗawējąn*
	OE Unstressed Long Vowel Shortening	**ʃḗawejąn*
	OE Weak Tail Reduction	**ʃḗawejan*
	OE Intervocalic J Vocalization	**ʃḗaweian*
	OE Unstressed EI Contraction	**ʃḗawian*

Outcome: *sċēawian*

Reconstruction and comparative evidence Orel and Kroonen cite a Class II verb of the type **skawōjan-*, with OE *scēawian* among the reflexes (Orel 2003; Kroonen 2013, 482). Brunner likewise records the Old English family as *scēawian, scāwian*, which places this entry in the ordinary show-verb set rather than in a special finite-cell workaround (Sievers 1965).

Old English evidence Bright lists *scēawian (W. II.)* and also the related form *scēawa* (Bright 1971). The source tradition therefore uses *scēawian*, while the target represented here is the normalized project spelling *sċēawian*.

Development to Old English From **skáwōjaną*, Old English *aw* before a following vowel yields *ēaw*, and the Class II suffix keeps **ō* between **w* and **j*. The development therefore runs regularly to *sċēawian*, without the direct **aw+j* problem seen in other verb types (Campbell 1959; Orel 2003).

Form note The difference between *scēawian* and *sċēawian* is orthographic normalization of initial , not a difference of lexeme or paradigm cell (Campbell 1959; Hogg 1992).

sleep — OE slǣpan

Derivation: **slḗpaną* > *slǣpan* (regular).

Derivation trace Proto input: **slḗpaną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**slǣpan*
[no change]		OE Secondary Nasalization	**slǣpąn*
Northwest Germanic		OE Weak Tail Reduction	**slǣpan*
NWGmc Long E Lowering	**slǣpaną*		

Outcome: *slǣpan*

Reconstruction and comparative evidence Kroonen preserves the comparative verb as **slēpan-*, and Fulk cites the same family under root **slēb-* (Kroonen 2013; Fulk 2018, 120). The selected input **slḗpaną* is the infinitive-style form used here for that inherited sleep-verb.

Old English evidence Clark Hall gives *slæpan* with preterite *slēp, slēap*, and Bright likewise lists *slæpan (slāpan), slēp slēpon slēpen* (Clark Hall 1960; Bright 1971, 435). The target represented here is therefore the normalized infinitive *slǣpan*, not the preterite forms and not the separate noun *slǣp*.

Development to Old English From **slḗpaną*, Northwest Germanic lowering gives **slǣpaną*. The later OE tail developments then yield *slǣpan* regularly. Brunner and Bülbring show that the OE tradition also has variant spellings such as West Saxon *slāpan/slæpan* and Anglian or Kentish *slēpan*, but those do not displace the infinitive chosen here (Sievers 1965; Bülbring 1902).

Form note The note concerns lemma type rather than a special derivational problem: this row represents the verb *slǣpan*, whereas *slǣp* belongs to noun or lookup background and *slēp/slēap* are preterite forms (Clark Hall 1960; Bright 1971, 435).

smear — OE smierwan

Derivation: **smérwijaną* > *smierwan* (regular).

Derivation trace Proto input: **smérwijaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Breaking	**sméorwijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**sméorwijan*
Northwest Germanic	OE Secondary Nasalization	**sméorwiją̨n*
[no change]	Sievers Law Syncope	**sméorwją̨n*
	OE I Umlaut	**smíerwją̨n*
	OE Weak Tail Reduction	**smíerwjan*
	OE J Loss After Heavy	**smíerwan*

Outcome: *smierwan*

Reconstruction and comparative evidence Kroonen gives the comparative headword as **smerwjan-* (Kroonen 2013). Ringe and Taylor instead cite a later-stage **smirwijana*, from which they derive West Saxon *smierwan*, Mercian *smirwan*, and Northumbrian *smiriga* (Ringe and Taylor 2014). The selected input **smérwijaną* therefore represents the Kroonen-aligned PGmc layer, while the later-stage dialect split belongs to a different chronological level.

Old English evidence The target represented here is the West Saxon citation form *smierwan*. Campbell's Anglian discussion explains the contrasting *smirwan*, and Clark Hall, Brunner, and Bright show that the same lexical family later also includes forms such as *smirian*, *smyrian*, and preterite *smyrode* (Campbell 1959; Clark Hall 1960; Sievers 1965; Bright 1971).

Development to Old English From **smérwijaną*, breaking before *r* + *consonant* yields *eo*, and later i-umlaut produces *ie*. The result is West Saxon *smierwan*. Anglian *smirwan* reflects the well-known failure of breaking in this environment, not a different lexeme (Campbell 1959; Ringe and Taylor 2014).

Dialect note The entry therefore represents the West Saxon member of a broader OE family: *smierwan* in West Saxon, *smirwan* in Anglian or Mercian, and related later class-II forms such as *smirian* or *smyrian* in the same lexical field (Ringe and Taylor 2014; Clark Hall 1960).

span — OE spannan

Derivation: **spánnaną* > *spannan* (regular).

Derivation trace Proto input: **spánnaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**spánnan*
[no change]	OE Secondary Nasalization	**spánną̨n*
Northwest Germanic	OE Weak Tail Reduction	**spánnan*
[no change]		

Outcome: *spannan*

Reconstruction and comparative evidence Kroonen cites the inherited verb as **spannan-* , with OE *spannan* among the reflexes (Kroonen 2013). The selected input **spánnaną* is the infinitive-style form used here for that same verbal lexeme.

Old English evidence Clark Hall keeps noun *spann* and verb *spannan* separate, and Brunner likewise records *sponnan, spannan stv.* (Clark Hall 1960; Sievers 1965). This entry treats the strong-verb infinitive, not the separate noun.

Development to Old English From **spánnaną*, the final nasal ending is lost and the regular OE weak-tail steps surface *spannan*. No paradigm-cell substitution is needed: the current derivation already lands on the infinitive directly.

Form note The note matters because English *span* can also reach noun *spann* in local lookup material. The entry represented here is the verb *spannan*, with the noun treated elsewhere (Clark Hall 1960).

spar — OE spearra

Derivation: **spárrô* > *spearra* (regular).

Derivation trace Proto input: **spárrô*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**spærrô*
[no change]	OE Breaking	**spearrô*
Northwest Germanic	OE Unstressed Long Vowel Shortening	**spearra*
[no change]		

Outcome: *spearra*

Reconstruction and comparative evidence Kroonen and Orel place this noun in the beam or rafter set **spar(r)an-*, with cognates such as Old Saxon and Old High German *sparro* (Kroonen 2013; Orel 2003). The selected input **spárrô* is the OE-facing nominal form used here for that same lexeme.

Old English evidence The noun represented here is *spearra*. The important lexical point is negative: English gloss overlap also reaches the unrelated verb *sperran* 'to bar', but that verb does not belong to this row.

Development to Old English From **spárrô*, Anglo-Frisian brightening gives **spærrô*, and OE breaking before geminate *rr* yields **spearrô*, later *spearra*. The development is therefore regular for a breaking-conditioned noun of this type (Luick 1914--1940).

Form note This entry concerns the noun *spearra* only. It should be kept separate from verb *sperran*, even though the Modern English glosses overlap (Kroonen 2013; Orel 2003).

still — OE stillan

Derivation: **stéllijaną* > *stillan* (regular).

Derivation trace Proto input: **stéllijaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**stéllijan*
[no change]	OE Secondary Nasalization	**stéllijąn*
Northwest Germanic	Sievers Law Syncope	**stélljąn*
[no change]	OE I Umlaut	**stilljąn*
	OE Weak Tail Reduction	**stilljan*
	OE J Loss After Heavy	**stillan*

Outcome: *stillan*

Reconstruction and comparative evidence The wider West Germanic family includes adjective *still* and verb *stillen* (Kluge 2011). The selected input **stéllijaną* represents the verbal j-formation used for the OE row.

Old English evidence Clark Hall gives *stillan* as the verb and separately *stille* as the adjective (Clark Hall 1960). Bosworth-Toller likewise preserves a substantial prefixed verbal family under *ge-stillan* and related forms (Bosworth and Toller 1898, 724). The selected target is the verb *stillan*, not the adjective.

Development to Old English As a heavy-stem Class I weak verb, **stéllijaną* undergoes the expected syncope and i-umlaut, and later loss of *j* after a heavy stem yields *stillan*. The development represented here is regular.

Form note The note concerns lexical framing rather than sound law: *stillan* is the verb represented here, while *stille* belongs to the related adjectival branch of the family (Clark Hall 1960; Kluge 2011).

summer — OE sumer

Derivation: **súmaraz* > *sumer* (regular).

Derivation trace Proto input: **súmaraz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**súmar*
[no change]		Anglo Frisian Brightening	**súmær*
Northwest Germanic		OE Unstressed AE Merger	**súmer*
PGmc Final Z Deletion	**súmara*		

Outcome: *sumer*

Reconstruction and comparative evidence Kroonen gives the lexeme as **sumara-*, and Ringe and Taylor likewise use **sumaraz*, while Orel preserves an alternate **sumeraz* (Kroonen 2013; Ringe and Taylor 2014; Orel 2003, 425). The selected input **súmaraz* follows the **a* vocalism that underlies the regular development represented here.

Old English evidence Clark Hall gives *sumor m., gs. sumeres, ds. sumera, sumere*, and Bright likewise lists *sumor (sumer)* with genitive *sumeres* (Clark Hall 1960; Bright 1971, 440). The tradition therefore preserves both *sumor* and *sumer*, with the oblique forms strongly supporting second-syllable *e*.

Development to Old English From **súmaraz*, loss of final *-az* is followed by fronting and merger in the unstressed second syllable, yielding *sumer*. The selected form is the regularized *e*-form, while the common citation form *sumor* remains part of the attested OE tradition (Ringe and Taylor 2014).

Form note The entry does not deny *sumor*. It represents *sumer* as the regular outcome chosen here, while *sumor* remains a common headword spelling and *sumeres/sumere* show that the *e*-vocalism was also real in Old English (Clark Hall 1960; Bright 1971, 440).

sunder — OE sundrian

Derivation: **súndrōjaną* > *sundrian* (regular).

Derivation trace Proto input: _*súndrōjaną_

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	_*súndrōjan_
[no change]		OE Secondary Nasalization	_*súndrōjąn_
Northwest Germanic		OE I Umlaut	_*súndrējąn_
[no change]		OE Unstressed Long Vowel Shortening	_*súndrejąn_
		OE Weak Tail Reduction	_*súndrejan_
		OE Intervocalic J Vocalization	_*súndreian_
		OE Unstressed EI Contraction	_*súndrian_

Outcome: _sundrian_

Reconstruction and comparative evidence Orel distinguishes three related formations: adverbial _*sunþraz_ > _sundor_, Class I verbal _*sunþrjanan_ > _syndrian_, and Class II verbal _*sunþrōjanan_ > _sundrian_ (Orel 2003). Kluge-Seebold aligns the cognate set with German _sondern_ and OE _gesundrian_, so this entry belongs with the Class II verb, not the adverb (Kluge 2011).

Old English evidence Clark Hall and Bosworth-Toller keep _sundrian_ and _syndrian_ separate from adverbial _sundor_, and both preserve the prefixed verbal family _ā-sundrian_ (Clark Hall 1960, 296; Bosworth and Toller 1898). The target represented here is therefore the weak verb _sundrian_.

Development to Old English From _*súndrōjaną_, the Class II weak-verb suffix yields regular OE _-ian_, producing _sundrian_. Because this is the _*-ōjan-_ verb and not the Class I _*-jan-_ formation, the form represented here does not belong to the umlauted _syndrian_ branch.

Form note The earlier confusion was lexical, not phonological: _sundor_ is the separate adverb, and _syndrian_ is a related but different verb. The verb treated here is the Class II verb _sundrian_ (Orel 2003; Clark Hall 1960, 296).

swallow — OE swealwe

Derivation: _*swálwōn_ > _swealwe_ (regular).

Derivation trace Proto input: _*swálwōn_

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	_*swœlwǭ_
[no change]		OE Breaking	_*swealwǭ_
Northwest Germanic		OE Unstressed Long Vowel Shortening	_*swealwœ_
NWGmc N Stem N Loss	_*swálwǭ_	OE Unstressed AE Merger	_*swealwe_

Outcome: _swealwe_

Reconstruction and comparative evidence Kroonen gives the bird name as _*swalwōn-_, and Ringe and Taylor cite the later West Germanic stage _*swalwa_, from which West Saxon _swealwe_ and Mercian _swalwe_ develop (Kroonen 2013, 535; Ringe and Taylor 2014, 200). The selected etymological comparison belongs to the swallow-bird family, not to the verb _swelgan_.

Old English evidence Clark Hall records _swealwe (a, o)_ as the noun headword (Clark Hall 1960). Campbell and Brunner also preserve later or oblique-family forms such as _swaluwe_, _swalewan_, and _swealuwe_, but those belong to wider variation around the noun rather than to the citation form represented here (Campbell 1959; Sievers 1965).

Development to Old English From **swálwōn*, brightening yields **swælw-*, and breaking before *lw* gives **swealw-*. The later noun ending develops regularly to *swealwe*. The relevant point is that the bird name has no inherited **g*: that consonant belongs to the separate verb *swelgan* (Ringe and Taylor 2014, 200; Kroonen 2013, 535).

Form note The final prose keeps the citation form *swealwe* separate from two different kinds of background material: the unrelated verb *swelgan*, and later or oblique spellings such as *swaluwe* or *swalewan* (Clark Hall 1960; Campbell 1959).

swine — OE swīn

Derivation: citation reconstruction **swī́ną*; selected input **swī́ną* > *swīn* (regular).

Derivation trace Proto input: **swī́ną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**swī́n*
[no change]		
Northwest Germanic		
[no change]		

Outcome: *swīn*

Reconstruction and comparative evidence Kroonen cites the noun as **swina-* ‘pig’ (Kroonen 2013). The selected comparative form here is the bare neuter citation cell **swī́ną*, which matches the singular noun aimed at in Old English rather than an oblique stem form.

Old English evidence Clark Hall records *swin (y)* as the ordinary noun headword (Clark Hall 1960). The target here is that singular citation form *swīn*, not a plural glossed in Modern English as *swine*.

Development to Old English From selected input **swī́ną*, loss of the final nasal vowel yields *swīn*. The outcome is therefore the regular monosyllabic noun with preserved long root *ī*.

Source note The selected input writes stressed long *ī* as **ī́*, so comparative **swī́ną* and derivational **swī́ną* represent the same lexical form.

think — OE þenċan

Derivation: **θánkijaną* > *þenċan* (regular).

Derivation trace Proto input: **θánkijaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**θánkijan*
[no change]	OE Secondary Nasalization	**θánkiją n*
Northwest Germanic	Sievers Law Syncope	**θánkjąn*
[no change]	OE Velar Palatalization	**θánʧjąn*
	OE I Umlaut	**θenʧjąn*
	OE Weak Tail Reduction	**θenʧjan*
	OE J Loss After Heavy	**θenʧan*

Outcome: *þenċan*

Reconstruction and comparative evidence Kroonen gives the verb as **þankjan-* ‘to think’, and Ringe and Taylor cite fully inflected **þankijaną* beside OE *þenċan* (Kroonen 2013; Ringe and Taylor 2014). The noun **þankaz* belongs only to the wider derivational background.

Old English evidence Bosworth-Toller preserves the verb under *þencan/geþencan*, and the citation form here is the ordinary infinitive *þenċan* (Bosworth and Toller 1898).

Development to Old English From **θánkijaną*, palatalization before **j* and i-umlaut produce *þenċan*. The infinitive is therefore a straightforward weak-verb outcome.

Lexical note Campbell’s assibilation discussion uses the same verb *þencan*; the class-III relic *hycgan* is a different lexeme (Campbell 1959; Hogg 1992).

thorn — OE þorn

Derivation: **θúrnaz* > *þorn* (regular).

Derivation trace Proto input: **θúrnaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**θórn*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**θórnaz*		
PGmc Final Z Deletion	**θórna*		

Outcome: *þorn*

Reconstruction and comparative evidence Kroonen gives **þurna-* ‘thorn, briar’, while Orel preserves the masculine pair **þurnuz* ~ **þurnaz* (Kroonen 2013; Orel 2003). The selected input **θúrnaz* belongs to that same comparative family.

Old English evidence Bright lists *þorn, m.*, and Clark Hall likewise treats *þorn* as the ordinary noun headword (Bright 1971; Clark Hall 1960).

Development to Old English The inherited stem shows regular lowering of *u* to *o* before *r*, and final loss yields *þorn*. The noun is therefore a regular Old English continuation of the Proto-Germanic thorn-family.

Source note The comparative sources preserve more than one stem formation, but the Old English target itself is simply the citation form *þorn*.

tide — OE tīd

Derivation: citation reconstruction **tī́diz*; selected input **tī́diz* > *tīd* (regular).

Derivation trace Proto input: **tī́diz*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**tī́d*
[no change]			
Northwest Germanic			
PGmc Final Z Deletion	**tī́di*		

Outcome: *tīd*

Reconstruction and comparative evidence Kroonen gives **tīdi-* 'time', and Orel's **tīđiz* points to the same feminine noun (Kroonen 2013; Orel 2003). The related verb *tīdan* is separate.

Old English evidence Bright records *tīd* with singular *tīde* and plural *tīda*, and Clark Hall treats *tīd* as the ordinary noun 'time, period, season' (Bright 1971; Clark Hall 1960, 309).

Development to Old English From **tī́diz*, final *z* is lost and the high final vowel drops, leaving *tīd*. The development is straightforward for a feminine i-stem.

Lexical note The note matters only because English *tide* can pull in the separate weak verb *tīdan*; the noun targeted here is *tīd*.

token — OE tācn

Derivation: **táiknq̨* > *tācn* (regular).

Derivation trace Proto input: **táikną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**tākn*
PWGmc Ai Monophthongization	**tākną*		
Northwest Germanic			
[no change]			

Outcome: *tācn*

Reconstruction and comparative evidence Kroonen cites **taikna-* and Orel **taiknan* for the noun 'sign, token' (Kroonen 2013; Orel 2003, 438). The selected input **táikną* is the simple citation-form noun used for the derivation.

Old English evidence Campbell and Sievers-Brunner preserve both unbroken *tācn* and broken *tācen*, with oblique *tācnes* remaining unbroken (Campbell 1959; Sievers 1965).

Development to Old English Monophthongization of *ai* yields *ā*, and loss of the final nasal vowel leaves *tācn*. The unbroken form is therefore a regular Old English outcome.

Form note *tācn* is the attested unbroken citation form selected here. Later West Saxon prose often prefers *tācen*, but that does not displace the older unbroken form (Campbell 1959; Sievers 1965).

town — OE tūn

Derivation: **tūną* > *tūn* (regular).

Derivation trace Proto input: **tūną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**tūn*
[no change]		
Northwest Germanic		
[no change]		

Outcome: *tūn*

Reconstruction and comparative evidence Kroonen cites **tūna-* 'fenced area', while Orel gives **tūnan* ~ **tūnaz* (Kroonen 2013; Orel 2003, 452). The selected input **tūną* is the simple citation-form noun used in the derivation.

Old English evidence Clark Hall records *tūn* as the ordinary headword 'enclosure, yard, village, town' (Clark Hall 1960).

Development to Old English The inherited long *ū* is preserved, and loss of the final nasal vowel yields *tūn* regularly (Sievers 1965).

Source note The comparative headwords vary, but the Old English target here is the direct citation form *tūn*, not an oblique **tūnăn*.

wade — OE wadan

Derivation: **wádaną* > *wadan* (regular).

Derivation trace Proto input: **wádaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**wædaną*
[no change]	OE A Restoration	**wadaną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**wadan*
	OE Secondary Nasalization	**wadąn*
[no change]	OE Weak Tail Reduction	**wadan*

Outcome: *wadan*

Reconstruction and comparative evidence Campbell and Ringe and Taylor describe A-restoration before a following back vowel, and Luick explicitly includes *wadan* among the standard open-syllable examples (Campbell 1959; Ringe and Taylor 2014; Luick 1914--1940, 239).

Old English evidence Clark Hall gives *wadan* as the verb 'to go, move, stride, advance', and Bright lists the same infinitive in the strong-verb paradigm (Clark Hall 1960; Bright 1971).

Development to Old English From **wádaną*, Anglo-Frisian brightening first gives **wædaną*. A-restoration before the back-vocalic infinitive ending then returns *a*, and later reduction yields *wadan* (Campbell 1959; Ringe and Taylor 2014).

Development note The note matters because this infinitive belongs to the A-restoration class. The citation form is therefore *wadan*, not a fronted *wæden*-type output.

warp — OE weorpan

Derivation: **wérpaną* > *weorpan* (regular).

Derivation trace Proto input: **wérpaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Breaking	**wéorpaną*
[no change]	OE Heavy Syllable Nasal Apocope	**wéorpan*
Northwest Germanic	OE Secondary Nasalization	**wéorpąn*
[no change]	OE Weak Tail Reduction	**wéorpan*

Outcome: *weorpan*

Reconstruction and comparative evidence Ringe and Taylor distinguish preterite **warp* from infinitive **werpana*, and the selected input here is the verbal form **wérpaną* (Ringe and Taylor 2014).

Old English evidence Clark Hall records *weorpan* as the strong verb headword and separately lists *wearp* as both noun and preterite. Bright gives the paradigm *weorpan, wearp, wurpon, worpen* (Clark Hall 1960; Bright 1971).

Development to Old English Breaking before *r* + *C* yields *weor-*, and the infinitive develops regularly to *weorpan* (Campbell 1959; Hogg 1992).

Lexical note The note matters because English *warp* also points to related *wearp* material. Here the target is specifically the infinitive *weorpan*.

wash — OE wascan

Derivation: **wáskaną* > *wascan* (regular).

Derivation trace Proto input: **wáskaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**wæskaną*
[no change]	OE A Restoration	**waskaną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**waskan*
[no change]	OE Secondary Nasalization	**waskąn*
	OE Weak Tail Reduction	**waskan*

Outcome: *wascan*

Reconstruction and comparative evidence Kroonen cites **waskan-*, Orel **waskanan*, and Ringe and Taylor likewise derive Old English *wascan* from the same verb family (Kroonen 2013; Orel 2003, 489; Ringe and Taylor 2014, 142).

Old English evidence Clark Hall heads the verb as *wascan*, while Sievers-Brunner also notes the variant *wæscan* (Clark Hall 1960; Sievers 1965).

Development to Old English From **wáskaną*, brightening gives **wæskaną*. A-restoration before the *sC* cluster restores *a*, and medial *sc* remains unpalatalized before the following back vowel, yielding *wascan* (Campbell 1959; Ringe and Taylor 2014, 142).

Form note The conservative citation form *wascan* is selected here. Spellings such as *wæscan* or *wasċan* belong to variant or normalized background rather than to the target of this entry.

wax — OE weaxan

Derivation: **wáxsaną* > *weaxan* (regular).

Derivation trace Proto input: **wáxsaną*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**wæxsaną*
[no change]		OE Breaking	**weaxsaną*
Northwest Germanic		OE Heavy Syllable Nasal Apocope	**weaxsan*
[no change]		OE Secondary Nasalization	**weaxsąn*
		OE Weak Tail Reduction	**weaxsan*
		OE Xs Merge	**weaXSan*

Outcome: *weaxan*

Reconstruction and comparative evidence Kroonen cites the verb as **wahs(j)an-*, Orel as **waxsanan*, and Ringe and Taylor discuss the prehistory of Old English *weaxan* within the same verbal family (Kroonen 2013; Orel 2003, 478; Ringe and Taylor 2014).

Old English evidence Clark Hall gives *weaxan* as the verb headword and separately records bare *wax* as a preterite form; Bright likewise treats *weaxan* as the infinitive (Clark Hall 1960; Bright 1971).

Development to Old English From **wáxsaną*, brightening and breaking yield *weax-*, and the infinitive develops regularly to *weaxan*. The cluster is preserved here, since *xs* > *s* belongs to forms where another consonant follows, such as *wæstm*, not to the infinitive itself (Campbell 1959; Sievers 1965).

Lexical note The target here is the infinitive *weaxan*. Noun *weax* and preterite *wax/wēox* belong to different lexical or paradigm slots.

way — OE weġ

Derivation: **wégaz* > *weġ* (regular).

Derivation trace Proto input: **wégaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**wég*
[no change]		OE Velar Palatalization	**wéʤ*
Northwest Germanic			
PGmc Final Z Deletion	**wéga*		

Outcome: *weġ*

Reconstruction and comparative evidence Kroonen cites the noun as **wega-* ‘way, road’, while the selected derivational form here is nominative-singular **wégaz* (Kroonen 2013). Campbell, Hogg, and Ringe and Taylor use the same word as the standard contrast between singular palatal *weġ* and inflected *wegas/wegum* (Campbell 1959; Hogg 1992; Ringe and Taylor 2014, 341).

Old English evidence The Old English singular is the ordinary noun *weg*, here normalized as *weġ* to show the palatal final. The contrasting plural and oblique forms *wegas, wegum* keep a velar stop before the following back vowel (Campbell 1959; Ringe and Taylor 2014, 341).

Development to Old English From **wégaz*, final **z* is lost and the weak tail apocopates, leaving word-final **g* after a front vowel. In that environment Old English palatalization yields *weġ*, whereas *wegas* remains velar because the following *a* blocks the same outcome (Campbell 1959; Hogg 1992; Ringe and Taylor 2014, 341).

Form note Normalized *weġ* and dictionary *weg* represent the same noun. *wē* is not supported in the checked Old English evidence for 'way'.

weapon — OE wǣpn

Derivation: **wḗpną* > *wǣpn* (regular).

Derivation trace Proto input: **wḗpną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**wǣpn*
[no change]			
Northwest Germanic			
NWGmc Long E Lowering	**wǣpną*		

Outcome: *wǣpn*

Reconstruction and comparative evidence Kroonen reconstructs a double-stem noun **wēbna-* ~ **wēpna-* and cites OE *wæpn* among its reflexes (Kroonen 2013, 617). The selected input **wḗpną* represents the unbroken citation-form noun rather than the later broken simplex.

Old English evidence Campbell's cluster-noun discussion preserves unbroken *wépn* beside broken *wépen*-type forms (Campbell 1959, 150; 1959, 226–27). Bright contrasts broken nominative *wǣpen/wapen* with unbroken oblique *wǣpnes*, while Clark Hall lemmatizes the noun under *wapen* and also preserves unbroken forms in compounds and related spellings (Bright 1971, 29; Clark Hall 1960, 355).

Development to Old English Northwest Germanic lowering gives *wǣpn*, and loss of the final nasal vowel leaves the unbroken cluster word-finally. The selected target is the attested unbroken form *wǣpn*.

Form note The ordinary late West Saxon simplex headword is *wǣpen*, but Campbell's noun-class discussion also preserves unbroken *wépn* beside broken *wépen*-type forms (Campbell 1959, 150; 1959, 226–27). *wǣpnes* remains the regular unbroken oblique comparator (Bright 1971, 29; Clark Hall 1960, 355).

will — OE willa

Derivation: **wéljô* > *willa* (regular).

Derivation trace Proto input: **wéljô*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	**willjô*
PWGmc J Gemination	**wélljô*	OE Unstressed Long Vowel Shortening	**willja*
Northwest Germanic		OE J Loss After Heavy	**willa*
[no change]			

Outcome: *willa*

Reconstruction and comparative evidence Kroonen separates noun **weljan- 2* 'will, wish' from verb **weljan- 1* 'to want', while Orel and Kluge represent the noun as **weljōn/*weljOn* (Kroonen 2013; Orel 2003; Kluge 2011). The selected derivational form **wéljô* is the noun-side input used for this row.

Old English evidence Clark Hall lemmatizes noun *willa m.* separately from verb *willan* (Clark Hall 1960, 368). The selected target is the noun citation form, not the related verb.

Development to Old English From **wéljô*, j-gemination yields a heavy stem, i-umlaut gives *will-*, and later shortening plus j-loss produce *willa*. The noun is therefore a regular weak masculine outcome.

Lexical note The target here is the noun *willa* 'will, wish'. Related verb *willan* belongs to a separate lexeme and should not be substituted for the noun row (Kroonen 2013; Clark Hall 1960, 368).

wind — OE windan

Derivation: **wíndaną* > *windan* (regular).

Derivation trace Proto input: **wíndaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**wíndan*
[no change]	OE Secondary Nasalization	**wíndąn*
Northwest Germanic	OE Weak Tail Reduction	**wíndan*
[no change]		

Outcome: *windan*

Reconstruction and comparative evidence Kroonen distinguishes noun **winda-* from verb **windan-*, and the present row belongs to the verb (Kroonen 2013). Later handbook discussion keeps the dental original from PIE **wendh-*, not a Verner alternant (Fulk 2018; Ringe and Taylor 2014).

Old English evidence Clark Hall and Bosworth-Toller record *windan* as the verb headword (Clark Hall 1960; Bosworth and Toller 1898, 101). The selected target is the ordinary infinitive of the strong verb.

Development to Old English The selected input **wíndaną* yields the regular infinitive *windan* by ordinary heavy-syllable apocope and weak-tail reduction. The form is therefore a straightforward strong-verb outcome.

Lexical note The note matters because English *wind* also names the noun. This row targets the class-III verb, not the noun (Kroonen 2013; Clark Hall 1960).

wold — OE weald

Derivation: **wálθuz* > *weald* (regular).

Derivation trace Proto input: **wálθuz*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**wældu*
PWGmc L Th Voicing	**wálduz*	OE Breaking	**wealdu*
Northwest Germanic		OE High Vowel Apocope	**weald*
PGmc Final Z Deletion	**wáldu*		

Outcome: *weald*

Reconstruction and comparative evidence Kroonen reconstructs the noun as **walþu-* and gives OE *weald* beside other West Germanic *wald* forms (Kroonen 2013). The selected input **wálθuz* is the nominative singular used for the derivation.

Old English evidence Clark Hall makes *weald* the main noun headword and cross-refers *wald* and *wold* to it (Clark Hall 1960). The Anglian-looking *wald* therefore remains variant background rather than the main target.

Development to Old English **lþ* voices to *ld*, Anglo-Frisian brightening yields *wæld-*, and breaking before the cluster gives *weald-*; apocope then yields *weald*. The noun is therefore a regular breaking outcome.

Dialect note The note matters because *wald* survives as an Anglian-type variant in the same family. The selected target is normalized *weald*, not the variant form (Clark Hall 1960; Ringe and Taylor 2014).

yarn — OE ġearn

Derivation: **gárną* > *ġearn* (regular).

Derivation trace Proto input: **gárną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**gærną*
[no change]	OE Breaking	**gearną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**gearn*
[no change]	OE Velar Palatalization	**ʤearn*

Outcome: *ġearn*

Reconstruction and comparative evidence Kroonen cites the noun as **garna-*, and Ringe and Taylor give the early chain **garna* > **geern* > **gearn* > *OE gearn* (Kroonen 2013; Ringe and Taylor 2014, 220). The selected input **gárną* is the nominal citation form used here, while oblique **garnăn* belongs only to comparative background.

Old English evidence Clark Hall records *gearn (e) n.* 'yarn, spun wool', and Bosworth-Toller glosses *gearn* as *filatum* (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English From **gárną*, brightening and breaking before *rn* yield *gearn*; palatalization of initial *g* before the resulting front-vocalic sequence gives normalized *ġearn*. The derivation is regular.

Form note Dictionary *gearn* and normalized *ġearn* refer to the same noun. The comparative stem **garna-* and oblique **garnăn* do not replace the selected input **gárną*.

Part II. Attested variants and selected comparison forms

These entries treat the selected Old English target as one member of an attested or historically documented variant set. The target is therefore anchored in the record, but the lexical comparison must account for variation rather than for a single unproblematic citation form.

cud — OE cwedu

Derivation: citation reconstruction **kwíθuz*; selected input **kwéðuz* > *cwedu* (attested variant).

Derivation trace Proto input: **kwéðuz*

Earlier Germanic changes		Old English changes
West Germanic		[no change]
PWGmc Dental Hardening	**kwéduz*	
Northwest Germanic		
PGmc Final Z Deletion	**kwédu*	

Outcome: *cwedu*

Reconstruction and comparative evidence Kroonen reconstructs the resin word as **kwedu-2* and gives Old English variants *cwidu*, *cweodu*, and *c(w)udu* (Kroonen 2013, 355). Orel likewise lists *cwidu* under the cognate set (Orel 2003, 266). The selected input **kwéðuz* therefore represents the older e-grade, voiced-dental form behind the chosen variant *cwedu*.

Old English evidence The Old English word survives in a wider variant set than one dictionary headword suggests. Ringe and Taylor discuss *cwidu* > *cwudu* > *cudu* and also note late West Saxon *cweodu*; Clark Hall gives *cwudu*, *cweodu*, and *cudu* (Ringe and Taylor 2014, 338; Clark Hall 1960, 84). Attested *cwedu* is treated here as the conservative variant within that set.

Development to Old English From **kwéðuz*, the West Germanic voiced dental hardens in the expected way and the regular Old English development yields *cwedu*. The other Old English spellings belong to the same lexical family, but reflect later leveling, back-umlaut, or further reduction rather than a need to replace the selected input.

Variant comparison

Variant type	Old English form	Comment
conservative target	*cwedu*	selected attested variant represented here
leveled i-grade form	*cwidu*	common lexical variant in the same family
back-umlauted forms	*cweodu*, *cwudu*	later developments within the same OE tradition
reduced form	*cudu*	further reduced member of the same variant set

ten — OE tēon

Derivation: **téxun* > *tēon* (attested variant).

Derivation trace Proto input: **téxun*

Earlier Germanic changes	Old English changes	
West Germanic	OE Med Unstressed U Lowering	**téxon*
[no change]	OE Breaking	**téoxon*
Northwest Germanic	OE H Loss	**téoon*
[no change]	OE Contraction	**tḗon*

Outcome: *tēon*

Reconstruction and comparative evidence Fulk states that Old English *tien* shows umlaut from the inflected forms, whereas the uninflected form without umlaut is reflected in *hund-tēontig* (Fulk 2018, sect. 10.2). Brunner gives the same contrast more broadly: *tēon* develops from **tĕhun*, while West Saxon *tien, tȳn* belong to a different, umlauted branch of the numeral history (Sievers 1965, sects 129.2, 129 Anm. 6, 234).

The selected comparison form *tēon* therefore represents the bare cardinal's un-umlauted line, not the later umlauted simplex tradition.

Old English evidence The attested simplex forms are varied. Campbell gives *tien*, northwestern West Saxon *tēn*, and late Northumbrian *tēo, tēa* (Campbell 1959, sect. 682). Brunner likewise lists West Saxon *tien, tȳn* beside *tēn, tēo, tēa* in other dialects (Sievers 1965, sect. 325).

Exact simplex *tēon* is weaker as a directly cited headword than those spellings. The un-umlauted stem is, however, explicit in *tēoða* and *-tēontig* (Sievers 1965, sect. 129.2; Fulk 2018, sect. 10.2). The comparison form *tēon* is therefore a **normalized spelling** of that un-umlauted base.

Development to Old English From **téxun*, lowering of medial unstressed *u* gives **téxon*, breaking gives **téoxon*, loss of intervocalic *h/x* yields **téoon*, and contraction produces **tḗon*, written *tēon*. This is the regular bare-cardinal path.

The umlauted forms *tien / tīen* belong to a different branch, created when the numeral was levelled from inflected forms with a front-vocalic trigger (Fulk 2018, sect. 10.2; Sievers 1965, sect. 129 Anm. 6).

Variant comparison The comparison below is manual. It distinguishes the normalized un-umlauted base from the attested simplex variants.

Form or branch	Status	Relevance to this entry
tēon	normalized un-umlauted comparison form; trace-supported	selected target
tien / tīen	attested West Saxon umlauted simplex forms	genuine OE variants, but not the bare-cardinal line modeled here
tēn / tēo / tēa	attested un-umlauted simplex variants in other dialects	support the same branch as the selected comparison form

three — OE þrīe

Derivation: **θréjez* > *þrīe* (attested variant).

Derivation trace Proto input: **θréjez*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	**θrije*
[no change]		OE Intervocalic J Vocalization	**θriie*
Northwest Germanic		OE Contraction	**θrīe*
PGmc Final Z Deletion	**θréje*		

Outcome: *þrīe*

Reconstruction and comparative evidence Kroonen cites the numeral under a broader stem-style reconstruction rather than under one Old English-ready paradigm cell (Kroonen 2013). The input **θréjez* is therefore best understood as the inherited masculine nominative-accusative singular.

That distinction matters because the Old English numeral does not have one uniform citation form across the paradigm. The masculine singular line must be kept apart from feminine-neuter *þrēo* and from later reduced spellings of the masculine form.

Old English evidence Campbell gives masculine nominative-accusative *þrīe*, feminine and neuter nominative-accusative *þrēo*, genitive *þrēora*, and dative *þrim*, adding that late West Saxon has *þry, þri* for *þrīe* (Campbell 1959, sect. 683). Fulk presents the same masculine *þrīe* beside the wider numeral paradigm (Fulk 2018, sect. 10.1).

The target is therefore an attested Old English paradigm cell. *þrī* belongs to later reduction or headword-style citation, whereas *þrīe* is the conservative masculine nominative-accusative form.

Development to Old English From **θréjez*, loss of final *-z* leaves a form of the **θréje* type. The following *j* fronts the stem vowel, then vocalizes between vowels, and contraction yields *þrīe*. The compact trace records the same sequence as **θrije* > **θriie* > *þrīe*.

Variant comparison The comparison below is manual. It separates the selected masculine cell from the later reduced form and from the rest of the numeral paradigm.

Form	Status	Relevance to this entry
þrīe	attested masculine nom./acc.; trace output	selected target
þrī / þry	later reduced masculine variant	genuine OE variant, but not the conservative comparison form
þrēo	attested feminine-neuter nom./acc.	same numeral, different paradigm cell
þrēora, þrim	attested genitive and dative forms	confirm the wider paradigm, not the selected cell

wasp — OE wæfs

Derivation: **wábsaz* > *wæfs* (attested variant).

Derivation trace Proto input: **wábsaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**wábs*
[no change]		Anglo Frisian Brightening	**wæbs*
Northwest Germanic		PGmc B Allophony	**wæβs*
PGmc Final Z Deletion	**wábsa*		

Outcome: *wæfs*

Reconstruction and comparative evidence The Proto-Germanic form **wábsaz* reaches Old English without any special change of stem or paradigm cell. The question in this entry is instead which attested Old English member of the variant set should serve as the comparison form.

Fulk presents the Old English forms together as *wæfs* with variants *wæsp* and *wæps* (Fulk 2018, sect. 6.5). Bülbring and Brunner then make the chronology more explicit by deriving later *wæps* and late West Saxon *wasp* from earlier *waefs* / *wæfs* through restricted metatheses (Bülbring 1902, sect. 484 Anm. 3; Sievers 1965, sects 193, 204).

Old English evidence The earliest directly cited Old English form is *wæfs*, written *waefs* in the Épinal-Corpus material discussed by Bülbring and Brunner (Bülbring 1902, sect. 484 Anm. 3; Sievers 1965, sect. 193). Later Old English also shows *wæps* and *wæsp* / *wasp*, and dictionary practice often favors *wæps* or later spellings as headwords (Clark Hall 1960).

This entry therefore distinguishes chronological priority from headword habit. *wæfs* is not a convenient reconstruction: it is an attested Old English form and also the one that matches the regular development most closely.

Development to Old English From **wábsaz*, the regular Old English path passes through loss of final *z*, Anglo-Frisian fronting, and the allophonic development of *b* to a fricative before *s*, yielding *wæfs*.

The later forms *wæps* and *wæsp* / *wasp* belong to subsequent, lexically restricted metatheses. They are genuine Old English forms, but they are later within the variant history.

Variant comparison The comparison below is manual. It separates the earliest attested and regular form from the later metathesized doublets.

Form	Status	Relevance to this entry
wæfs	earliest attested OE form; regular trace output	selected target
wæps	later attested metathesized variant	genuine OE doublet, but secondary
wæsp / *wasp*	later West Saxon metathesized variant	genuine OE doublet, but not the selected form

Part III. Early analogy and pre-Old-English input selection

These entries involve a distinction between the lexeme-level citation reconstruction and the form selected as input to the Old English derivation. The issue is upstream of Old English: the selected input represents the pre-Old-English form that gives the attested target under the current cascade.

bottom — OE botm

Derivation: citation reconstruction **búdmaz*; selected input **búttmaz* > *botm* (early analogy).

Derivation trace Proto input: **búttmaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**bóttm*
[no change]		OE Preconsonantal Degemination	**bótm*
Northwest Germanic			
NWGmc U Lowering	**bóttmaz*		
PGmc Final Z Deletion	**bóttma*		

Outcome: *botm*

Reconstruction and comparative evidence Kroonen reconstructs the word as a stem complex **budmō, gen. *buttaz*, summarized as **budman-* ~ **buttman-*, and gives Old English *botm* as the reflex (Kroonen 2013, 120). The comparative label **búdmaz* names the lexeme-level stem complex, while the selected input **búttmaz* represents the pre-Old-English form with oblique **butt-* generalized into the nominative formation.

Orel likewise preserves both sides of the comparison under **budmaz *butmaz* (Orel 2003, 100). The selected input is thus a historical stem choice, not an arbitrary respelling.

Old English evidence The Old English noun itself is secure. Clark Hall gives *botm* (Clark Hall 1960, 63). Bosworth-Toller cross-references *bodan* to *botm*, showing the wider reflex family without weakening the attested lemma (Bosworth and Toller 1898, 112).

Development to Old English Once the oblique **butt-* stem has been generalized, the selected input **búttmaz* develops regularly to *botm*. The analogical step is therefore early: it belongs to pre-Old-English stem formation rather than to a later choice among Old English paradigm cells.

brand — OE brandes

Derivation: citation reconstruction **brándaz*; selected input **brándas* > *brandes* (early analogy).

Derivation trace Proto input: **brándas*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**brándæs*
[no change]		OE Unstressed AE Merger	**brándes*
Northwest Germanic			
[no change]			

Outcome: *brandes*

Reconstruction and comparative evidence The inherited noun is the masculine a-stem **brándaz*, continued by Old English *brand* and its continental cognates (Orel 2003; Kroonen 2013). The selected input **brándas* is not a different lexeme but the genitive singular of that same a-stem noun.

What matters here is therefore not a stem-class disagreement but the difference between the citation form and a specific inherited inflectional cell. The selected input preserves the same root and declension as the headword while making the oblique ending explicit.

Old English evidence Old English dictionaries lemmatize the noun as *brand* (Clark Hall 1960; Bosworth and Toller 1898, 116). Bosworth-Toller also records inflectional forms such as *brandas*, *branda*, and *brandum* under the same entry (Bosworth and Toller 1898, 116).

The specific comparison form in this entry, *brandes*, is the expected genitive singular of that a-stem noun. It is therefore an inferred Old English paradigm form rather than the ordinary dictionary headword.

Development to Old English From **brándas*, the regular Old English development passes through the usual unstressed-vowel weakening of the inflectional ending, yielding *brandes*. Nothing in the stem itself requires a special repair. The root consonants and the stressed vowel are the same as in the citation lemma *brand*.

The analytical weight of the entry lies in the ending. By choosing the oblique singular rather than the nominative citation form, the entry presents the same lexeme in a different inherited cell.

Form comparison The comparison below is manual. It separates the citation lemma from the selected oblique singular.

Form / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	*brándaz	expected OE lemma *brand*	brand	regular headword-level outcome
selected genitive singular	*brándas	compact-trace output: *brandes*	brandes	exact match for the chosen oblique cell

The noun itself is straightforwardly inherited. The main point of the entry is that *brandes* belongs to the same regular a-stem paradigm as *brand*, even though the citation lemma remains the nominative singular.

breast — OE brēost

Derivation: citation reconstruction **brústz*; selected input **bréustą* > *brēost* (early analogy).

Derivation trace Proto input: **bréustą*

Earlier Germanic changes	Old English changes	
West Germanic	OE Diphthong Leveling	**brēostą*
[no change]	OE Heavy Syllable Nasal Apocope	**brēost*
Northwest Germanic		
[no change]		

Outcome: *brēost*

Reconstruction and comparative evidence The word family shows two related but distinct Proto-Germanic formations. The root noun **brust-* lies behind forms such as Gothic *brusts*, whereas Old English *brēost* belongs to a thematic formation **breusta-*, alongside Old Norse *brjóst* and Old Saxon *briost* (Kroonen 2013; Orel 2003, 95; Ringe and Taylor 2014, 43).

The selected input **bréustą* therefore differs from the citation label **brústz* because Old English reflects the thematic branch rather than the root noun. The morphological choice comes before the Old English sound changes themselves.

Old English evidence Old English dictionaries record the noun as *brēost* / *breóst* (Bosworth and Toller 1898; Clark Hall 1960, 65). The form is an established Old English lexeme, not a reconstructed target assembled from comparative evidence alone.

What requires explanation is not the Old English attestation but the relation between that attested noun and the broader Germanic word family. The relevant comparison form is therefore the thematic Old English noun *brēost*.

Development to Old English From **bréustą*, the regular Old English development gives *brēost*, with the expected *eu* > *ēo* vowel history (Campbell 1959). No special repair is needed once the correct thematic formation is chosen.

The earlier mismatch arose only if the word was forced into the root-noun line. The Old English noun itself continues the thematic branch cleanly and directly.

Formation comparison The comparison below is manual. It separates the broader root-noun family label from the thematic formation actually continued in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader root-noun family	*brústz	root-noun type outcomes outside OE	non-OE comparanda	useful family label, but not the direct source of *brēost*
selected thematic formation	*bréustą	compact-trace output: *brēost*	brēost	exact match between formation and attested OE noun

The relevant point is the formation split. *brēost* is the regular Old English outcome of the thematic **breusta-* branch, not of the root noun **brust-*.

craft — OE cræft

Derivation: citation reconstruction **kráftiz*; selected input **kráftaz* > *cræft* (early analogy).

Derivation trace Proto input: **kráftaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**kráft*
[no change]		Anglo Frisian Brightening	**krǽft*
Northwest Germanic			
PGmc Final Z Deletion	**kráfta*		

Outcome: *cræft*

Reconstruction and comparative evidence Comparative sources disagree about the older stem class. Kroonen gives a u-stem **kraftu-*, while Orel prints **kraftiz* ~ **kraftuz* (Kroonen 2013, 340; Orel 2003, 259). The comparative label **kráftiz* remains in view as a lexeme-level shorthand, while **kráftaz* is the pre-Old-English form used for the Old English derivation.

Old English evidence The Old English noun itself is secure. Clark Hall and Bosworth-Toller both give *crœft* as the headword (Clark Hall 1960, 19; Bosworth and Toller 1898, 145).

Development to Old English The comparison is between possible pre-Old-English inputs. The i-stem comparator **kráftiz* gives *creft*, while the u-stem comparator **kráftuz* gives *craft*. The a-stem-shaped input **kráftaz* yields *crœft* and is therefore the form used for the Old English derivation. This does not require treating the comparative dictionaries as identical; it shows the narrower point that the Old English derivation needs a pre-Old-English form without the i-umlaut trigger of **-iz* and without the back-vowel outcome associated with the u-stem comparator.

Form comparison

Candidate input	OE output	Result
*kráftiz	*creft*	non-match; i-stem comparator
*kráftuz	*craft*	non-match; u-stem comparator
*kráftaz	*crœft*	exact match; selected pre-OE input

dill — OE dile

Derivation: citation reconstruction **déljaz*; selected input **déliz* > *dile* (early analogy).

Derivation trace Proto input: **déliz*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	**dili*
[no change]		OE Med Unstressed I Lowering1	**dile*
Northwest Germanic			
PGmc Final Z Deletion	**déli*		

Outcome: *dile*

Reconstruction and comparative evidence Kroonen treats the word as preserving evidence for both an i-stem and a ja-stem formation, with Old English *dile* on one side and continental forms such as Old Saxon *dilli* and Old High German *tilli* on the other (Kroonen 2013). The selected input **déliz* therefore represents the i-stem side of the paradigm, whereas the citation label **déljaz* is a broader comparative headword.

That stem-class distinction matters for the Old English consonant shape. A ja-stem with **-lj-* would be expected to produce gemination, but the Old English noun shows a single *l*. Fulk's discussion of ja-stems transferred to the i-stems provides the relevant morphological background for the OE side (Fulk 2018).

Old English evidence Old English dictionaries record the plant name as *dile*, alongside the variant *dili* (Bosworth and Toller 1898, 164; Clark Hall 1960). The form discussed here is therefore an attested Old English noun with single *l*.

The Old English evidence is the relevant point. Whatever broader comparative headword is chosen for the family, the inherited form reflected in OE is the i-stem type *dile*, not a geminated *dill* outcome.

Development to Old English From **déliz*, regular loss of final *z* and the later lowering of unstressed *i* yield *dile*. The stem itself remains ungeminated throughout that path.

The important contrast is negative rather than phonological. If the word were forced through a ja-stem **-lj-* pathway, the expected result would show *ll*. The attested Old English noun instead matches the i-stem development.

Formation comparison The comparison below is manual. It separates the broader comparative headword from the stem class actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative ja-stem label	*déljaz	ja-stem type outcome with gemination	dill-type comparison	useful comparative label, but not the OE form
selected i-stem formation	*déliz	compact-trace output: *dile*	dile	exact match between formation and attested OE noun

The single *l* is the decisive diagnostic. It identifies *dile* with the i-stem formation rather than with the continental ja-stem branch.

fast — OE festan

Derivation: citation reconstruction **fastēną*; selected input **fástijaną* > *festan* (early analogy).

Derivation trace Proto input: **fástijaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**fæstijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**fæstijan*
Northwest Germanic	OE Secondary Nasalization	**fæstiją̄n*
[no change]	Sievers Law Syncope	**fæstją̄n*
	OE I Umlaut	**festją̄n*
	OE Weak Tail Reduction	**festjan*
	OE J Loss After Heavy	**festan*

Outcome: *festan*

Reconstruction and comparative evidence Kroonen places the verb under a comparative headword **fastēną*, the wider Germanic family behind meanings such as 'make firm' and, in Old English, 'fast' (Kroonen 2013). Ringe and Taylor, however, distinguish the Old English verb more closely: they treat OE 'to fast' as originally a class-I weak verb that later acquired the stative meaning through lexical association with that wider family (Ringe and Taylor 2014).

The selected input **fástijaną* therefore represents the inherited class-I formation reflected in Old English, whereas the citation label **fastēną* belongs to the broader comparative presentation of the lexeme.

Old English evidence Old English dictionaries record forms such as *festan*, alongside related *fæstan* / *fǣstan* spellings and meanings (Bosworth and Toller 1898, 213; Clark Hall 1960). The form selected here is *festan*, which fits the regular class-I phonological development.

The *æ*-forms remain relevant, but they do not control the entry. As Ringe and Taylor argue, their vowel belongs to later analogical leveling under the adjective *fæst*, whereas *festan* reflects the regular inherited class-I verb (Ringe and Taylor 2014).

Development to Old English From **fástijaną*, Anglo-Frisian brightening and subsequent i-umlaut produce the fronted vowel seen in *festan*. The later weak-tail reductions and loss of *j* after a heavy syllable complete the regular Old English outcome.

What makes the entry non-regular is not the phonology of *festan* itself, but the choice of formation. Old English continues the class-I verb, even though the comparative headword is often given under the parallel **fastēn-* family.

Class comparison The comparison below is manual. It distinguishes the comparative class-III headword from the class-I formation actually reflected in Old English.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative class-III headword	*fastēną	class-III type outcome, not *festan*	wider family context	useful family label, but not the direct source of the target
selected class-I weak verb	*fástijaną	compact-trace output: *festan*	festan	exact match between formation and attested OE verb
later analogical reshaping	adjective-driven *fæst* influence	*fæstan* / *fǣstan* type spellings	fæstan-type evidence	genuine later OE reshaping, but secondary to the selected target

The relevant point is the class split. *festan* is the regular Old English outcome of the class-I formation, while the better-known *æ*-forms belong to a later analogical layer.

flask — OE flasce

Derivation: citation reconstruction **flaskō*; selected input **fláskōn* > *flasce* (early analogy).

Derivation trace Proto input: **fláskōn*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**flæskǭ*
[no change]		OE A Restoration	**flaskǭ*
Northwest Germanic		OE Unstressed Long Vowel Shortening	**flaskæ*
NWGmc N Stem N Loss	**fláskǭ*	OE Unstressed AE Merger	**flaske*

Outcome: *flasce*

Reconstruction and comparative evidence The wider Germanic family is often cited under a form such as **flaskō*, but the evidence relevant for Old English points instead to a weak feminine formation **fláskōn* / **flaskǭ* (Orel 2003; Kroonen 2013). That distinction is crucial for the suffixal history of the noun.

The selected input therefore differs from the citation label in stem class. Old English *flasce* belongs with the weak feminine line, and the plural or oblique forms *flascan* support that analysis (Ringe and Taylor 2014).

Old English evidence Old English dictionaries record the noun as *flasce*, with inflectional support from forms such as *flascan*; a later West Saxon *flaxe* is also noted as a secondary variant (Bosworth and Toller 1898, 235; Clark Hall 1960, 121).

The relevant comparison form is therefore the weak feminine noun *flasce*. The plural and oblique evidence matters because it helps explain why the vowel and ending are preserved as they are in the singular.

Development to Old English From **fláskōn*, the weak feminine passes through the expected loss of *n* and the later Old English development of the unstressed ending, reaching *flasce*. Once the weak feminine formation is chosen, the noun follows a regular path to its Old English shape (Campbell 1959).

The decisive issue is morphological rather than phonological. A simple strong feminine citation form does not capture the OE weak noun as cleanly as the selected **fláskōn* does.

Formation comparison The comparison below is manual. It separates the broader comparative headword from the weak feminine formation actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader comparative headword	*flaskō	broader family label	wider family context	useful lexeme label, but not the cleanest OE-facing derivation
selected weak feminine formation	*fláskōn	compact-trace output: *flasce*	flasce	exact match between formation and attested OE noun

The weak feminine suffix is the relevant point. It aligns the inherited form with attested *flasce* and its supporting paradigm forms.

follow — OE fylġan

Derivation: citation reconstruction **fulgēną*; selected input **fúlgijaną* > *fylġan* (early analogy).

Derivation trace Proto input: **fúlgijaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**fúlgijan*
[no change]	OE Secondary Nasalization	**fúlgiją̄n*
Northwest Germanic	Sievers Law Syncope	**fúlgjąn*
[no change]	OE Velar Palatalization	**fúldʒjąn*
	OE I Umlaut	**fyldʒjąn*
	OE Weak Tail Reduction	**fyldʒjan*
	OE J Loss After Heavy	**fyldʒan*

Outcome: *fylġan*

Reconstruction and comparative evidence Kroonen keeps the verb under **fulgen-* and gives Old English *fylgan*, *folgian*, adding that Old Norse *fylgja* and Old English *fylg(e)an* continue a formation **fulgjan-* (Kroonen 2013). The comparative headword and the class-I formation are therefore related but not identical.

Ringe and Taylor make the split explicit as *PNWGmc *fulgija- ~ *fulgai- > OE fylgan ~ folgian* and describe it as a dual formation that probably reflects an older alternation between j-present and e-stative (Ringe and Taylor 2014, 293–94). This is a stem-class choice, not a spelling choice. The selected input **fúlgijaną* belongs to the class-I **fulgija- / *fulgjan-* branch; the citation form **fulgēną* belongs to the parallel class-II history behind *folgian*.

Old English evidence The Old English evidence preserves both formations. Clark Hall gives *folgian* and cross-refers to *fylgan*, while also listing *fylgan* with variant spellings *fylgian* and *fyligan* (Clark Hall 1960). Bosworth-Toller likewise has separate entries for *folgian* and *fylgean* (Bosworth and Toller 1898).

Bright treats *fylg(e)an* as a survival of the older conjugation and contrasts it with forms that have conformed to the Second Conjugation, *folgian, folgode* (Bright 1971). The relevant comparison form in this entry is therefore the class-I verb *fylgan / fylgean*, here normalized as *fylġan*. The spelling with represents the palatalized velar before a front-vocalic environment.

Development to Old English **fúlgijaną* is a class-I weak-verb formation. In the class-I branch the **j* blocks NWGmc lowering of *u* to *o*, since Ringe and Taylor formulate that lowering for environments in which no **j* intervened (Ringe and Taylor 2014, 96). The same front-vocalic environment then triggers i-umlaut, so *u* becomes *y* (Ringe and Taylor 2014, sect. 6.6.2).

The subsequent Old English developments are palatalization of the velar, weak-tail reduction, and loss of *j* after a heavy syllable, yielding *fylġan*. This is the regular outcome of the class-I formation. The class-II form *folgian* belongs to the parallel **-ē- / *-ai-* branch and is not the form modeled here.

Class comparison A class comparison identifies which inherited formation corresponds to the established Old English form under discussion. The comparison below is manual; no full automatic class probe is presented here.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation class-II formation	*fulgēną	probe output: *folgon*	folgian	mismatch: the regular output is not the remodeled infinitive *folgian*
parallel class-II branch	PNWGmc *fulgai-	Ringe-Taylor: OE *folgian*	folgian	documents the separate class-II branch, but not the target of this entry
selected class-I formation	*fúlgijaną	compact-trace output: *fylġan*	fylġan / fylgan	exact match between input, output, and class

The relevant point is the class split. *fylġan* is the regular Old English outcome of the class-I **fulgija- / *fulgjan-* formation, whereas *folgian* belongs to the parallel class-II branch.

gall — OE ġealla

Derivation: citation reconstruction **gállą*; selected input **gállô* > *ġealla* (early analogy).

Derivation trace Proto input: **gállô*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**gællô*
[no change]	OE Breaking	**geallô*
Northwest Germanic	OE Velar Palatalization	**ʤeallô*
[no change]	OE Unstressed Long Vowel Shortening	**ʤealla*

Outcome: *ġealla*

Reconstruction and comparative evidence The wider cognate family can be presented under a form such as **gállą*, but the Old English noun itself belongs with a weak noun **gallōn-*, cited here as **gállô* (Kroonen 2013). The selected input therefore differs from the broader comparative headword in stem class.

That stem-class distinction matters directly for the Old English shape. The weak masculine pathway preserves the ending needed for *ġealla*, whereas a simple strong-noun headword does not align as closely with the attested OE noun.

Old English evidence Old English dictionaries record the noun as *gealla*, and Bright also gives the dative *geallan*, confirming a weak-noun paradigm (Bosworth and Toller 1898, 297; Clark Hall 1960, 145; Bright 1971, 372). The form used here, *ġealla*, is a normalized spelling with macrons omitted and palatal made explicit.

Campbell also notes dialectal variation, contrasting West Saxon or Kentish *gealla* with Anglian *galla* (Campbell 1959). The target of this entry is the West Saxon type *ġealla*.

Development to Old English From **gállô*, the weak noun develops through the expected Old English history of the suffix and the regular breaking environment before *ll*, yielding *ġealla* (Campbell 1959). Once the weak masculine input is chosen, the noun follows a regular path to its attested Old English form.

The decisive issue is therefore morphological. Old English reflects the weak noun, while the broader family label belongs to a different way of presenting the cognate set.

Stem comparison The comparison below is manual. It separates the broader comparative headword from the weak noun formation actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader family label	*gállą	broader cognate-set headword	wider family context	useful lexeme label, but not the direct source of *ġealla*
selected weak noun	*gállô	compact-trace output: *ġealla*	ġealla	exact match between formation and attested OE noun
dialectal Anglian continuation	weak noun branch	Anglian *galla* type	galla	genuine OE variant, but not the selected West Saxon target

The weak-noun stem class is the relevant point. It gives a direct route to attested *ġealla*, while the broader comparative label serves only as a family heading.

knight — OE cniht

Derivation: citation reconstruction **kníxtaz*; selected input **knéxtaz* > *cniht* (early analogy).

Derivation trace Proto input: **knéxtaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**knéxt*
[no change]		OE Breaking	**knéoxt*
Northwest Germanic		OE Ws Palatal Umlaut	**knixt*
PGmc Final Z Deletion	**knéxta*		

Outcome: *cniht*

Reconstruction and comparative evidence The comparative sources align on an *e*-grade reconstruction for this noun: Ringe and Taylor cite **kneht*, Orel gives **knextaz*, and Kluge-Seebold likewise points to **knehta-* (Ringe and Taylor 2014; Orel 2003, 256; Kluge 2011). The selected input **knéxtaz* follows that comparative evidence.

A competing citation reconstruction **kníxtaz* remains possible as a label for the word family, but it is not the reconstruction followed here. The Old English development discussed below is based on **knéxtaz*.

Old English evidence Old English dictionaries record the noun as *cniht* (Clark Hall 1960; Bosworth and Toller 1898, 71). Campbell's discussion of related forms such as plural *cneohtas* helps show the same vowel environment from another point in the paradigm (Campbell 1959).

The target is therefore an ordinary attested Old English noun. No reconstructed OE comparator is needed here.

Development to Old English From **knéxtaz*, the relevant Old English changes include breaking before the velar cluster and then the later reduction that yields *cniht* (Campbell 1959; Sievers 1965). With that corrected input, the derivation is straightforward.

Stem comparison The comparison below is manual. It separates the handbook-supported *e*-grade input from a competing citation reconstruction.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing citation reconstruction	*kníxtaz	not the reconstruction followed here	broader citation tradition	useful as a competing label, but not the source-based choice used for the OE derivation
handbook-supported reconstruction	*knéxtaz	compact-trace output: *cniht*	cniht	exact match between comparative reconstruction and attested OE noun

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
related plural evidence	same stem family	plural *cneohtas* type background	cneohtas	supports the vowel environment, but not the selected target cell

lade — OE hladan

Derivation: citation reconstruction **laθōjanq̨*; selected input **xláðanq̨* > *hladan* (early analogy).

Derivation trace Proto input: **xláðanq̨*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**xlædaną*
PWGmc Dental Hardening	**xládaną*	OE A Restoration	**xladaną*
Northwest Germanic		OE Heavy Syllable Nasal Apocope	**xladan*
[no change]		OE Secondary Nasalization	**xladąn*
		OE Weak Tail Reduction	**xladan*

Outcome: *hladan*

Reconstruction and comparative evidence The wider Germanic family can be cited under a weak-verb label such as **laθōjaną*, but the Old English verb *hladan* belongs to a strong-verb line with voiced consonantism in the selected input **xláðaną* (Ringe and Taylor 2014; Kroonen 2013). The two forms are related as members of one word family, but they do not play the same role in the OE derivation.

The selected input therefore marks an early stem choice. The entry follows the strong Verner-grade form that reaches Old English *hladan* directly.

Old English evidence Old English dictionaries record the verb as *hladan* and preserve the expected strong-verb paradigm material around it (Bosworth and Toller 1898, 559; Clark Hall 1960). The target is an attested infinitive rather than a reconstructed paradigm cell.

For this entry the relevant comparison form is the infinitive *hladan* itself. The question is how that attested strong verb relates to the broader comparative family.

Development to Old English From **xláðaną*, the verb passes through the expected early voiced stop stage, Anglo-Frisian brightening, and the later A-restoration that returns the root vowel to *a* before the full infinitival ending (Campbell 1959; Ringe and Taylor 2014). The resulting Old English infinitive is *hladan*.

Class comparison The comparison below is manual. It separates the wider weak-verb family label from the strong verb actually reflected in Old English.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative weak-verb label	*laθōjaną	wider family background	broader family context	useful family label, but not the direct source of *hladan*
selected strong Verner-grade input	*xláðaną	compact-trace output: *hladan*	hladan	exact match between formation and attested OE infinitive

lap — OE lappa

Derivation: citation reconstruction **lábbaz*; selected input **láppô* > *lappa* (early analogy).

Derivation trace Proto input: **láppô*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**læppô*
[no change]	OE A Restoration	**lappô*
Northwest Germanic	OE Unstressed Long Vowel Shortening	**lappa*
[no change]		

Outcome: *lappa*

Reconstruction and comparative evidence The comparative sources point to a weak noun with *pp*: Orel gives **lappōn*, Kroonen places the word in a weak **lappan-* family, and Kluge-Seebold cites West Germanic *lappa* forms alongside Old English *læppa* (Orel 2003; Kroonen 2013; Kluge 2011). The selected input **láppô* follows that evidence.

A competing comparative label **lábbaz* has also circulated for the word family, but the cited handbooks do not make it the direct source of the Old English weak noun. The form relevant to the OE development is the weak masculine input **láppô*.

Old English evidence Campbell cites *lappa* as a case of restored *a*, while Sievers-Brunner records *lappa*, variant *læppa*, and plural or oblique *leappan* (Campbell 1959; Sievers 1965). The dictionary tradition also preserves *læppa* (Clark Hall 1960; Bosworth and Toller 1898, 613).

The target of this entry is the restored singular *lappa*. The variant *læppa* and the oblique or plural *leappan* remain part of the Old English record and help frame the noun's vowel history.

Development to Old English From **láppô*, Anglo-Frisian brightening first yields *æ*, and later A-restoration returns the root vowel to *a* before the back-vocalic ending, after which shortening of the final vowel gives *lappa* (Campbell 1959; Sievers 1965). With the weak masculine input chosen, the OE development is regular.

Stem comparison The comparison below is manual. It separates the weak masculine formation from a competing voiced comparative label.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing voiced comparative label	*lábbaz	not the form followed for the OE weak-noun derivation	broader comparative background	useful as a competing label, but not the source-based choice used here
selected weak masculine noun	*láppô	compact-trace output: *lappa*	lappa	exact match between formation and attested OE noun
attested OE variant line	same noun family	*læppa*, *leappan*	læppa / leappan	useful control forms within the same OE tradition

laugh — OE hliehhan

Derivation: citation reconstruction **lákaną*; selected input **xláxjaną* > *hliehhan* (early analogy).

Derivation trace Proto input: **xláxjaną*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**xlæxxjaną*
PWGmc J Gemination	**xláxxjaną*	OE Breaking	**xleaxxjaną*
Northwest Germanic		OE Velar Fricative Palatalization	**xleaxçjaną*
[no change]		OE Heavy Syllable Nasal Apocope	**xleaxçjan*
		OE Secondary Nasalization	**xleaxçjąn*
		OE I Umlaut	**xliexçjąn*
		OE Weak Tail Reduction	**xliexçjan*
		OE J Loss After Heavy	**xliexçan*

Outcome: *hliehhan*

Reconstruction and comparative evidence The wider Germanic family includes a non-j branch represented here by the citation label **lákaną*, but Kroonen and Ringe-Taylor both distinguish a j-present formation behind Old English *hliehhan* (Kroonen 2013; Ringe and Taylor 2014). The selected input **xláxjaną* reflects that j-present line.

This branch choice matters because it brings with it the geminate fricative and the vowel development characteristic of the Old English verb. The comparative family label and the OE-facing input are therefore related but not identical.

Old English evidence Old English dictionaries and readers record the verb as *hliehhan*, while also preserving variants such as *hlæhhan* and *hlehhan* (Bosworth and Toller 1898; Clark Hall 1960; Bright 1971). The target of this entry is the West Saxon *hliehhan*.

The variant set matters as background, but the argument of the entry rests on the attested lemma *hliehhan* itself.

Development to Old English From **xláxjaną*, regular j-gemination yields the doubled fricative, and the subsequent Old English vowel developments lead to *hliehhan* (Fulk 2018; Campbell 1959). Ringe and Taylor discuss the broken vowel of the Old English form as part of this same history, with possible support from the related noun *hleahtor* (Ringe and Taylor 2014).

Branch comparison The comparison below is manual. It separates the wider non-j family label from the j-present branch actually reflected in Old English.

Formation / branch	Candidate input	Expected or documented OE outcome	OE comparison form	Result
wider non-j family	*lákaną	comparative background outside the selected OE line	wider family context	useful family label, but not the direct source of *hliehhan*
selected j-present branch	*xláxjaną	compact-trace output: *hliehhan*	hliehhan	exact match between branch and attested OE lemma
attested OE variants	same OE verb line	*hlæhhan*, *hlehhan*	hlæhhan / hlehhan	genuine variant evidence, but secondary to the selected form

loam — OE lām

Derivation: citation reconstruction **laimōn*; selected input **láimą* > *lām* (early analogy).

Derivation trace Proto input: **láimą*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**lām*
PWGmc Ai Monophthongization	**lāmą*		
Northwest Germanic			
[no change]			

Outcome: *lām*

Reconstruction and comparative evidence The inherited comparative noun is given as **laimōn* or **laiman-*, and both Orel and Kroonen identify Old English *lām* as a neuter reflex of that family (Orel 2003; Kroonen 2013, 363). The selected input **láimą* differs from the comparative headword because it represents the stem class that matches the Old English noun most directly.

This is therefore a class shift within the history of the English branch rather than a dispute about the OE target itself.

Old English evidence Old English dictionaries record the noun as *lām*, a neuter word for ‘loam, clay, mud’ (Bosworth and Toller 1898; Clark Hall 1960, 196). The target is an attested citation form rather than a reconstructed comparator.

The relevant question is not whether *lām* is Old English, but which inherited formation best accounts for that attested neuter noun.

Development to Old English From **láimą*, regular monophthongization of *ai* and the later loss of the final nasal syllable yield *lām*. With that OE-facing input, the phonological development is straightforward.

Class comparison The comparison below is manual. It separates the inherited comparative n-stem label from the OE-facing stem class used to derive the attested noun.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
inherited comparative noun	*laimōn	comparative family background	wider family context	useful headword, but not the direct OE-facing input
selected OE-facing stem class	*láimą	compact-trace output: *lām*	lām	exact match between input and attested OE noun

lung — OE lungen

Derivation: citation reconstruction **lungō*; selected input **lúnganjō* > *lungen* (early analogy).

Derivation trace Proto input: **lúnganjō*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	**lúngennju*
PWGmc J Gemination	**lúngannjō*	OE High Vowel Apocope	**lúngennj*
Northwest Germanic		OE J Loss After Heavy	**lúngenn*
NWGmc Final Long O Raising	**lúngannju*	OE Final Geminate Simplification	**lúngen*

Outcome: *lungen*

Reconstruction and comparative evidence Kroonen treats the basic noun as **lungōn-* and also cites an OE-facing derivative **lungunjō-*, continued by Old English *lungen* and close West Germanic cognates (Kroonen 2013). The selected input **lúnganjō* models that derived feminine formation rather than the base noun. The notation differs slightly from Kroonen's **lungunjō-*, but both point to the same derived feminine line.

The difference between the citation label and the selected input is therefore derivational. Old English *lungen* is not a direct reflex of the bare base noun **lungō*; it belongs to an expanded feminine formation.

Old English evidence Old English dictionaries record the noun as *lungen*, with inflected forms such as *lungenne* and *lungena* (Bosworth and Toller 1898, 634). Clark Hall also preserves a small family of compounds such as *lungenādl*, *lungensealf*, and *lungenwyrt* (Clark Hall 1960).

The target is an attested Old English lexeme with its own paradigm, not a rescued inflectional cell.

Development to Old English From the selected derived input, the expected derivational consonant and vowel adjustments lead to *lungen*. Once the expanded feminine formation is chosen, the Old English outcome is regular.

Formation comparison The comparison below is manual. It separates the base noun from the derived feminine formation reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
base noun	*lungō	base-noun outcome without the OE derivative suffix	broader family context	useful headword, but not the direct source of *lungen*
derived OE-facing formation	*lúnganjō	compact-trace output: *lungen*	lungen	exact match between selected formation and attested OE noun
Kroonen’s cited derivative	**lungunjō-*	comparative support for the same OE-facing formation	lungen and cognate set	supports the derived feminine formation, with notation differing from the normalized input form used here

navel — OE nafola

Derivation: citation reconstruction **nablô*; selected input **nábulô* > *nafola* (early analogy).

Derivation trace Proto input: **nábulô*

Earlier Germanic changes	Old English changes	
West Germanic	OE Med Unstressed U Lowering	**nábolô*
[no change]	Anglo Frisian Brightening	**nǽbolô*
Northwest Germanic	OE A Restoration	**nabolô*
[no change]	PGmc B Allophony	**naβolô*
	OE Unstressed Long Vowel Shortening	**naβola*

Outcome: *nafola*

Reconstruction and comparative evidence Kroonen lemmatizes the word with a syncopated comparative headword **nablô*, while Ringe and Taylor give the derivational pathway **nabulō* > **nœbula* > *nafola* (Kroonen 2013; Ringe and Taylor 2014). The difference is one of stage and notation rather than of lexeme identity: the selected input **nábulô* is the pre-syncope form needed for the Old English development.

The literature also differs on the older history of the medial *u*, whether it is inherited or secondary (Streitberg 1896; Ringe 2006; Mayrhofer 1992--2001; Sievers 1965). For the Old English comparison, however, both lines place a medial vowel in the pre-OE form.

Old English evidence The Old English record includes *nafola*, *nafela*, and Corpus *nabula* (Ringe and Taylor 2014; Campbell 1959). The target of this entry is the nominative singular *nafola*, the form that matches the selected derivational pathway most directly.

nafela is the better-known later West Saxon spelling, while *nabula* preserves a less reduced medial vowel. These forms belong to the same lexical history, but this entry is centered on *nafola*.

Development to Old English From **nábulô*, the trace gives **nábolô* by lowering of unstressed *u*, then **nǽbolô* by Anglo-Frisian brightening, followed by A-restoration to **nabolô* (Ringe and Taylor 2014). Intervocalic *b* then surfaces as *f*, and final weak-tail shortening gives *nafola*.

The medial vowel is still present when A-restoration applies. That is why the selected pre-syncope input differs from the syncopated comparative headword.

Stage comparison The comparison below is manual. It separates the comparative citation form from the pre-syncope input and from the later OE spellings.

Formation / stage	Candidate input	Expected or documented OE outcome	OE comparison form	Result
syncopated comparative headword	*nablô	reduced *næfla*-type outcome rather than *nafola*	not the selected target	useful citation form, but too reduced for the pathway modeled here
selected pre-syncope input	*nábulô	compact-trace output: *nafola*	nafola	exact match between selected input and target
later OE reduction stages	same lexical history	attested *nafela*; Corpus *nabula*	nafela / nabula	related OE spellings, but not the chosen comparator

neck — OE hnecca

Derivation: citation reconstruction **xnákkaz*; selected input **xnékkô* > *hnecca* (early analogy).

Derivation trace Proto input: **xnékkô*

Earlier Germanic changes	Old English changes	
West Germanic	OE Unstressed Long Vowel Shortening	**xnékka*
[no change]		
Northwest Germanic		
[no change]		

Outcome: *hnecca*

Reconstruction and comparative evidence The noun belongs to an ablauting n-stem family. Kroonen reconstructs a paradigm with nominative **hnekkō*, genitive **hnukkaz*, and accusative plural **hnakkuns*, and he places Old English *hnecca* among the e-grade descendants (Kroonen 2011). Kluge-Seebold likewise identifies *ae. hnecca* as an ablaut partner of the a-grade *Nacken* family (Kluge 2011).

A competing comparative label **xnákkaz* remains useful for the wider family, and Orel also gives an a-grade headword line (Orel 2003). The selected input **xnékkô*, however, is the form that matches the Old English branch.

Old English evidence Old English dictionaries record the weak masculine noun *hnecca* (Clark Hall 1960; Bosworth and Toller 1898, 567). The target is therefore an attested citation form, not an oblique cell or a reconstructed lemma.

The phonological question is upstream of the Old English evidence. The attested noun already shows that the branch continued an e-grade form rather than the a-grade seen in much of the continental material.

Development to Old English From **xnékkô*, the derivation is straightforward. The trace shortens the final long vowel to **xnékka*, and Old English orthography gives *hnecca*.

The derivation depends on the earlier selection of the e-grade weak-noun form continued by Old English.

Stem comparison The comparison below is manual. It separates the wider a-grade family from the selected e-grade Old English branch.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing comparative label	*xnákkaz	broader a-grade family rather than the selected OE source	continental *Nacken* line	useful family label, but not the input followed for the Old English derivation
weak noun with a-grade	*xnakkô	expected *hnacca* type outcome	hnacca	fixes the class, but not the vowel grade
selected e-grade nominative	*xnékkô	compact-trace output: *hnecca*	hnecca	exact match between selected input and attested OE noun
oblique paradigm background	**hnukkaz*, **hnakkuns*	ON/OHG/German a-grade continuation	hnakki / Nacken	shows the wider ablaut family, but not the chosen OE branch

needle — OE nǣdl

Derivation: citation reconstruction **nḗθlō*; selected input **nḗ̃ðlō* > *nǣdl* (early analogy).

Derivation trace Proto input: **nḗ̃ðlō*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**nǣdl*
PWGmc Dental Hardening	**nḗdlō*		
Northwest Germanic			
NWGmc Final Long O Raising	**nḗdlu*		
NWGmc Long E Lowering	**nǣdlu*		

Outcome: *nǣdl*

Reconstruction and comparative evidence Ringe and Taylor treat the word as a voiced/voiceless alternant, and Ringe also cites the pair **nēþlō-* ~ **nēdlō-* (Ringe and Taylor 2014; Ringe 2006). The selected input **nḗ̃ðlō* is the voiced Verner-grade form used for the Old English comparison, while the citation form **nḗθlō* remains the broader lexeme label.

Kroonen and Orel preserve different comparative headwords for the family (Kroonen 2013; Orel 2003). The development discussed here follows the Ringe-Taylor alternant framework.

Old English evidence Old English has the attested citation form *nǣdl* (Clark Hall 1960). Campbell lists *nédl* among the expected unbroken forms after *t* and *d*, and Hogg also includes *nidi* / *nǣdl* in the same broader cluster history (Campbell 1959; Hogg 1992).

The target is therefore an attested citation form. No oblique-cell substitution is involved in this entry.

Development to Old English From **nḗðlō*, the trace gives **nḗdlō* by dental hardening, then **nḗdlu*, **nǣdlu*, and finally **nǣdl*, hence *nǣdl* (Ringe and Taylor 2014; Campbell 1959).

The essential choice lies in the PGmc alternant selected for the derivation. Once the voiced form is chosen, the rest of the pathway is regular.

Alternant comparison The comparison below is manual. It separates the broader citation headword from the voiced alternant used for Old English.

Formation / stage	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative voiceless headword	*nḗθlō	broader word-family label rather than the OE-facing alternant	**nēþlō* line	useful citation form, but not the selected input for the Old English derivation
selected voiced Verner alternant	*nḗðlō	compact-trace output: *nǣdl*	nǣdl	exact match between selected input and attested OE noun
later hardening stage	*nḗdlō	intermediate pre-OE stage in the same derivation	nǣdl	genuine stage in the pathway, but not the selected PGmc input

nose — OE nosu

Derivation: citation reconstruction **nasō*; selected input **núsō* > *nosu* (early analogy).

Derivation trace Proto input: **núsō*

Earlier Germanic changes		Old English changes
West Germanic		[no change]
[no change]		
Northwest Germanic		
NWGmc U Lowering	**nósō*	
NWGmc Final Long O Raising	**nósu*	

Outcome: *nosu*

Reconstruction and comparative evidence Kroonen reconstructs a Germanic ablaut pair **nasō-* ~ **nusō-* and adds that the root **nus-* is likely to have arisen as a secondary zero grade after a remodeling of the older paradigm (Kroonen 2013). Campbell is more specific for Old English, citing *nosu* < **nusō* (Campbell 1959).

The citation reconstruction **nasō* is therefore best treated as the full-grade comparative headword, while the selected input **núsō* represents the remodeled zero-grade line continued by the

Old English form discussed here. Orel's **nasō … OE nasu* preserves the competing full-grade notation and shows that the two lines should not be collapsed without comment (Orel 2003).

Old English evidence *Nosu* is an attested Old English noun. Ringe and Taylor list it among the few surviving early Old English feminine u-stems (Ringe and Taylor 2014). Clark Hall likewise gives *nosu f.*, with genitive-dative singular *nosa*, and cross-refers *nasu* to *nosu* (Clark Hall 1960).

The selected OE target is therefore an attested *nosu*, not a reconstructed placeholder. At the same time, the lexicographical record keeps *nasu* visible as a parallel notation belonging to the full-grade side of the tradition.

Development to Old English From **núsō*, the regular path is the one documented by the current trace: **núsō* > **nósō* > **nósu* > *nosu*. The early special step lies in the choice of the zero-grade input, not in any late Old English repair.

With that input chosen, the OE development is straightforward. The full-grade line behind **nasō* instead points toward *nasu*, not to the form treated here.

Stem comparison The comparison below is manual. It separates the full-grade comparative line from the remodeled zero-grade input that yields the Old English form.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
full-grade comparative line	*nasō	expected full-grade continuation *nasu*	nasu	useful comparative background, but not the selected OE-facing input
remodeled zero-grade line	*núsō	compact-trace output: *nosu*	nosu	exact match between selected input and attested OE noun

sap — OE sæp

Derivation: citation reconstruction **sapōn*; selected input **sápą* > *sæp* (early analogy).

Derivation trace Proto input: **sápą*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**sæpą*
[no change]	OE Heavy Syllable Nasal Apocope	**sæp*
Northwest Germanic		
[no change]		

Outcome: *sæp*

Reconstruction and comparative evidence The comparative sources do not give one uniform inherited stem. Kroonen derives the word family from material pointing to dialectal dissolution of a primary n-stem **safō, gen. *sappaz* (Kroonen 2013). Orel preserves the comparative notation **sapōn* ~ **sapan* (Orel 2003), while Kluge-Seebold instead gives West Germanic **sapi-* and still cites Old English *sæp n.* (Kluge 2011).

The selected input **sápą* therefore does not replace those comparative labels. It identifies the OE-facing stem shape that yields the attested noun treated here.

Old English evidence Clark Hall records *sæp (e) n.* (Clark Hall 1960), and Kluge-Seebold likewise cites *ae. sæp n.* (Kluge 2011). The target is therefore an attested neuter Old English noun. Orel's plain *sap* notation belongs to comparative normalization, not to the spelling adopted here for the Old English form (Orel 2003).

Development to Old English From **sápą*, Anglo-Frisian brightening yields *sæ*, and heavy-syllable nasal apocope then produces *sæp*. That is the regular path documented by the current trace.

The competing comparative lines do not give the same result. The inherited n-stem notation **sapōn* yields *sape*, while an i-stem continuation from the **sapi-* line leads to *sep / sepe* rather than to *sæp*. The special step in this entry is therefore the early stem choice, not a late OE paradigm-cell selection.

Stem comparison The comparison below is manual. It separates the competing comparative stem lines from the selected OE-facing input.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative n-stem line	*sapōn	local comparator output: *sape*	sape	useful comparative background, but not the source of attested *sæp*
inferred i-stem comparator from **sapi-*	*sapiz	local comparator output: *sepe*	sepe	confirms that an i-triggering stem does not reach the target
selected a-stem input	*sápą	compact-trace output: *sæp*	sæp	exact match between selected input and attested OE noun

sea — OE sǣ

Derivation: citation reconstruction **sái*; selected input **sáiwiz* > *sǣ* (early analogy).

Derivation trace Proto input: **sáiwiz*

Earlier Germanic changes		Old English changes	
West Germanic		OE W Loss Before I	**sāi*
PWGmc Ai Monophthongization	**sāwiz*	OE I Umlaut	**sǣi*
Northwest Germanic		OE High Vowel Apocope	**sǣ*
PGmc Final Z Deletion	**sāwi*		

Outcome: *sǣ*

Reconstruction and comparative evidence Kroonen gives the noun in stem notation as **saiwi-*, an i-stem whose English reflex is cited as OE *sæ* (Kroonen 2013). Ringe and Taylor write the fuller form **saiwiz* and derive it through **sawi* > **sei* > *OE sǣ* (Ringe and Taylor 2014, sect. 6.7.1). The comparative headword is therefore shorter than the form required for the English history: **sái* names the lexeme, but **sáiwiz* preserves the medial **w* and the final high vowel that control the later development.

Old English evidence The Old English noun is the ordinary word for ‘sea’. Kroonen cites it as *sœ*; the normalized form here is *sǣ* (Kroonen 2013). Campbell likewise treats *sea* as continuing the same **saiui-* > **sǣi* history, with loss of *u/w* before *i* (Campbell 1959, sect. 406).

Development to Old English Once the fuller i-stem input is chosen, the development is regular. After Proto-West-Germanic monophthongization **sáiwiz* > **sāwiz* and final **-z* loss **sāwiz* > **sāwi*, the non-initial **w* disappears before unstressed **i*, and the following high vowel fronts the root vowel before final apocope. The documented chain is **sáiwiz* > **sāwiz* > **sāwi* > **sāi* > **sǣi* > *sǣ* (Ringe and Taylor 2014, sect. 6.7.1; Campbell 1959, sect. 406).

Stem and stage comparison The comparison below is manual. It separates the abbreviated comparative headword from the fuller i-stem input that yields the Old English form.

Formation / label	Candidate input	OE output or comparison	OE comparison form	Result
abbreviated comparative headword	*sái	too short to preserve the **w* ... **i* environment needed for the documented chronology	sǣ	useful comparative label, but not the selected OE-facing input
selected i-stem input	*sáiwiz	documented trace output: *sǣ*	sǣ	exact match between selected input and Old English target

sieve — OE sife

Derivation: citation reconstruction **síbaz*; selected input **síbi* > *sife* (early analogy).

Derivation trace Proto input: **síbi*

Earlier Germanic changes	Old English changes	
West Germanic	PGmc B Allophony	**síβi*
[no change]	OE Med Unstressed I Lowering1	**síβe*
Northwest Germanic		
[no change]		

Outcome: *sife*

Reconstruction and comparative evidence Kluge-Seebold gives *wg. *sibi- n. ... ae. sife*, and Campbell groups *sife* with short neuter i-stems such as *spere* (Kluge 2011; Campbell 1959, sect. 609). The older morphological background is the s-stem **sib-iz*, but the selected input is the normalized i-stem form **síbi*.

Kroonen’s relevant nearby entry is not ‘sieve’ but **sebjō-* ‘kinship’, the source of Old English *sibb* (Kroonen 2013). Orel’s **sibaz ... OE sife* preserves a broader handbook notation, but that a-stem shape does not fit the Old English form treated here (Orel 2003).

Old English evidence Clark Hall gives *sibi (GL) ...* = *sife* and also *sife n. ‘sieve’* (Clark Hall 1960). Campbell likewise cites Corpus Glossary *sibi* and treats *sife* as a short neuter i-stem (Campbell 1959, sects 444, 609). The normalized Old English target is therefore *sife*, while *sibi* is an attested earlier spelling rather than a separate lexeme.

Development to Old English From **síbi*, the documented trace gives **síβi* > **síβe* > *sife*. Medial *b* is realized as a spirant and later written *f*, while the final unstressed *i* lowers to *e*. The older s-stem background **sib-iz* explains the morphology, but the selected input **síbi* is the immediate pre-Old-English form.

Stem comparison The comparison below is manual. It distinguishes the accepted i-stem line from its rejected competitors.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
ja-stem kinship line	Kroonen **sebjō-* / comparator **sibja*	OE *sibb*	sibb	separate lexeme, not the target treated here
a-stem handbook line	*síbaz	expected *sif*	sif	wrong ending for the attested noun
selected i-stem line from older **sib-iz*	*síbi	documented trace output: *sife*	sife; early spelling *sibi*	exact match between selected input and Old English evidence

spare — OE sparian

Derivation: citation reconstruction **sparēną*; selected input **spárōjaną* > *sparian* (early analogy).

Derivation trace Proto input: **spárōjaną*

Earlier Germanic changes

West Germanic

[no change]

Northwest Germanic

[no change]

Old English changes

Anglo Frisian Brightening	**spærōjaną*
OE A Restoration	**sparōjaną*
OE Heavy Syllable Nasal Apocope	**sparōjan*
OE Secondary Nasalization	**sparōjąn*
OE I Umlaut	**sparējąn*
OE Unstressed Long Vowel Shortening	**sparejąn*
OE Weak Tail Reduction	**sparejan*
OE Intervocalic J Vocalization	**spareian*
OE Unstressed EI Contraction	**sparian*

Outcome: *sparian*

Reconstruction and comparative evidence Kroonen and Orel keep the inherited verb under class-III **sparēn- / *sparēnan* (Kroonen 2013; Orel 2003). Ringe and Taylor, however, reconstruct **sparai- ~ *sparja-* for the English branch and derive the citation verb from a class-II line (Ringe and Taylor 2014, 162, 191). The selected input **spárōjaną* therefore represents the refashioned class-II formation behind Old English *sparian*, while the citation reconstruction **sparēną* remains the inherited comparative headword.

Old English evidence Campbell says that *sparian* does not show the ordinary class-III characteristics, but the Ritual forms, normalized here as *spæria*, *spær*, and *spærede*, together with Vespasian Psalter *spearad*, point to primitive Old English forms both with and without back vowels (Campbell 1959, sect. 764). Brunner likewise records Northumbrian *spæria*, *spærede* beside common Old English *sparian* and Vespasian Psalter *spearad* (Sievers 1965, sect. 364 Anm. 11). The citation form treated here is *sparian*; the Anglian forms are relics of the older formation, not alternative headwords of equal status.

Development to Old English Once the class-II formation **spárōjaną* is chosen, the remaining development is regular. The documented trace shows brightening, restoration of *a* before the back vocalism of the suffix, later i-mutation within the weak ending, weak-tail reduction, and contraction to *sparian*. By contrast, Brunner's rule against further apocope of final *-e* explains why Ritual *spær* cannot be the regular continuation of inherited **spárē* (Sievers 1965, sect. 150).

Formation comparison The comparison below is manual. It contrasts the inherited class-III formation with the refashioned class-II one that yields the citation verb.

Formation / comparison	Candidate input	Expected or documented OE outcome	OE comparison form	Result
inherited class-III infinitive	*spárēną	manual comparison / probe output: *sparen*	sparian	wrong class and wrong ending for the citation verb
inherited class-III imperative singular	*spárē	manual comparison / probe output: *spære*	Ritual *spær*	loss of final *-e* is not regular, so the relic form cannot control the entry
inherited class-III finite present	*spárēθi	manual comparison / probe output: *spæreþ*	*spearad*	attested form is mixed, not a direct continuation of the inherited cell
selected class-II formation	*spárōjaną	documented trace output: *sparian*	sparian	exact match between selected input and Old English citation form

staff — OE stæf

Derivation: citation reconstruction **stábiz*; selected input **stábaz* > *stæf* (early analogy).

Derivation trace Proto input: **stábaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**stáb*
[no change]		Anglo Frisian Brightening	**stæb*
Northwest Germanic		PGmc B Allophony	**stæβ*
PGmc Final Z Deletion	**stába*		

Outcome: *stæf*

Reconstruction and comparative evidence The comparative dictionaries do not give one uniform stem class. Kroonen reconstructs an a-stem **staba-* (Kroonen 2013), Orel writes **stabiz* ~ **stabaz* (Orel 2003), and Kluge-Seebold explicitly marks *g.* **stabi-/a-* (Kluge 2011). That disagreement matters because a direct i-stem input in **-iz* would predict i-mutation in Old English, whereas the attested noun keeps *æ*.

Old English evidence The Old English noun itself is the ordinary citation form *stæf*. Luick lists *stæf* among closed monosyllables with *æ* (Luick 1914--1940, 176), and Ringe and Taylor pair singular *stæf* with plural *stafas* (Ringe and Taylor 2014). The normalized form here is therefore *stæf*; later English *staff* with *a* belongs to a later stage of the word's history.

Development to Old English With the selected a-stem input, the development is regular. Final *-z* disappears, bare final *-a* is lost, Anglo-Frisian brightening gives *æ* in the closed monosyllable, and medial *b* surfaces as a fricative written *f*. The documented chain is **stábaz* > **stába* > **stáb* > **stæb* > *stæf*. A direct continuation of **stábiz*, by contrast, would produce i-mutated *stefe* rather than the attested singular.

Formation comparison The comparison below is manual. It separates the rejected i-stem line from the selected a-stem input.

Formation / label	Candidate input	OE output or comparison	OE comparison form	Result
comparative i-stem line	*stábiz	expected *stefe* after i-mutation	stæf	wrong vowel for the attested singular
mixed comparative notation	Orel **stabiz* ~ **stabaz*; Kluge **stabi-/a-*	source-level stem-class uncertainty	stæf	useful comparative background, but not a single OE-facing input
selected a-stem input	*stábaz	documented trace output: *stæf*	stæf	exact match between selected input and Old English target

stem — OE stefn

Derivation: citation reconstruction **stámnaz*; selected input **stébnō* > *stefn* (early analogy).

Derivation trace Proto input: **stébnō*

Earlier Germanic changes		Old English changes	
West Germanic		PGmc B Allophony	**stéβnu*
[no change]		OE High Vowel Apocope	**stéβn*
Northwest Germanic			
NWGmc Final Long O Raising	**stébnu*		

Outcome: *stefn*

Reconstruction and comparative evidence The source tradition behind *stefn* is not the same as the comparative label **stámnaz*. Ringe and Taylor cite **stebnō* for the noun continued by Gothic *stibna* and Old English *stebn* > *stefn* > *stemn* (Ringe and Taylor 2014, 330). Orel likewise gives **stebnō* ~ **stemnō*, whereas Kroonen prefers **stimnō-*, and Fulk describes the etymology of *stefn, stemn* as insecure (Orel 2003, 374; Kroonen 2013, 480; Fulk 2018, sect. 6.11 n. 6).

These forms belong to the Old English noun *stefn* 'voice, sound'. The selected input **stébnō* is therefore best treated as the OE-facing transponent supported by that source tradition. It does not settle the deeper comparative reconstruction implied by the citation label **stámnaz*.

Old English evidence Clark Hall records *stefn* as the noun 'voice, sound' and cross-refers *stemn* to the same word (Clark Hall 1960). Ringe and Taylor give the OE chronology directly as *stebn* > *stefn* > *stemn* (Ringe and Taylor 2014, 330).

Bülbring and Luick treat *stemn* as a later West Saxon development from older *stefn*, produced by *fn* > *mn* only after the earlier period of nasal influence on *e* (Bülbring 1902, sect. 62 Anm. 3, 445; Luick 1914--1940, sect. 75 Anm. 1). The relevant comparison form is therefore the conservative *stefn*, not the later West Saxon doublet *stemn*.

Development to Old English From **stébnō*, raising of final long *ō* gives a **stébnu* stage. Regular fricativization of *b* before *n* then yields **stéβnu*, and loss of the final high vowel leaves **stéβn*, written *stefn* in Old English. The later form *stemn* belongs to a separate West Saxon assimilation after this stage (Ringe and Taylor 2014, 330; Bülbring 1902, sect. 445).

Source comparison The comparison below is manual. It keeps apart the broader comparative label, the OE-facing transponent, and the later West Saxon variant history.

Form or label	Status	OE relation	Result
**stámnaz*	comparative citation label for the broader stem/trunk family	does not itself control the *stefn* derivation discussed here	broader lexical label only
**stébnō*	voice-noun transponent	trace output: *stefn*	selected OE-facing input
stemn	later attested West Saxon doublet	secondary form from *stefn* by *fn* > *mn*	real OE variant, but not the selected comparator

swan — OE swanes

Derivation: citation reconstruction **swánaz*; selected input **swánas* > *swanes* (early analogy).

Derivation trace Proto input: **swánas*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**swánæs*
[no change]	OE Unstressed AE Merger	**swánes*
Northwest Germanic		
[no change]		

Outcome: *swanes*

Reconstruction and comparative evidence The Germanic noun is ordinarily cited as the masculine a-stem **swanaz* (Orel 2003, 367). The selected input **swánas* is not a competing lexeme reconstruction. It is the **genitive singular** of the same paradigm.

The question here is therefore one of paradigm cell rather than stem history. The citation form remains **swanaz* > *swan*; the selected comparison form is the genitive singular **swánas* > *swanes*.

Old English evidence Old English dictionaries give the ordinary headword as *swan* (Clark Hall 1960). Bright's glossary, however, also records the exact inflected form *swanes*, glossing *swan, m., swan: gs. swanes* and citing the phrase *swanes feðre* (Bright 1971).

The target is therefore an **attested Old English genitive singular**, not a reconstruction. It is also not the ordinary citation lemma. The entry must keep those two facts distinct.

Development to Old English From **swánas*, Anglo-Frisian brightening gives **swánæs*, and subsequent merger of unstressed *æ* with *e* yields *swanes*. The comparison is straightforward once the genitive singular is chosen as the relevant cell.

Paradigm-cell comparison The comparison below is manual. It separates the ordinary citation form from the selected inflected cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*swánaz	OE headword *swan*	swan	ordinary lexeme line
selected genitive singular	*swánas	trace output: *swanes*	swanes	selected attested cell

thousand — OE þūsend

Derivation: citation reconstruction **θūs-undī*; selected input **θūs-èndi* > *þūsend* (early analogy).

Derivation trace Proto input: **θūsèndi*

Earlier Germanic changes		Old English changes	
West Germanic		OE Strip Secondary Stress	**θūsend*
PWGmc Early I Apocope	**θūsènd*		
Northwest Germanic			
[no change]			

Outcome: *þūsend*

Reconstruction and comparative evidence Kroonen reconstructs the Germanic numeral as **þūsundī-* and cites Old English *þūsend* among its continuations (Kroonen 2013, 554). The selected input **θūs-èndi* is not the same claim. It is an OE-oriented transponent with the second-member vowel already resolved to *e* and the final high vowel already shortened for apocope.

The important question is therefore chronological. Why does Old English show *þūsend*, while related languages such as Old Saxon and Old High German keep *u* in the second syllable? (Kroonen 2013, 554).

Old English evidence Old English *þūsend* is an ordinary citation form, not a selected oblique or paradigm cell. Campbell treats it as a neuter noun with normal case forms (Campbell 1959, sect. 689). The problem lies in the internal history of the word, not in its lexical status.

Development to Old English If the old final *-ī* had remained long enough to trigger ordinary double umlaut, Campbell's rule would point toward a form of **þȳsend* type rather than attested *þūsend* (Campbell 1959, sect. 203). Preserved root *ū* therefore argues that the umlaut-triggering vowel was lost or neutralized before the ordinary OE umlaut outcome could develop.

That early loss, however, does not by itself explain the medial *e*. Luick compares the word with *ærende* and later groups *thousand* with forms reshaped on that pattern (Luick 1914--1940, sects 198, 492). Viredaz is more cautious, arguing that Old English *e* in this weak position may simply write schwa and so need not prove a unique *ærende*-type analogy (*Germanic, Slavic, and Baltic 'Thousand'* 2025, sect. 2.1.4).

The selected transponent **θūs-èndi* captures the OE-side state from which the documented trace reaches *þūsend*.

Stage comparison The comparison below is manual. It separates the secure chronology from the more interpretive account of the second-syllable vowel.

Stage / interpretation	Candidate form	OE relation	Result
surviving -*ī* with ordinary double umlaut	**þūsundī-* treated as still umlaut-active in OE	would point toward **þȳsend*	excluded by preserved *ū*
early loss of the trigger without further reshaping	**þūsund-* type	explains *ū*, but not why OE alone has medial *e*	incomplete account
selected OE-oriented transponent	**θūs-èndi*	trace output: *þūsend*	selected modeling input

timber — OE timber

Derivation: citation reconstruction **tímrą*; selected input **tímbrą* > *timber* (early analogy).

Derivation trace Proto input: **tímbrą*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**tímbr*
[no change]	OE Epenthetic Vowel	**tímber*
Northwest Germanic		
[no change]		

Outcome: *timber*

Reconstruction and comparative evidence Kroonen reconstructs the noun as **timbra-* and cites Old English *timber* among its continuations (Kroonen 2013). Ringe and Taylor instead state the history from *PGmc *timra* through West Germanic **timbr* to Old English *timber* (Ringe and Taylor 2014).

The difference is therefore not over the Old English noun itself. It concerns whether medial *b* belongs in the comparative citation form or appears in an early pre-Old-English stage of the cluster.

Old English evidence Clark Hall lemmatizes the noun as *timber* and also records *timbor* as a variant spelling (Clark Hall 1960). The Old English form is thus an ordinary citation noun, not a selected oblique cell or a reconstructed convenience form.

Development to Old English With the consonantal frame *timbr-* in place, the rest of the development is straightforward. Loss of final -*ą* leaves **tímbr*, and epenthetic *e* in the final cluster yields *timber*. Ringe and Taylor's **timra* > **timbr* > *OE timber* and the handbook treatment of this epenthetic vowel point to the same Old English result (Ringe and Taylor 2014; Campbell 1959, sects 463–464).

Formation comparison The comparison below is manual. It separates the comparative headword from the OE-facing consonantal input.

Formation or notation	Candidate form	OE relation	Result
Kroonen’s comparative citation	**timbra-*	already matches the consonantal frame of OE *timber*	closest comparative support for the selected input
Ringe-Taylor citation line	**timra* > **timbr*	reaches the same OE noun through early cluster expansion	compatible comparative background
modeled input	**tímbrą*	trace output: *timber*	selected OE-facing input

wake — OE wacan

Derivation: citation reconstruction **wakēną*; selected input **wákaną* > *wacan* (early analogy).

Derivation trace Proto input: **wákaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**wækaną*
[no change]	OE A Restoration	**wakaną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**wakan*
[no change]	OE Secondary Nasalization	**wakąn*
	OE Weak Tail Reduction	**wakan*

Outcome: *wacan*

Reconstruction and comparative evidence Kroonen gives the strong verb as **wakan-* with Old English *wacan* (Kroonen 2013). Ringe and Taylor separately derive Old English *wacian* from weak **wakai-* ~ **wakja-* (Ringe and Taylor 2014, sect. 3.3.2).

The difference is therefore lexical and class-based, not graphic. Strong *wacan* ‘wake up, arise’ and weak *wacian* ‘be awake, watch’ belong to related but distinct histories.

Old English evidence Clark Hall keeps *wacan* and *wacian* as separate headwords (Clark Hall 1960). Bosworth-Toller adds an important caution under *wacan*: the simplex infinitive itself does not occur, its place seeming to be taken by *wæcnan* (Bosworth and Toller 1898, 226).

The target *wacan* is therefore best understood as a normalized strong headword for the verb family, not as a directly quoted simplex infinitive. It still remains the correct Old English comparison form for the strong branch.

Development to Old English With strong **wákaną*, Anglo-Frisian brightening first gives a form of the **wækaną* type. A-restoration then returns *a*, and the ordinary tail reductions yield *wacan*. The weak verb *wacian* belongs to a different prehistory and is not the expected outcome of this input.

Class comparison The comparison below is manual. It separates the strong and weak verb lines.

Formation / class	Candidate input	OE outcome or comparison	Result
weak class-III / class-II branch	**wakēną*, **wakai-* ~ **wakja-*	OE *wacian* and related weak forms	related lexeme, but not the target of this entry

Formation / class	Candidate input	OE outcome or comparison	Result
strong class-VI branch	**wákaną*	trace output: *wacan*	selected OE-facing input
strong normalized headword	*wacan*	dictionary comparison form beside attested strong-family forms	correct Old English comparator, though not a directly quoted simplex infinitive

water — OE wæter

Derivation: citation reconstruction **wátną*; selected input **wátōr* > *wæter* (early analogy).

Derivation trace Proto input: **wátōr*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**wætær*
PWGmc Final Or Lowering	**wátar*	OE Unstressed AE Merger	**wæter*
Northwest Germanic			
[no change]			

Outcome: *wæter*

Reconstruction and comparative evidence Kroonen reconstructs a heteroclitic noun **watar-* ~ **watan-* and states that the Proto-Germanic material points to **watōr, *watenaz* (Kroonen 2013, 616). Ringe and Taylor likewise start from singular **wator* before the Old English branch (Ringe and Taylor 2014, sect. 3.1.4).

The generalized comparative label is therefore broader than the singular form that actually corresponds to Old English *wæter*. The relevant comparator is the inherited nominative-accusative singular **wátōr*.

Old English evidence Bright gives the noun as *wæter* with the regular paradigm *wæteres, wætere, wæter(u), wætera, wæterum* (Bright 1971). Ringe and Taylor add the dialectal contrast between West Saxon *weeter* and Mercian *weter* (Ringe and Taylor 2014, sect. 6.5.2).

The target is therefore an attested Old English citation form within a normal paradigm. The complication lies on the comparative side of the lexeme, not in Old English attestation.

Development to Old English From **wátōr*, pre-final **ō* becomes *a* before final *r*, yielding **watar* (Ringe and Taylor 2014, sect. 3.1.4). Anglo-Frisian brightening then gives **wætær*, and merger of unstressed *æ/e* yields *wæter*.

Stage comparison The comparison below is manual. It separates the generalized lexeme label from the singular input that matches the Old English citation form.

Stage or notation	Candidate form	OE relation	Result
generalized comparative label	**wátną*	broader lexeme shorthand, not the singular that corresponds directly to *wæter*	useful background only

Stage or notation	Candidate form	OE relation	Result
heteroclitic stem notation	**watar-~*watan-*	source-faithful comparative reconstruction	explains why a singular comparator is needed
inherited singular input	**wátōr*	trace output: *wæter*	selected OE-facing input

whale — OE hwæl

Derivation: citation reconstruction **wálaz*; selected input **xwálaz* > *hwæl* (early analogy).

Derivation trace Proto input: **xwálaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**xwál*
[no change]		Anglo Frisian Brightening	**xwæl*
Northwest Germanic			
PGmc Final Z Deletion	**xwála*		

Outcome: *hwæl*

Reconstruction and comparative evidence The comparative sources are not uniform. Orel gives **xwalaz* and notes some mixed **xwaliz* evidence (Orel 2003). Kroonen instead cites **hwali-* (Kroonen 2013).

Both notations agree on inherited initial *hw-/xw-*, but they differ in stem label. The a-stem-like input followed here is closer to Orel's notation than to Kroonen's citation form.

Old English evidence Clark Hall lemmatizes the noun as *hwal*, and Bosworth-Toller preserves the plural *hwalas* (Clark Hall 1960; Bosworth and Toller 1898, 326). The comparison form is normalized here as *hwæl* for the singular citation form with Anglo- Frisian fronting.

The plural *hwalas* remains important control evidence. It shows the same lexeme with *a* in an open syllable, beside singular *hwæl* in the closed monosyllable.

Development to Old English From **xwálaz*, final *-z* disappears and bare final *-a* is lost. Anglo-Frisian fronting then yields *æ* in the closed monosyllable, and Old English orthography writes *hwæl*.

Formation comparison The comparison below is manual. It separates the competing comparative notations from the normalized Old English singular.

Comparative line	Candidate form	OE relation	Result
Orel's citation	**xwalaz*	same stem notation as the modeled singular line	closest comparative support for the selected input
Kroonen's citation	**hwali-*	same initial cluster, different stem label	important comparative rival, but not the notation followed here
modeled input	**xwálaz*	trace output: *hwæl*	selected OE-facing input

Comparative line	Candidate form	OE relation	Result
plural control	*hwalas*	attested open-syllable plural beside singular *hwæl*	confirms that the lexeme also preserves an *a*-vocalism branch

whine — OE hwīnan

Derivation: citation reconstruction **wainōjaną*; selected input **xwī́naną* > *hwīnan* (early analogy).

Derivation trace Proto input: **xwī́naną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**xwī́nan*
[no change]	OE Secondary Nasalization	**xwī́nąn*
Northwest Germanic	OE Weak Tail Reduction	**xwī́nan*
[no change]		

Outcome: *hwīnan*

Reconstruction and comparative evidence The citation reconstruction preserved in the header belongs to the lament-family verb seen in German *weinen* and Old English *wānian*. Kroonen instead separates Old English *hwīnan* under **hwinan-*, and Orel likewise distinguishes strong **xwinanan* from weak **wainōjanan* (Kroonen 2013; Orel 2003). Ringe and Taylor make the same split at the Northwest Germanic level, linking Old Norse *hvina* and Old English *hwinan* to the same strong verb (Ringe and Taylor 2014).

The difference affects both phonology and morphology. The lament family has initial *w-*, diphthongal *ai*, and weak-II morphology, whereas the verb behind Old English *hwīnan* has initial *hw-/xw-*, long *ī*, and strong-verb inflection (Kroonen 2013; Orel 2003). The selected input **xwī́naną* therefore represents a competing comparative identification rather than a hidden cell of **wainōjaną*.

Old English evidence Clark Hall records *hwinan* with the gloss ‘to hiss, whizz, whistle’ (Clark Hall 1960). Seebold keeps the verb among the strong verbs and notes that only a present-tense attestation is directly preserved, while Sievers-Brunner likewise lists *hwinan stv.* (Seebold 1970; Sievers 1965).

The Old English form is normalized here as *hwīnan*. That normalization adds the usual vowel length marking to the dictionary spelling *hwinan*; it does not turn an unattested verb into a reconstructed one.

Development to Old English Once the strong-verb input **xwī́naną* is selected, the path to Old English is straightforward. The compact trace shows heavy-syllable nasal apocope, secondary nasalization, and weak-tail reduction, after which the form surfaces as *hwīnan*.

No special paradigm maneuver is needed for this verb. The comparison is between two different Germanic verb families: the Old English form belongs with the strong verb **hwīnan-*, not with the weak lament verb.

Verb-family comparison The comparison below is manual. It separates the competing comparative labels that stand behind the inherited Old English forms.

Verb family / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lament-family weak verb	**wainōjaną*	comparative continuation in OE *wānian*	wānian	competing citation reconstruction, but not the source of *hwīnan*
selected strong verb	**xwī́naną*	compact-trace output: *hwīnan*	hwīnan	exact match between selected input and OE verb
comparative North Germanic cognate	Northwest Germanic strong verb behind ON *hvina* / OE *hwinan*	ON *hvina* / OE *hwinan*	hwīnan	supports the strong-verb identification

withy — OE wīþiġ

Derivation: citation reconstruction **wáiθiz*; selected input **wī́θagą* > *wīþiġ* (early analogy).

Derivation trace Proto input: **wī́θagą*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**wī́θægą*
[no change]	OE Heavy Syllable Nasal Apocope	**wī́θæg*
Northwest Germanic	OE Velar Palatalization	**wī́θæʤ*
[no change]	OE Unstressed AE Merger	**wī́θeʤ*
	OE Late Unstressed Ag Suffix	**wī́θiʤ*

Outcome: *wīþiġ*

Reconstruction and comparative evidence The comparative evidence groups the word with Germanic forms of the **wīþja/ō-* or **wiþ-* type (Kluge 2011; Orel 2003). That material is useful for the cognate set, but it does not by itself explain the Old English suffix of *wīþiġ*.

For Old English, the relevant point is the suffix history. Campbell's account of OE *-ig*, including forms such as *hunig*, supports an analysis in which the *-iġ* of *wīþiġ* continues a derivational **-ag-* sequence rather than a heavy ja-stem **-ij-* (Campbell 1959, sects 275, 376). The earlier ja-stem pathway remains useful as a comparative possibility, but the Campbell-Adamczyk line on heavy ja-stems points instead toward *-e* or zero type outcomes (Campbell 1959; Adamczyk 2001). The selected input **wī́θagą* is thus a formation choice rather than a mere respelling of the comparative headword.

Old English evidence Clark Hall records the noun as *wiðig*, with related inflected forms of the same lexical base (Clark Hall 1960). The form used here, *wīþiġ*, is a normalized Old English spelling with macrons and palatal .

The relevant comparison form is therefore not a reconstructed dictionary convenience but an established Old English noun. What requires explanation is why the selected Proto-Germanic input is **wī́θagą* rather than a comparative **wīþja-*-type headword.

Development to Old English From **wī́θagą*, Anglo-Frisian brightening gives a fronted vowel in the suffixal syllable, and, on the Campbell analysis adopted here, the later Old English development of **-ag-* yields *-iġ* (Campbell 1959, sects 275, 376). Palatalization supplies the final *ġ*, and the full development reaches *wīþiġ*.

This derivation is regular for the selected formation. The central claim of the entry is therefore morphological: Old English *wīþiġ* belongs with an **-ag-* derivative, whereas the comparative **wīþja-* label belongs to a different way of presenting the cognate family.

Formation comparison The comparison below is manual. It distinguishes the comparative headword from the Old English-facing formation that actually yields the attested noun.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative family label	*wáiθiz	broader cognate-set headword	OE family context	useful lexeme label, but not the direct source of *wīþiġ*
heavy ja-stem analysis	**wīþja-* type	Campbell/Adamczyk-style heavy ja-stem *-e* / zero outcome	*wīþiġ*	does not account cleanly for the OE suffix
selected **-ag-* derivative	*wī́θagą	compact-trace output: *wīþiġ*	wīþiġ	exact match between formation and target

world — OE weorold

Derivation: citation reconstruction **wíra-àldiz*; selected input **wír-àldu* > *weorold* (early analogy).

Derivation trace Proto input: **wíràldu*

Earlier Germanic changes		Old English changes	
West Germanic		OE Inter Stress Raising	**wéruldu*
[no change]		OE Med Unstressed U Lowering	**wéroldu*
Northwest Germanic		OE Back Mutation	**wéoroldu*
NWGmc I Lowering	**wéràldu*	OE High Vowel Apocope	**wéorold*

Outcome: *weorold*

Reconstruction and comparative evidence The word is the old compound ‘age of men’. Comparative sources preserve two slightly different views of its first element. Orel and the **wira-* tradition keep the older *i*-vocalism, while Ringe and Taylor discuss the lowered form **weraldiz* and its pre-Old-English chain **weraldu* > **weruld* (Orel 2003; Ringe and Taylor 2014, 341). Kluge-Seebold preserves both tendencies by giving compound **wira-aldō* beside simplex **wera-* (Kluge 2011).

The selected input **wír-àldu* therefore differs from the citation label in two ways. It keeps the older **wir-* vowel of the comparative headword, but it also presupposes the early shift of the compound into the ō-stems that Ringe and Taylor note for this lexeme (Ringe and Taylor 2014, 341). The early analogical step lies in that stem-class reassignment; the later phonological developments can then run regularly.

Old English evidence Old English does not preserve a single isolated form. Ringe and Taylor give West Saxon *weorold* ~ *worold*, Mercian *weoruld*, Northumbrian *woruld*, and Kentish *wiarald* (Ringe and Taylor 2014, 341). Sievers-Brunner and Bright present the same wider set, including the syncopated *world* and later rounded *wurold* (Sievers 1965; Bright 1971, 465).

The selected target here is the West Saxon form *weorold*. It is an attested Old English form within that broader variant cluster, not the only form the lexeme ever shows.

Development to Old English From the selected input **wír-àldu*, Northwest Germanic *i*-lowering gives **wér-àldu*. Inter-stress raising then changes the medial *a* to *u*, producing **wér-uldu*. In the Old English branch that unstressed *u* lowers to *o*, and back mutation yields **wéor-oldu*; final high-vowel apocope then gives *weorold*.

This sequence matches the comparative background in Ringe and Taylor's **weraldiz* > **weraldu* > **weruld* chain while preserving the **wir-* notation of the selected comparative label (Ringe and Taylor 2014, 341). The modeled Old English form therefore stands at the meeting point of an early stem-class reshaping and later regular sound change.

Stage comparison The comparison below is manual. It separates the comparative headword from the OE-facing stage chosen for the derivation.

Stage / interpretation	Candidate form	Old English outcome or comparison	Relevance to this entry
comparative compound with older first-element vowel	**wíra-àldiz*	citation reconstruction / lexeme label	preserves the older **wir-* tradition of the compound
literature-stage lowered compound after early stem-class shift	**weraldiz* > **weraldu* > **weruld*	Ringe-Taylor background chain to OE *weorold* ~ *worold*	explains the older comparative literature cited for the word
selected OE-facing input	**wír-àldu*	compact-trace output: *weorold*	exact match for the selected West Saxon target
broader OE variant cluster	—	*worold*, *weoruld*, *woruld*, *wiarald*, *world*	real attested comparanda that remain outside the selected target line

youth — OE ġeoguþ

Derivation: citation reconstruction **júgunθiz*; selected input **júgunθ* > *ġeoguþ* (early analogy).

Derivation trace Proto input: **júgunθ*

Earlier Germanic changes		Old English changes	
West Germanic		OE Diphthong Leveling	**jéogūθ*
[no change]		OE Unstressed Long Vowel Shortening	**jéoguθ*
Northwest Germanic			
OE Ws Palatal Glide	**jéugunθ*		
NWGmc Nasal Spirant Lengthening	**jéugūnθ*		
NWGmc Nasal Spirant Loss	**jéugūθ*		

Outcome: *ġeoguþ*

Reconstruction and comparative evidence The wider etymological tradition reconstructs an earlier form of the word as **ju(w)unþi-* (Kroonen 2013; Fulk 2018). The selected comparative label **júgunθiz* already stands at a later Germanic stage with *g*, and the chosen input **júgunθ* is later again: it represents the form after final *-i* has been lost.

That staging matters because Ringe and Taylor explicitly give the sequence **jugunþi* > **juguþ* > *OE geoguþ* ~ *iuguþ* (Ringe and Taylor 2014). Campbell makes the same point with the parallel *duguþ* < **dugunþ-*, adding *geoguþ* to the same history (Campbell 1959). The selected input therefore differs from the broader comparative headword because the Old English development must begin after early loss of final *-i*.

Old English evidence The Old English noun is attested with varying spellings. Campbell records forms of the *iuguþ* / *gioguð* / *geoguð* type, and Ringe and Taylor likewise cite *geoguþ* ~ *iuguþ* (Campbell 1959; Ringe and Taylor 2014). The form is normalized here as *ġeoguþ*: the initial palatal is written with *ġ*, and the attested spelling variation is treated as orthographic rather than lexical.

Nothing in the source stack suggests that a different paradigm cell should be chosen. The relevant Old English comparison form is the noun *ġeoguþ* itself.

Development to Old English The decisive early step is the loss of final *-i* before the Old English umlaut stage. If that high vowel remained, the word would develop an over-umlauted *y*-type vowel instead of the attested form (Ringe and Taylor 2014).

From the selected input **júgunθ*, the later development is regular: palatal fronting yields **jéugunθ*; nasal-spirant lengthening and loss give **jéogūθ*; unstressed long-vowel shortening then produces **jéoguθ*, which surfaces as *ġeoguþ*. Medial *u* remains preserved because the handbooks treat this environment as one of stem-*u* harmony after stressed *u*, citing forms such as *munuc*, *duguþ*, and *iuguþ* (Campbell 1959; Sievers 1965; Luick 1914--1940, 397).

Stage comparison The comparison below is manual. It separates the broader comparative headword from the later stages relevant to the Old English noun.

Stage / interpretation	Candidate form	Old English outcome or comparison	Relevance to this entry
earlier etymological headword	**ju(w)unþi-*	comparative family background	older comparative reconstruction of the lexeme
later g-bearing comparative label	**júgunθiz*	citation reconstruction / lexeme label	preserves the later Germanic stage behind the selected entry
selected OE-facing input	**júgunθ*	compact-trace output: *ġeoguþ*	exact match for the chosen Old English form
full *-i* stage retained too long	**jugunþi*	expected over-umlauted *y*-type result	negative control showing why early *-i* loss must precede the OE umlaut stage

Part IV. Late analogy and paradigm-cell selection

These entries involve a later paradigm-cell or analogical comparison. The citation reconstruction remains relevant to the lexeme, but the selected target is best explained through a particular inflectional or analogical form rather than through the citation form alone.

ban — OE bannes

Derivation: citation reconstruction **bánną*; selected input **bánnas* > *bannes* (late analogy).

Derivation trace Proto input: **bánnas*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**bánnæs*
[no change]	OE Unstressed AE Merger	**bánnes*
Northwest Germanic		
[no change]		

Outcome: *bannes*

Reconstruction and comparative evidence Orel cites a bann-noun under **bannan*, while Seebold distinguishes bann-stems of both masculine and neuter type and gives Old English *gebann* as the noun reflex (Orel 2003; Seebold 1970). The citation reconstruction **bánną* therefore names the lexeme, but the selected input **bánnas* is a specific genitive singular cell.

That distinction matters because the analysis depends on medial, not final, gemination.

Old English evidence Old English lexicographic evidence securely supports the noun itself. Clark Hall gives +*bann*, and Bosworth-Toller records *ge-bann* with oblique usage such as *gebanne* (Clark Hall 1960; Bosworth and Toller 1898, 303). The exact unprefixed genitive *bannes* is less directly cited in the reviewed material, so it is best treated here as the selected regular genitive comparison form rather than as a dictionary headword.

Development to Old English From **bánnas*, the geminate remains medial before the case ending and the unstressed vowel develops regularly to give *bannes*. By contrast, citation **bánną* loses its final vowel and simplifies the word-final geminate, so the regular nominative outcome is *ban* (Campbell 1959).

Paradigm comparison The comparison below is manual. It shows why the genitive singular is the conservative cell used for the entry.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*bánną	compact-trace output: *ban*	ban	regular nominative outcome, but not the selected target
selected genitive singular	*bánnas	compact-trace output: *bannes*	bannes	exact match for the chosen conservative cell

berry — OE berġes

Derivation: citation reconstruction **bázją*; selected input **bázjas* > *berġes* (late analogy).

Derivation trace Proto input: **bázjas*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**bærjæs*
[no change]	OE I Umlaut	**berjæs*
Northwest Germanic	OE Unstressed AE Merger	**berjes*
[no change]		

Outcome: *berġes*

Reconstruction and comparative evidence Kroonen reconstructs the berry noun as **basja-* ~ **bazja-* (Kroonen 2013). The selected input **bázjas* is therefore not a rival lexeme headword, but a specific genitive singular cell drawn from that paradigm.

The relevant point is that **rj* did not geminate in Proto-West Germanic. Ringe and Taylor's *here, herges* comparison shows the same *rj* environment in an Old English paradigm without any hidden gemination repair (Ringe and Taylor 2014).

Old English evidence Campbell cites feminine *berige* 'berry' and notes that *-j-* is retained after *r* in this type (Campbell 1959, 250). The reviewed evidence therefore supports the citation form more directly than the exact genitive *berġes*, which is best read here as the selected regular genitive comparison form rather than as a dictionary headword.

Development to Old English Citation **bázją* gives *bere*, not the selected target. The genitive singular **bázjas*, however, gives *berġes*, with medial *-rġ-* preserved in the same way that Ringe and Taylor cite *herges* beside *here* (Ringe and Taylor 2014). This points to paradigm choice rather than to an extra phonological rule.

Paradigm comparison The comparison below is manual. It shows the contrast between the citation form and the selected genitive singular cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*bázją	compact-trace output: *bere*	berige / berġe	useful citation-form background, but not the selected target
selected genitive singular	*bázjas	compact-trace output: *berġes*	berġes	exact match for the chosen conservative cell

bow — OE bēag

Derivation: citation reconstruction **béuganą*; selected input **báug* > *bēag* (late analogy).

Derivation trace Proto input: **báug*

Earlier Germanic changes	Old English changes	
West Germanic	OE Au Fronting	**báeug*
[no change]	OE Diphthong Leveling	**bēag*
Northwest Germanic		
[no change]		

Outcome: *bēag*

Reconstruction and comparative evidence The inherited verb belongs to the class-II strong-verb family **béuganą* (Ringe and Taylor 2014, 55). Within that paradigm, however, the infinitive and the singular preterite continue different ablaut grades. The selected input **báug* is the singular preterite cell, whereas the citation form **béuganą* is the infinitive.

Campbell's account of Old English class-II strong verbs treats the singular preterite *au* > *ēa* development as regular in this environment (Campbell 1959, 53). That is the phonological path relevant for *bēag*, whereas the analogical *ū* of the present stem belongs to the separate history behind *būgan* (Ringe and Taylor 2014, 55).

Old English evidence Bosworth-Toller and Clark Hall both record *bēag* as a preterite form of *būgan* (Bosworth and Toller 1898, 122; Clark Hall 1960, 45). The form discussed here is therefore an attested Old English verbal form, not a reconstructed substitute for the infinitive.

The ordinary dictionary headword remains *būgan*, but the relevant comparison form for this entry is the singular preterite *bēag*. That is the paradigm cell in which the inherited **au* grade is preserved most directly.

Development to Old English From **báug*, Anglo-Frisian fronting and the later leveling of the diphthong produce *bēag* (Campbell 1959, 53). No special analogical repair is needed for that cell. The form is the regular Old English outcome of the singular-preterite grade.

The analogical element in the wider lexeme belongs instead to the present stem seen in *būgan*. The selected input differs from the citation form because the regular inherited pathway survives more transparently in the preterite than in the infinitive.

Paradigm comparison The comparison below is manual. It distinguishes the regular singular preterite from the more familiar infinitival citation form.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	*béuganą	inherited present-stem history behind *būgan*	būgan	establishes the lexeme, but not the selected target
singular preterite	*báug	compact-trace output: *bēag*	bēag	exact match between input, output, and attested cell
past participial branch	participial **bugan-* type	later participial outcomes	bogen-type evidence	relevant to the paradigm, but not the clearest match for this entry

The singular preterite is the relevant comparison form. It gives a direct lautgesetzlich path to attested *bēag*, while the citation form *būgan* belongs to a paradigm whose present stem has already undergone later leveling.

cow — OE cȳ

Derivation: citation reconstruction **kōz*; selected input **kūi* > *cȳ* (late analogy).

Derivation trace Proto input: **kūi*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	**kȳi*
[no change]		OE High Vowel Apocope	**kȳ*
Northwest Germanic			
[no change]			

Outcome: *cȳ*

Reconstruction and comparative evidence Kroonen reconstructs a root noun with the inherited alternation **kō-* ~ **ku-*, explicitly *nom. *kōz, obl. *kū-* (Kroonen 2013, 299). The citation form therefore belongs to the nominative singular, whereas the selected input **kūi* belongs to the oblique stem.

Ringe and Taylor also posit a later PNWGmc nominative **kūaz* > **kūz* behind Old English *cū* (Ringe and Taylor 2014, sect. 3.1.3). That nominative history matters for the headword, but the form *cȳ* depends on the oblique **kū-* stem and a following **i*.

Old English evidence Clark Hall lemmatizes the noun under *cū* and records gen.sg. *cū(e), cȳ, cūs*, dat.sg. *cȳ*, nom.-acc.pl. *cȳ*, and dat.pl. *cūm* (Clark Hall 1960). Ringe and Taylor likewise state that the root-noun *cū* exhibits dat.sg., nom.-acc.pl. *cȳ* < **cūi*, dat.pl. *cūm* < **cūm*, and apparently gen.sg. *cā* < **cūiz* (Ringe and Taylor 2014, sect. 6.6.1).

The dictionary headword is therefore *cū*, but *cȳ* is an established Old English paradigm form rather than a convenient reconstruction. It serves as datative singular and also as nominative-accusative plural. The genitive singular is less stable, with *cā*, *cȳ*, *cū(e)*, and *cūs* all appearing in the local source record.

Development to Old English **kūi* is the dative singular of the oblique **kū-* stem. The following **i* triggers i-umlaut, so *ū* becomes *ȳ*, and loss of the final high vowel leaves *cȳ*. The development is **kūi* > **kȳi* > **kȳ* > *cȳ*.

This is the regular oblique-cell path recognized by Ringe and Taylor's paradigm statement (Ringe and Taylor 2014, sect. 6.6.1). It also explains why *cȳ* is the cleanest comparison form for the present entry, even though the ordinary headword is *cū*.

Paradigm comparison A paradigm comparison identifies the Proto-Germanic inflectional cell that corresponds to an established Old English paradigm form. The comparison below is manual; no full automatic paradigm-generation run is presented here.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*kōz	OE headword *cū* belongs to the nominative history of the lexeme	cū	useful background, but not the chosen comparison for *cȳ*
later generalized nominative	PNWGmc *kūaz* > kūz	inferred nominative *cū*	cū	explains the leveled headword, not the oblique target
dative singular oblique	*kūi	compact-trace output: *cȳ*	cȳ	exact match between input, output, and paradigm cell

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
genitive singular oblique	*kūiz	Ringe-Taylor: apparently *cā*; Hall: *cū(e), cȳ, cūs*	gen.sg. variable	too unstable to control the entry

The dative singular is the relevant comparison form. It gives a regular path to attested *cȳ*, while the broader Old English paradigm shows how far the oblique **kū-* grade spread beyond that one cell.

find — OE fundene

Derivation: citation reconstruction **fínθaną*; selected input **fúnðanǭ* > *fundene* (late analogy).

Derivation trace Proto input: **fúnðanǭ*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Fronting Early	**fúndænǭ*
PWGmc Dental Hardening	**fúndanǭ*	OE Unstressed Long Vowel Shortening	**fúndænœ*
Northwest Germanic		OE Unstressed AE Merger	**fúndene*
[no change]			

Outcome: *fundene*

Reconstruction and comparative evidence The inherited verb is the strong verb **fínθaną*, continued by Old English *findan* (Ringe and Taylor 2014). The selected input **fúnðanǭ*, however, belongs to the past-participial paradigm rather than to the infinitive. It represents an oblique singular form of the participle.

That distinction matters because the familiar dictionary form *funden* is not the cleanest inherited comparison. Campbell, Luick, and Brunner treat the better-known nominative participial forms as analogically leveled from inflected cases of the paradigm (Campbell 1959; Luick 1914--1940; Sievers 1965). The selected input therefore targets the cell in which the regular development is most transparent.

Old English evidence Bosworth-Toller records *fundene* under *findan* as an Old English participial form (Bosworth and Toller 1898). Clark Hall likewise preserves the participial stem in forms such as *funden* and *tō-fundennes* (Clark Hall 1960).

The ordinary dictionary headword for the participle is *funden*, but the relevant comparison form for this entry is the attested oblique *fundene*. It is an Old English form in its own right, not a merely convenient probe.

Development to Old English From **fúnðanǭ*, the participial oblique develops through regular loss and weakening of the final ending, yielding *fundene*. In that cell both the consonantism and the medial vowel history remain regular.

The broader participial paradigm then matters for interpretation. The more familiar nominative *funden* represents a later analogical leveling, whereas the selected oblique form preserves the inherited pathway more directly.

Paradigm comparison The comparison below is manual. It distinguishes the attested oblique participle from the more familiar but analogically leveled participial forms.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	*fínθaną	inherited verb *findan*	findan	establishes the lexeme, but not the selected target
nominative participial line	*fúnðanaz	later leveled *funden* type	funden	important paradigm background, but not the cleanest regular cell
selected oblique participle	*fúnðanǭ	compact-trace output: *fundene*	fundene	exact match between input, output, and attested cell

The oblique participle is the relevant comparison form. It preserves the inherited development most directly, while the nominative participial headword belongs to a later analogical leveling within the paradigm.

fright — OE fyrhte

Derivation: citation reconstruction **furxtīn*; selected input **fúrxtīnaz* > *fyrhte* (late analogy).

Derivation trace Proto input: **fúrxtīnaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**fúrxtīn*
[no change]		OE I Umlaut	**fyrxtīn*
Northwest Germanic		NWGmc In Stem N Loss	**fyrxtī*
PGmc Final Z Deletion	**fúrxtīna*	OE Unstressed Long Vowel Shortening	**fyrxti*
		OE Med Unstressed I Lowering1	**fyrxte*

Outcome: *fyrhte*

Reconstruction and comparative evidence The noun belongs to the inherited in-stem abstract **furxtīn*, the same family as Gothic *faurhtei* (Orel 2003). The selected input **fúrxtīnaz* is not a different lexeme but an oblique singular cell within that in-stem paradigm.

Ringe and Taylor treat the later nominative forms with *-u* or *-o* as analogically remodeled, whereas the oblique in-stem forms continue the older history more directly (Ringe and Taylor 2014). The selected input therefore differs from the citation form because the oblique cell preserves the inherited pathway more clearly than the better-known lemma forms do.

Old English evidence Bosworth-Toller records *fyrhte* with multiple textual attestations, and it also records nominative forms such as *fyrhtu* and *fyrhto* (Bosworth and Toller 1898). The noun must be kept distinct from the adjective *forht*, a distinction also reflected in Clark Hall (Clark Hall 1960).

The relevant comparison form is therefore the attested oblique *fyrhte*. The nominative lemma forms remain part of the Old English evidence, but they are not the cleanest inherited comparison for this entry.

Development to Old English From **fúrxtīnaz*, the oblique in-stem develops through the loss and weakening of the final ending and the regular OE history summarized by Campbell as *-e* < *-i* < *-in* in this class of abstracts (Campbell 1959). That yields *fyrhte*.

The later nominative forms with *-u* or *-o* belong to a subsequent analogical reshaping of the paradigm. The selected target is earlier in that sense: it is the attested OE cell in which the inherited in-stem development remains most transparent.

Paradigm comparison The comparison below is manual. It separates the attested oblique form from the later remodeled nominative line.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation in-stem headword	*furxtīn	broader noun-class label	wider family context	useful lexeme label, but not the selected cell
remodeled nominative line	nominative in-stem forms	*fyrhtu* / *fyrhto* type lemma forms	fyrhtu / fyrhto	genuine OE evidence, but later remodeled
selected oblique singular	*fúrxtīnaz	compact-trace output: *fyrhte*	fyrhte	exact match between input, output, and attested cell

The oblique in-stem form is the relevant comparison form. It yields attested *fyrhte* directly, while the more familiar nominative forms belong to a later analogical layer.

hammer — OE hameres

Derivation: citation reconstruction **xámaraz*; selected input **xámaras* > *hameres* (late analogy).

Derivation trace Proto input: **xámaras*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**xámæræs*
[no change]	OE Unstressed AE Merger	**xámeres*
Northwest Germanic		
[no change]		

Outcome: *hameres*

Reconstruction and comparative evidence The inherited noun is the masculine a-stem **xámaraz*, reflected in Old English citation forms such as *hamor* and *hamer* (Kroonen 2013; Orel 2003, 197; Clark Hall 1960, 160). The selected input **xámaras* is the genitive singular of that same noun rather than a different lexeme.

The genitive matters because the citation tradition is already mixed in its unstressed vowel, while the oblique singular gives a cleaner comparison form. This is a cell choice within one paradigm, not a change of stem class.

Old English evidence Bosworth-Toller directly records *hameres* in an Old English genitival phrase (Bosworth and Toller 1898). The same dictionary tradition and Clark Hall also preserve the simplex headword as *hamor* or *hamer* (Bosworth and Toller 1898; Clark Hall 1960, 160).

Sievers-Brunner gives a paradigm line *hamor* — *hamores*, which shows that the oblique tradition itself was not entirely uniform (Sievers 1965). The relevant comparison form here is the attested genitive singular *hameres*.

Development to Old English From **xámaras*, Anglo-Frisian brightening and the subsequent merger of unstressed *æ* with *e* yield *hameres*. The derivation of that oblique form is straightforward once the genitive singular cell is selected.

The noun as a whole retains a mixed citation tradition in *hamor* and *hamer*, and the selected oblique cell avoids making that variation carry the argument of the entry.

Paradigm comparison The comparison below is manual. It separates the attested genitive singular from the less stable citation tradition.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	*xámaraz	regular citation form *hamer* / *hamor*	hamor / hamer	good lexical background, but not the selected target
selected genitive singular	*xámaras	compact-trace output: *hameres*	hameres	exact match between input, output, and attested cell
later oblique tradition	oblique a-stem forms	*hamores* type evidence	hamores	attested background variant, but not the chosen comparison form

have — OE hæfeþ

Derivation: citation reconstruction **xabēną*; selected input **xábēθi* > *hæfeþ* (late analogy).

Derivation trace Proto input: **xábēθi*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**xæbǣθ*
PWGmc Early I Apocope	**xábēθ*	OE Velar Fricative Palatalization	**çæbǣθ*
Northwest Germanic		PGmc B Allophony	**çæβǣθ*
NWGmc Long E Lowering	**xábǣθ*	OE Unstressed Long Vowel Shortening	**çæβæθ*
		OE Unstressed AE Merger	**çæβeθ*

Outcome: *hæfeþ*

Reconstruction and comparative evidence The verb belongs to the inherited class-III weak paradigm usually cited under **xabēną* and Old English *habban* (Kroonen 2013; Ringe and Taylor 2014). Within that paradigm, however, the infinitive and the singular present indicative do not continue the same stem. Ringe and Taylor distinguish a *-ja-* stem in the infinitive from a non-geminating *-ai-* / *-ē-* stem in the 2sg and 3sg present forms (Ringe and Taylor 2014).

The selected input **xábēθi* is therefore the 3sg present cell rather than a rephrasing of the infinitive. Fulk's discussion of *habban* treats the ordinary citation form as analogically leveled, which is why the finite cell is the cleaner comparator here (Fulk 2018).

Old English evidence The ordinary Old English headword is *habban*, while the present paradigm also shows forms of the *hæf-* type (Bosworth and Toller 1898; Clark Hall 1960, 157). Campbell notes occasional unsyncopated forms that support the normalized target *hæfeþ* (Campbell 1959).

The target form is therefore a normalized finite cell rather than the ordinary dictionary lemma. It represents the inherited non-geminating present stem more directly than *habban* does.

Development to Old English From **xábēθi*, early apocope of final *-i*, subsequent vowel changes, and the regular fricative outcome of *b* in this environment yield *hæfeþ* (Ringe and Taylor 2014; Campbell 1959). The derivation of the finite form itself is regular.

The wider lexeme is less straightforward only because the infinitive *habban* shows later leveling. That difference in paradigm history is what makes the 3sg present cell the more useful comparison form.

Paradigm comparison The comparison below is manual. It separates the analogically leveled citation form from the regular 3sg present line.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	*-ja-* stem of **xabēną*	citation form *habban*	habban	important headword, but shaped by later leveling
selected 3sg present	*xábēθi	compact-trace output: *hæfeþ*	hæfeþ	exact match between input, output, and selected finite cell
syncopated finite tradition	same present stem	*hæfþ* type evidence	hæfþ	genuine later OE finite form, but not the normalized target used here

heaven — OE heofon

Derivation: citation reconstruction **xémenaz*; selected input **xémonų* > *heofon* (late analogy).

Derivation trace Proto input: **xémonų*

Earlier Germanic changes		Old English changes	
West Germanic		OE Med Unstressed U Lowering	**xéβonų*
[no change]		OE Velar Fricative Palatalization	**çéβonų*
Northwest Germanic		OE Back Mutation	**çéoβonų*
NWGmc Unstressed O Raising	**xémunų*	OE High Vowel Apocope	**çéoβon*
NWGmc Mn Dissimilation	**xéβunų*		

Outcome: *heofon*

Reconstruction and comparative evidence The inherited noun belongs to the mn-stem family cited by Kroonen as **hemina-* ~ *hemna-** (Kroonen 2013). The selected input **xémonų* is an oblique singular form within that paradigm rather than the lexeme-level citation form **xémenaz*.

That difference matters for the West Saxon target. The back-vocalic oblique stem provides the environment for the diphthong seen in *heofon*, whereas a front-vocalic stem yields Anglian or Mercian *hefen* / *heofen* (Campbell 1959; Ringe and Taylor 2014).

Old English evidence Old English dictionaries record the standard West Saxon noun as *heofon*, alongside Anglian or Mercian *hefen* material (Clark Hall 1960; Bosworth and Toller 1898, 43). Campbell also cites an earlier stage *hefzen* in the history of the word (Campbell 1959).

The target of this entry is the West Saxon citation form *heofon*. Its vowel history points toward the oblique stem rather than the front-vocalic nominative line.

Development to Old English From **xémonų*, the relevant pathway includes early *o*-raising before *u*, dissimilation in the m/n cluster, and later lowering of unstressed *u* (Fulk 2018; Campbell 1959). Back mutation then yields *heo-* before the labial plus back-vowel sequence, and loss of the final high vowel gives *heofon*.

The front-vocalic nominative line remains important as background because it explains the dialectal *hefen* type. West Saxon *heofon* reflects the oblique stem that was generalized into the nominative position.

Paradigm comparison The comparison below is manual. It distinguishes the front-vocalic nominative line from the oblique stem selected for West Saxon *heofon*.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	*xémenaz	front-vocalic *hefen* type outcome	hefen / heofen	useful control, but not the selected West Saxon target
selected oblique singular	*xémonų	compact-trace output: *heofon*	heofon	exact match between input, output, and target
older pre-OE stage	inherited oblique line	earlier *hefzen* stage	hefzen	historical background for the same West Saxon development

live — OE lifeþ

Derivation: citation reconstruction **libēną*; selected input **líbēθi* > *lifeþ* (late analogy).

Derivation trace Proto input: **líbēθi*

Earlier Germanic changes		Old English changes	
West Germanic		PGmc B Allophony	**líβǣθ*
PWGmc Early I Apocope	**líbēθ*	OE Unstressed Long Vowel Shortening	**líβæθ*
Northwest Germanic		OE Unstressed AE Merger	**líβeθ*
NWGmc Long E Lowering	**líbǣθ*		

Outcome: *lifeþ*

Reconstruction and comparative evidence The inherited verb belongs to the class-III weak family cited by Kroonen under **libēn-*, reflected in Old English *libban* (Kroonen 2013). Ringe and Taylor show that the paradigm also contained a separate 3sg present stem, continued in late Northumbrian *lifed*, which they treat as an archaism (Ringe and Taylor 2014).

The selected input **líbēθi* therefore represents a finite present cell rather than the citation infinitive. That distinction matters because the ordinary later lemma tradition also includes remodeled forms such as *lifian*.

Old English evidence The ordinary dictionary headwords are *libban* and, in later remodeling, *lifian* (Bosworth and Toller 1898; Clark Hall 1960). For this entry, however, the relevant comparison form is the archaic 3sg present attested as *lifed*, here normalized as *lifeþ* (Ringe and Taylor 2014; Campbell 1959).

The target is thus a normalized finite form, not the ordinary dictionary lemma. Its value lies in preserving the older present-stem history more clearly than the remodeled lemma tradition does.

Development to Old English From **líbēθi*, regular reduction of the final syllable and later weakening of the unstressed vowel yield *lifeþ*. The attested spelling *lifed* belongs to the same finite form in late Northumbrian orthography (Campbell 1959; Ringe and Taylor 2014).

Paradigm comparison The comparison below is manual. It separates the archaic finite cell from the ordinary infinitival and later remodeled lemma lines.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive line	*libēną	OE *libban* headword tradition	libban	establishes the lexeme, but not the selected target
selected 3sg present	*líbēθi	compact-trace output: *lifeþ*; attested *lifed*	lifed, normalized here as lifeþ	selected archaic finite cell
later remodeled present tradition	later class-II-type forms	*lifian* and related finite remodeling	lifian	genuine OE development, but secondary to the selected cell

man — OE mannes

Derivation: citation reconstruction **mánnaz*; selected input **mánnas* > *mannes* (late analogy).

Derivation trace Proto input: **mánnas*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**mánnæs*
[no change]	OE Unstressed AE Merger	**mánnes*
Northwest Germanic		
[no change]		

Outcome: *mannes*

Reconstruction and comparative evidence The lexeme-level reconstruction is not uniform. Kroonen cites **mannan-*, Orel has **mannz*, and Ringe and Taylor summarize the inherited noun as **mann-* (Kroonen 2013; Orel 2003, 299; Ringe and Taylor 2014). The selected input **mánnas* belongs to a different level: it is the genitive-singular cell chosen for the Old English comparison.

That distinction matters because the target of this entry is not the ordinary citation form. The selected cell is the one that keeps the geminate medial before the ending.

Old English evidence Campbell gives the paradigm *mann, man / mannes / menn* (Campbell 1959). Sievers-Brunner likewise cites *man mannes* and explains that word-final simplification underlies forms such as *man* beside inflected *monnes* (Sievers 1965). Clark Hall keeps the dictionary headword under *mann* (Clark Hall 1960).

The relevant comparison form is therefore the attested genitive singular *mannes*, not the citation lemma *mann*.

Development to Old English From **mánnas*, Anglo-Frisian brightening yields **mánnæs*, and the later unstressed merger gives **mánnes*, hence *mannes* (Campbell 1959). In this cell the geminate remains medial before *-es*. The citation form behaves differently because word-final gemination was simplified in Old English (Sievers 1965).

Paradigm comparison The comparison below is manual. It separates the citation-form line from the selected genitive singular.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	*mannăz	expected citation-form outcome *man*	mann / monn	establishes the lexeme, but not the selected target
accusative singular	*mannį̨	expected *man*	man	same word-final simplification as the nominative
dative singular	*mannăi	expected *manne*	manne	preserves medial *nn*, but not the chosen cell
selected genitive singular	*mánnas	compact-trace output: *mannes*	mannes	exact match between input, output, and attested comparator

meed — OE meorde

Derivation: citation reconstruction **mizdō*; selected input **mízdai* > *meorde* (late analogy).

Derivation trace Proto input: **mízdai*

Earlier Germanic changes		Old English changes	
West Germanic		OE Breaking	**méordē*
PWGmc Ai Monophthongization	**mírdē*	OE Unstressed Long Vowel Shortening	**méorde*
Northwest Germanic			
NWGmc I Lowering	**mérdē*		

Outcome: *meorde*

Reconstruction and comparative evidence The lexeme-level reconstruction is **mizdō*, but the selected input **mízdai* is a dative-singular cell rather than the citation form. That distinction is important because the Old English evidence for the *meord* side is oblique.

The wider history of competing *mēd* remains disputed. Crist, Kroonen, Ringe and Taylor, and Fulk explain it through some form of *z*-loss and compensatory lengthening (Crist 2002; Kroonen 2013; Ringe and Taylor 2014; Fulk 2018), Orel keeps a doublet analysis (Orel 2003), and Kilday instead argues that West Saxon *mēd* is a Saxono-Frisian loan (Kilday 2024). The comparison here concerns the attested oblique line *meorde*.

Old English evidence The directly attested forms are obliques: *meorde* as a dative singular and *meorda* as a genitive plural (Bright 1971; Bosworth and Toller 1898). Lexicographers reconstruct a bare nominative *meord* from those obliques, while West Saxon prose more commonly shows the competing doublet *mēd* (Clark Hall 1960; Bosworth and Toller 1898).

The target of this entry is therefore the attested oblique *meorde*, not the reconstructed lemma *meord* and not the better-known West Saxon citation form *mēd*.

Development to Old English From **mízdai*, rhotacism gives **mírdai*. Proto-West-Germanic monophthongization then yields **mírdē*, Northwest-Germanic lowering gives **mérdē*, Old English breaking yields **méordē*, and unstressed shortening gives **méorde*, hence *meorde* (Ringe and Taylor 2014).

This is the regular oblique-cell path modeled by the current trace. The entry therefore depends on the attested oblique line rather than on a full decision about the history of the competing *mēd* tradition.

Paradigm comparison The comparison below is manual. It distinguishes the attested oblique target from the broader lemma history.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	*mizdō	inferred lemma outcome *meord*	meord	useful background, but the bare lemma is reconstructed rather than directly attested
selected dative singular	*mízdai	compact-trace output: *meorde*	meorde	exact match between selected input and attested target
genitive singular	*mizdōz	compact-trace output: *meorde*	meorde	converges on the same attested string, but the dat.sg. has the clearest direct support
genitive plural control	plural oblique line	attested *meorda*	meorda	confirms the broader oblique tradition, but not the chosen singular target

night — OE niht

Derivation: citation reconstruction **náxtz*; selected input **náxti* > *niht* (late analogy).

Derivation trace Proto input: **náxti*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**næxti*
[no change]	OE Breaking	**neaxti*
Northwest Germanic	OE I Umlaut	**niexti*
[no change]	OE Ws Palatal Umlaut	**nixti*
	OE High Vowel Apocope	**nixt*

Outcome: *niht*

Reconstruction and comparative evidence Ringe and Taylor distinguish the high-vowel oblique and plural side of the word from the nominative citation form: they cite *gen.sg. *nahtiz*, *dat.sg. *nahti*, and *nom.pl. *nahtiz*, and derive West Saxon *niht* from that side of the paradigm (Ringe and Taylor 2014). The citation reconstruction **náxtz* therefore belongs to the nominative-like headword, while the selected input **náxti* represents the dative-singular cell.

That distinction matters because the word later became the model for endingless datives. Ringe and Taylor explicitly explain forms such as *dæg* by analogy with *dat. sg. niht* < **nahti* (Ringe and Taylor 2014).

Old English evidence Clark Hall lemmatizes *niht* and records the spelling range *æ, e, ea, ie, y*, while cross-referencing forms such as *neaht*, *neht*, and *nieht* (Clark Hall 1960). Campbell likewise preserves the fluctuation between *neaht* and *niht*, giving genitive *nihte, nihtes*, dative *niht, nihte*, nominative plural *niht*, and the contrasting plural-side forms represented by *neahtas* (Campbell 1959).

The comparison form used here is therefore an attested Old English *niht*, not a reconstructed substitute. The broader lexical record still preserves the non-umlauted side of the paradigm in *neaht*-type forms.

Development to Old English From **náxti*, Anglo-Frisian brightening first fronts the root vowel, and the following high vowel then triggers i-umlaut. In West Saxon, the sequence before *ht* yields *niht*, whereas plural forms with a following back vowel preserve the non-umlauted *neahtas* type (Ringe and Taylor 2014; Campbell 1959; Sievers 1965).

The modeled path is therefore **náxti* > **næxti* > **neaxti* > **niexti* > **nixti* > *niht*.

Paradigm comparison The comparison below is manual. It identifies the inherited cell that matches the attested Old English form.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*náxtz	expected non-umlauted outcome *neaht*	neaht	useful background, but not the selected comparison for *niht*
selected dative singular	*náxti	compact-trace output: *niht*	niht	exact match between input, output, and paradigm cell

rest — OE ræste

Derivation: citation reconstruction **rastō*; selected input **rástōz* > *ræste* (late analogy).

Derivation trace Proto input: **rástōz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Surviving Bimoric O Unrounding	**rástā*
[no change]		Anglo Frisian Brightening	**ræstǣ*
Northwest Germanic		OE Unstressed Long Vowel Shortening	**ræstæ*
PGmc Final Z Deletion	**rástō*	OE Unstressed AE Merger	**ræste*

Outcome: *ræste*

Reconstruction and comparative evidence Kroonen treats the noun as a feminine ō-stem **rastō-*, continued by Old English *ræst* (Kroonen 2013). The selected input **rástōz* therefore does not replace the lexeme-level headword. It identifies one oblique singular cell on the side of the paradigm that yields *ræste*.

The source tradition used here labels that cell specifically as genitive singular, but the broader local synthesis of the ō-stem paradigm shows that the oblique singulars converge on the same front-vocalic *ræste* side, in contrast to a nominative singular that would remain *rast*.

Old English evidence The ordinary Old English citation form is *ræst* (Kroonen 2013; Clark Hall 1960). Bosworth-Toller also preserves oblique uses of *ræste*, including prepositional examples such as *on ræste* and *tó ræste* (Bosworth and Toller 1898, 121).

The comparison form used here is therefore an attested oblique *ræste*, not a reconstructed surrogate. The dictionary headword *ræst* remains an equally real part of the Old English record.

Development to Old English Once final **z* is lost, the selected input moves through the front-vocalic oblique side of the paradigm rather than the back-vocalic nominative side. In the modeled derivation, the surviving final long vowel is first exposed, unrounded, fronted, shortened, and then reduced to the final *-e* of *ræste*.

The key point is the paradigm split. Nominative **rastō* yields a regular *rast*, whereas the selected oblique input yields *ræste*. The later citation form *ræst* is best explained as leveling from that oblique *ræst-* stem.

Paradigm comparison The comparison below is manual. It distinguishes the nominative citation form from the oblique singular chosen here.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*rastō	expected regular outcome *rast*	ræst	useful background, but not the cell that matches attested oblique *ræste*
selected oblique singular	*rástōz	compact-trace output: *ræste*	ræste	exact match between selected input and attested OE oblique form

shoulder — OE sċuldrum

Derivation: citation reconstruction **skuldrō*; selected input **skúldramiz* > *sċuldrum* (late analogy).

Derivation trace Proto input: **skúldramiz*

Earlier Germanic changes		Old English changes	
West Germanic		OE Sk Palatalization	**ʃúldrum*
NWGmc A To U Before M	**skúldrumiz*		
PWGmc Early I Apocope	**skúldrumz*		
Northwest Germanic			
PGmc Final Z Deletion	**skúldrum*		

Outcome: *sċuldrum*

Reconstruction and comparative evidence The handbooks do not agree on the reconstruction of the Germanic word. Orel gives **skuldr(j)ō*, a feminine ō-/jō-stem, and explicitly notes that Old English *sculdor* is masculine beside OFrisian *skulder*, Middle Low German *schulder*, and Old High German *scultra, scultirra* (Orel 2003, 345). Kroonen reconstructs **skuldra-*, a masculine a-stem, and derives the Old High German feminine forms from **skuldrjōn-* (Kroonen 2013, 478). Ringe and Taylor cite PWGmc **skuldru* for the Old English branch (Ringe and Taylor 2014, 142).

These forms imply different stem classes and different expectations for the Old English inflection. The question is which inflectional cell best aligns with the Old English evidence.

A dative/instrumental plural form **skúldramiz* aligns with the inherited plural ending that later yields Old English *-um*, and it corresponds directly to the attested dative plural discussed below.

Old English evidence The ordinary Old English headword is *sculdor*. Bosworth-Toller and Clark Hall both lemmatize *sculdor* as the normal dictionary form, and the Bosworth-Toller material also preserves plural and oblique forms such as *sculdru*, *sculdra*, and *sculdrum* (Bosworth and Toller 1898; Clark Hall 1960). The dative plural *sculdrum* is directly attested in the dictionary material (Bosworth and Toller 1898).

Bosworth-Toller's Supplement records a weak-feminine *sculdra, an*, so *sculdra* belongs to the Old English record beside the stronger masculine paradigm headed by *sculdor* (Bosworth and Toller 1898). Brunner and Luick also record later spellings such as *sceoldor* and the i-mutated dative plural *scyldrum*, which reflect secondary phonological and analogical reshaping within Old English (Sievers 1965, sect. 92.2.a; Luick 1914--1940, 230).

The singular and plural evidence point to different parts of the paradigm. The relevant comparison form here is the attested dative plural *sċuldrum*. The spelling with *sċ-* is a normalized representation of the same Old English initial cluster.

Development to Old English Proto-Germanic **skúldramiz* can be interpreted as a dative/instrumental plural form. In this environment the post-tonic *a* before *m* is raised to *u*, giving a form of the **skúldrumiz* type. Unstressed *u* is regularly preserved before *m*, especially in the dative plural ending *-um*: Campbell states this explicitly, and Hogg formulates the same condition for the dative plural inflexion (Campbell 1959, sect. 373; Hogg 1992, sect. 3.3.1.3). Brunner points in the same direction by excluding *m* from the environments in which medial *o* became general in West Saxon (Sievers 1965, sect. 44 Anm. 7).

Subsequent reduction of the ending removes the final **i* and **z*, so that the inflectional ending appears in Old English as *-um*. The initial cluster is written here as *sċ-*, and the development is **skúldramiz* > **skúldrumiz* > **skúldrum* > *sċuldrum*.

Paradigm comparison A paradigm comparison identifies the Proto-Germanic inflectional cell that corresponds to an established Old English paradigm form. The comparison below is manual; no full automatic paradigm-generation run is presented here.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
singular-oriented citation input	*skúldrō	probe output: *sċoldor*	sculdor	fails: the singular output has root *o*, not the attested *u*
serious plural-based singular alternative	*skúldru	probe output: *sċuldor*	sculdor	close formally, but it compares a plural-stage input with a singular form
dat./inst.pl. input	*skúldramiz	compact-trace output: *sċuldrum*	sculdrum	matches both the output and the dative plural comparison form
later weak-feminine singular	—	OE *sculdra*	sculdra	secondary doublet, useful as a control rather than the inherited target

The dative plural line is decisive because it matches both the output and the paradigm cell of Old English *sculdrum*. Singular-oriented candidates either lower the root vowel or compare unlike cells.

shove — OE sċēaf

Derivation: citation reconstruction **skéubaną*; selected input **skáub* > *sċēaf* (late analogy).

Derivation trace Proto input: **skáub*

Earlier Germanic changes	Old English changes	
West Germanic	OE Au Fronting	**skáeub*
[no change]	OE Diphthong Leveling	**skēab*
Northwest Germanic	PGmc B Allophony	**skēaβ*
[no change]	OE Sk Palatalization	**ʃēaβ*

Outcome: *sċēaf*

Reconstruction and comparative evidence Kroonen reconstructs the strong verb as **skeuban-* ~ **skūban-* and cites Old English present forms *scēofan, scūfan* (Kroonen 2013). Ringe and Taylor also show that the English present system belongs to a wider class-II split that is not identical with the preterite grade (Ringe and Taylor 2014). The selected input **skáub* is therefore not a spelling variant of the infinitive but the singular preterite cell.

Old English evidence The ordinary dictionary verb is *scūfan/scēofan*, but the preterite itself is well attested. Bright gives the principal parts *scufan, sceaf, scufon, scofen* and also quotes *he sceaf þa mid þam scylde*; Sweet gives the same paradigm (Bright 1971; Sweet 1953). The normalized form here is *sċēaf*, regularizing the attested spellings *sceaf* and prefixed *āsceaf*.

Development to Old English From **skáub*, the documented trace is straightforward. **au* fronts and levels to *ēa*, final **b* becomes a fricative and is written *f*, and initial **sk-* undergoes the usual Old English palatalized spelling in this environment. The trace therefore gives **skáub* > **skáeub* > **skēab* > **skēaβ* > *sċēaf*.

Paradigm comparison A paradigm comparison is required here because the ordinary citation verb and the selected Old English target belong to different cells of the same strong paradigm. The comparison below is manual.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation infinitive	*skéubaną	inherited infinitive line *sċēofan*; present system also leveled *scūfan*	scēofan / scūfan	necessary background, but not the selected comparison for *sċēaf*
1/3 sg. preterite	*skáub	documented trace output: *sċēaf*	sċēaf	exact match between selected input and Old English target
preterite plural	*skúbun	later leveled plural *scufon* beside expected *sċufun* under the corrected cascade	scufon	poorer comparison for the singular-preterite target
past participle	*skúbanaz	attested participial line *scofen*	scofen	valid clean cell, but not the chosen one

span — OE spanne

Derivation: citation reconstruction **spannō*; selected input **spánnai* > *spanne* (late analogy).

Derivation trace Proto input: **spánnai*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Long Vowel Shortening	**spánne*
PWGmc Ai Monophthongization	**spánnē*		
Northwest Germanic			
[no change]			

Outcome: *spanne*

Reconstruction and comparative evidence Orel reconstructs the noun as **spannō*, and Seebold likewise gives Old English *spann* under the same noun family (Orel 2003; Seebold 1970, 450). The selected input **spánnai* is therefore not a rival headword, but a specific dative singular cell of the feminine ō-stem paradigm (Sievers 1965).

Old English evidence The reviewed lexicographic evidence more directly supports the citation noun *spann* than the exact form *spanne*. Clark Hall gives *spann*, and *spanne* is accordingly treated as the selected regular dative singular comparison form rather than as a dictionary headword (Clark Hall 1960).

Development to Old English Citation **spannō* yields *span*. The oblique cell **spánnai* therefore supplies the conservative comparison form: it preserves the medial geminate and yields *spanne*, while citation **spannō* gives the nominative background form.

Paradigm comparison The comparison below is manual. It shows the contrast between the citation form and the selected dative singular cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*spannō	compact-trace output: *span*	spann	useful citation-form background, but not the selected target
selected dative singular	*spánnai	compact-trace output: *spanne*	spanne	exact match for the chosen conservative cell

thistle — OE þistles

Derivation: citation reconstruction **θéstilaz*; selected input **θístilas* > *þistles* (late analogy).

Derivation trace Proto input: **θístilas*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**θístilæs*
[no change]	OE L Adjacent Syncope	**θístlæs*
Northwest Germanic	OE Unstressed AE Merger	**θístles*
[no change]		

Outcome: *þistles*

Reconstruction and comparative evidence The comparative tradition is divided. Orel prints **þe(x)stilaz*, while Kluge-Seebold gives **þistila-* (Orel 2003, 458; Kluge 2011). The comparative label **θéstilaz* therefore remains in view as the lexeme-level headword, while the selected input **θístilas* is a specific genitive singular cell.

Old English evidence The ordinary simplex headword tradition is broken *þistel* / *ðistel*. Clark Hall gives *ðistel* as the noun headword (Clark Hall 1960, 326). The selected target here is the genitive singular *þistles*, which preserves the same stem in an oblique form where the cluster is medial.

Development to Old English Campbell's discussion of cluster nouns shows the contrast clearly. Simplex forms often develop a parasite vowel in word-final obstruent + sonorant clusters, while comparable medial clusters remain unbroken; his examples include *hrefn*, *tacn*, *wépn*, and *botm* beside forms with parasitic vowels elsewhere in the same lexical class (Campbell 1959, 151). The selected genitive singular **θístilas* therefore supplies the conservative comparison form: the cluster is medial and the regular development yields *þistles*, while the simplex nominative belongs to the broken headword tradition *þistel*.

Paradigm comparison The comparison below is manual. It shows the contrast between the citation form and the selected genitive singular cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*θéstilaz	computed output: *þistl*	þistel	useful citation-form background, but not the selected target
selected genitive singular	*θístilas	computed output: *þistles*	þistles	exact match for the chosen conservative cell

make (iptv.2sg) — OE maca

Derivation: citation reconstruction **makōną*; selected input **mákô* > *maca* (late analogy).

Derivation trace Proto input: **mákô*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**mækô*
[no change]	OE A Restoration	**makô*
Northwest Germanic	OE Unstressed Long Vowel Shortening	**maka*
[no change]		

Outcome: *maca*

Reconstruction and comparative evidence The make-family belongs to the Old English class-II weak verbs. Campbell cites *macian* beside *lapian* and similar verbs with restored *a*, and Ringe and Taylor place the Germanic verb in the same class, comparing West Germanic continuants such as Old Frisian *makia*, Old Saxon *makon*, and Old High German *mahhon* (Campbell 1959; Ringe and Taylor 2014).

The selected input **mákô* is not the citation form of the lexeme but a finite paradigm cell. Ringe and Taylor's account of the class-II weak paradigm is the reason for choosing it: the imperative singular belongs to the small set of finite cells that preserve the trimoric **ō* directly, whereas the ordinary infinitive continues the remodelled **-ōja-* formation behind *macian* (Ringe and Taylor 2014).

Old English evidence The dictionary headword is *macian* (Clark Hall 1960). The selected form in this entry is therefore not the lemma but the imperative singular *maca*, chosen as a paradigm form beside the headword *macian* and the related finite form *macaþ*.

That distinction matters for the comparison. The lexical history of the verb is still the history of *macian*, but the finite cell isolates the regular outcome of trimoric **ō* more cleanly than the citation form does.

Development to Old English From **mákô*, Anglo-Frisian brightening first gives **mækô*. In this class-II environment A-restoration returns the stem vowel to *a*, and unstressed long vowel shortening then gives **maka*, whence *maca* (Campbell 1959; Ringe and Taylor 2014).

The same development explains why earlier fronted forms of the *mæċa* type do not control the entry. Once trimoric **ô* is treated as a back-vocalic trigger for restoration, the imperative singular falls into line with the broader *macian* family.

Paradigm comparison A paradigm comparison identifies which finite cell of the make-family matches the Old English form chosen here. The comparison below is manual.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**mákōjaną*	comparative continuation *macian*	macian	ordinary headword of the verb, but not the selected finite cell
selected imperative singular	**mákô*	compact-trace output: *maca*	maca	exact match between input, output, and selected paradigm form
present third singular companion	**mákōθi*	comparative companion *macaþ*	macaþ	useful family control, but not the target of this entry

make (3sg) — OE macaþ

Derivation: citation reconstruction **makōną*; selected input **mákōθi* > *macaþ* (late analogy).

Derivation trace Proto input: **mákōθi*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**mækōθ*
PWGmc Early I Apocope	**mákōθ*	OE A Restoration	**makōθ*
Northwest Germanic		OE Late O Shortening	**makaθ*
[no change]			

Outcome: *macaþ*

Reconstruction and comparative evidence Kroonen derives the Old English verb from **makōjan-* on the make-family base **maka-* (Kroonen 2013). Ringe and Taylor likewise place the verb among the class-II weak verbs and note that the present 2sg, 3sg, and imperative singular preserve suffixal **ō* rather than the remodelled infinitival formation (Ringe and Taylor 2014).

The selected input **mákōθi* is therefore a finite 3sg cell of the same family, not the citation form of the verb.

Old English evidence Clark Hall lemmatizes the verb as *macian* (Clark Hall 1960). The relevant comparison form here is the normalized present-third-singular *macaþ*, set beside the dictionary headword and the related imperative singular *maca*.

Campbell's class-II paradigm makes the ordinary 3sg ending *-aþ*, while his dialect survey allows secondary *-e-* spellings in some traditions (Campbell 1959, sect. 355.4; 1959, sect. 757). *Macaþ* is thus the regular comparison form for the non-*j* 3sg cell.

Development to Old English After early loss of final *-i*, **mákōθi* yields **mákōθ*. Anglo-Frisian brightening gives **mækōθ*, but Campbell lists *macian* among the class-II verbs with restored *a*, so the stem returns to *mak-* before the ending is reduced (Campbell 1959, sect. 159).

The ending then follows the ordinary class-II 3sg development. Campbell's *lufas, -aþ (< -ōsi, -ōþi)* and Ringe and Taylor's discussion of stable *a* in the finite non-*j* cells point to _*makōθ_ > _*makaθ_ > *macaþ* (Campbell 1959, sect. 355.4; Ringe and Taylor 2014).

Paradigm comparison The comparison below is manual. It distinguishes the selected 3sg cell from the make-family lemma and from the companion imperative form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	_*makōjaną_	dictionary headword *macian*	macian	family background, but not the selected cell
selected 3sg present	_*mákōθi_	trace output *macaþ*	macaþ	exact match
imperative singular companion	_*mákô_	related finite form *maca*	maca	useful control, but not the target

bore (iptv.2sg) — OE bora

Derivation: citation reconstruction _*burōną_; selected input _*búrô_ > *bora* (late analogy).

Derivation trace Proto input: _*búrô_

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Long Vowel Shortening	_*bóra_
[no change]			
Northwest Germanic			
NWGmc U Lowering	_*bórô_		

Outcome: *bora*

Reconstruction and comparative evidence Kroonen reconstructs the bore-family verb as _*burojan-_ and cites Old English *borian* among its continuants (Kroonen 2013). In Ringe and Taylor's account of class-II weak verbs, the imperative singular belongs to the finite cells that preserve inherited suffixal _*ō_, unlike the remodelled infinitive (Ringe and Taylor 2014).

The selected input _*búrô_ is therefore an imperative cell of the same family, not the citation form of the verb.

Old English evidence Clark Hall lemmatizes the verb as *borian* (Clark Hall 1960). The comparison form here is the normalized imperative singular *bora*, used beside the headword and the related 3sg form *boraþ*.

The imperative is thus a paradigm form rather than a replacement for the dictionary lemma. It is the most direct Old English comparator for the non-*j* finite cell represented by _*búrô_.

Development to Old English Northwest Germanic lowering first gives _*bórô_ from _*búrô_, and late shortening of the unstressed long vowel then yields _*bóra_, whence *bora*.

Ringe and Taylor's class-II imperative singular *-a* < _*-ō_ points to exactly this type of outcome (Ringe and Taylor 2014). The selected form therefore isolates the regular finite-cell development more cleanly than the remodelled infinitive does.

Paradigm comparison The comparison below is manual. It distinguishes the selected imperative cell from the bore-family lemma and from the companion 3sg form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**burōjaną*	dictionary headword *borian*	borian	family background, but not the selected cell
selected imperative singular	**búrô*	trace output *bora*	bora	exact match
3sg present companion	**búrōθi*	related finite form *boraþ*	boraþ	useful control, but not the target

bore (3sg) — OE boraþ

Derivation: citation reconstruction **burōną*; selected input **búrōθi* > *boraþ* (late analogy).

Derivation trace Proto input: **búrōθi*

Earlier Germanic changes		Old English changes	
West Germanic		OE Late O Shortening	**bóraθ*
PWGmc Early I Apocope	**búrōθ*		
Northwest Germanic			
NWGmc U Lowering	**bórōθ*		

Outcome: *boraþ*

Reconstruction and comparative evidence Kroonen reconstructs the bore-family verb as **burojan-* and cites Old English *borian* among its reflexes (Kroonen 2013). The selected form isolates the finite 3sg cell **búrōθi* rather than the infinitive.

Campbell's class-II pattern *lufas, -aþ (< -ōsi, -ōþi)* and Ringe and Taylor's account of stable *a* in the class-II 2sg and 3sg make this finite cell the relevant comparison form for the ending (Campbell 1959, sect. 355.4; Ringe and Taylor 2014).

Old English evidence Clark Hall lemmatizes the verb as *borian* (Clark Hall 1960). The relevant comparison form here is the normalized present-third-singular *boraþ*, used beside the headword and the imperative singular *bora*.

Campbell's dialect survey allows secondary *-e-* and *-o-* spellings in 2sg and 3sg class-II forms, but the basic ending remains *-aþ* (Campbell 1959, sect. 757). *Boraþ* is therefore the regular comparison form for this non-*j* 3sg cell.

Development to Old English Early loss of final *-i* first gives **búrōθ* from **búrōθi*. Northwest Germanic lowering then produces **bórōθ*, and late shortening of unstressed *ō* yields **bóraθ*, whence *boraþ*.

Campbell's class-II ending evidence and Ringe and Taylor's discussion of stable *a* in the finite non-*j* cells support exactly this sequence (Campbell 1959, sect. 355.4; Ringe and Taylor 2014).

Paradigm comparison The comparison below is manual. It distinguishes the selected 3sg cell from the bore-family lemma and from the companion imperative form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**burōjaną*	dictionary headword *borian*	borian	family background, but not the selected cell
selected 3sg present	**búrōθi*	trace output *boraþ*	boraþ	exact match
imperative singular companion	**búrô*	related finite form *bora*	bora	useful control, but not the target

learn (iptv.2sg) — OE liorna

Derivation: citation reconstruction **liznōjaną*; selected input **líznô* > *liorna* (late analogy).

Derivation trace Proto input: **líznô*

Earlier Germanic changes	Old English changes	
West Germanic	OE Breaking	**líornô*
[no change]	OE Unstressed Long Vowel Shortening	**líorna*
Northwest Germanic		
[no change]		

Outcome: *liorna*

Reconstruction and comparative evidence Ringe and Taylor give Old English *liornian* ~ *leornian* from a learn-family base of the **lizn-* type, and Kroonen likewise keeps the weak verb as **liznōn-* (Ringe and Taylor 2014; Kroonen 2013, 380). Fulk cites the same Old English family from **liznō-* (Fulk 2018).

The selected input **líznô* is a finite imperative cell of that family, not the citation form of the verb.

Old English evidence Clark Hall gives the ordinary headword as *leornian* (Clark Hall 1960). Brunner, however, explicitly records *leornian, nordh. auch liorna*, and Campbell notes that beside *leornian* Northumbrian forms with *io* occur where original *eo* and *io* remain distinct (Sievers 1965; Campbell 1959, sect. 123 n. 2).

Liorna can therefore be treated as an attested Northumbrian finite form, while *leornian* remains the better-known dictionary headword.

Development to Old English The selected form develops regularly as **líznô* > **lírnô* by rhotacism, then **líornô* by breaking before *rn*, and finally **líorna* by late shortening of the unstressed long vowel.

Campbell's Northumbrian *io* evidence and Ringe and Taylor's explicit statement that no form of *liornian* stood in an i-umlauting environment support this stem shape (Campbell 1959, sect. 123 n. 2; Ringe and Taylor 2014). The West-Saxon-looking *eo* forms belong to a different dialectal presentation of the same family.

Paradigm comparison The comparison below is manual. It distinguishes the selected imperative cell from the learn-family infinitive and from the companion 3sg form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**liznōjaną*	Northumbrian *liornian*; dictionary headword often *leornian*	liornian / leornian	family background, but not the selected cell
selected imperative singular	**líznô*	trace output and Brunner's Northumbrian *liorna*	liorna	exact match
3sg present companion	**líznōθi*	related finite form *liornaþ*	liornaþ	useful control, but not the target

learn (3sg) — OE liornaþ

Derivation: citation reconstruction **liznōjaną*; selected input **líznōθi* > *liornaþ* (late analogy).

Derivation trace Proto input: **líznōθi*

Earlier Germanic changes		Old English changes	
West Germanic		OE Breaking	**líornōθ*
PWGmc Early I Apocope	**lírnōθ*	OE Late O Shortening	**líornaθ*
Northwest Germanic			
[no change]			

Outcome: *liornaþ*

Reconstruction and comparative evidence Ringe and Taylor give Old English *liornian* ~ *leornian* from a learn-family base of the **lizn-* type, and Kroonen likewise keeps the weak verb as **liznōn-* (Ringe and Taylor 2014; Kroonen 2013, 380). The selected input **líznōθi* is the finite 3sg cell of that family, not the citation form of the verb.

For the ending, Campbell's *lufas, -aþ (< -ōsi, -ōþi)* and Ringe and Taylor's discussion of stable *a* in the class-II 2sg and 3sg make the non-*j* 3sg cell the relevant comparison point (Campbell 1959, sect. 355.4; Ringe and Taylor 2014).

Old English evidence Clark Hall gives the ordinary headword as *leornian* (Clark Hall 1960). Brunner records Northumbrian finite forms in *liorn-*, including *liorna* and the 3sg *liornes*, beside the West-Saxon-looking *leornian* tradition (Sievers 1965). Campbell likewise notes Northumbrian forms with *io* beside *leornian* (Campbell 1959, sect. 123 n. 2).

The relevant comparison form here is the normalized 3sg *liornaþ*. The directly cited Old English evidence supports the finite stem *liorn-*; the exact *-aþ* ending follows the regular class-II 3sg pattern.

Development to Old English The selected form develops as **líznōθi* > **lírnōθi* by rhotacism, then **lírnōθ* after early apocope of final *-i*, then **líornōθ* by breaking before *rn*, and finally **líornaθ* > *liornaþ* by late shortening of the unstressed long vowel.

Campbell's Northumbrian *io* evidence and Ringe and Taylor's statement that no form of *liornian* stood in an i-umlauting environment support the stem, while Campbell's class-II ending evidence supports the final *-aþ* (Campbell 1959, sect. 123 n. 2; Campbell 1959, sect. 355.4; Ringe and Taylor 2014).

Paradigm comparison The comparison below is manual. It distinguishes the selected 3sg cell from the learn-family infinitive and from the companion imperative form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**liznōjaną*	Northumbrian *liornian*; dictionary headword often *leornian*	liornian / leornian	family background, but not the selected cell
selected 3sg present	**líznōθi*	trace output *liornaþ*	liornaþ	exact match
imperative singular companion	**líznô*	trace output and Brunner's Northumbrian *liorna*	liorna	useful control, but not the target

lick (iptv.2sg) — OE licca

Derivation: citation reconstruction **likkōną*; selected input **líkkô* > *licca* (late analogy).

Derivation trace Proto input: **líkkô*

Earlier Germanic changes	Old English changes	
West Germanic	OE Unstressed Long Vowel Shortening	**líkka*
[no change]		
Northwest Germanic		
[no change]		

Outcome: *licca*

Reconstruction and comparative evidence Ringe and Taylor place the verb among the West Germanic class-II weak verbs, with PWGmc **li/ekkōn* continuing as Old English *liccian*, Old Saxon *likkon*, and Old High German *lecchon* (Ringe and Taylor 2014). Orel gives the fuller weak-verb reconstruction **likkōjanan* with the same Old English continuation (Orel 2003).

The form treated here is not that remodeled infinitive but a finite cell in bare trimoric **-ō*. Campbell's weak class-II discussion and Ringe and Taylor's account of the paradigm both distinguish those finite singular cells from the ordinary **-ōja-* citation formation (Campbell 1959; Ringe and Taylor 2014).

Old English evidence Bosworth-Toller lemmatizes the verb as *liccian*, and Campbell and Brunner likewise cite *liccian* with preserved geminate *cc* (Bosworth and Toller 1898; Campbell 1959; Sievers 1965). The Old English evidence therefore establishes the verbal headword and its consonantal frame securely.

The selected target in this entry is the imperative singular *licca*. It is a paradigm form chosen beside the headword *liccian* and the related present *liccaþ*, not a separately lemmatized citation word.

Development to Old English With the stem *licc-* established, the remaining development is brief. Trimoric **-ô* shortens late to *-a* in this finite class-II cell, so **líkkô* yields **líkka* and then *licca* (Campbell 1959; Ringe and Taylor 2014). The same stem consonantism that appears in *liccian* is preserved here, giving *cc* throughout the finite form.

Paradigm comparison The comparison below is manual.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**líkkōjaną*	manual probe output *liccian*	liccian	ordinary dictionary headword of the verb, but not the selected finite cell
selected imperative singular	**líkkô*	manual probe output *licca*	licca	exact match between the chosen input and the selected target
present third singular companion	**líkkōθi*	manual probe output *liccaþ*	liccaþ	useful family control, but not the target of this entry

lick (3sg) — OE liccaþ

Derivation: citation reconstruction **likkōną*; selected input **líkkōθi* > *liccaþ* (late analogy).

Derivation trace Proto input: **líkkōθi*

Earlier Germanic changes		Old English changes	
West Germanic		OE Late O Shortening	**líkkaθ*
PWGmc Early I Apocope	**líkkōθ*		
Northwest Germanic			
[no change]			

Outcome: *liccaþ*

Reconstruction and comparative evidence Ringe and Taylor place the verb among the West Germanic class-II weak verbs, with PWGmc **li/ekkōn* continuing as Old English *liccian*, Old Saxon *likkon*, and Old High German *lecchon* (Ringe and Taylor 2014). Orel gives the fuller weak-verb reconstruction **likkōjanan* with the same Old English continuation (Orel 2003).

The selected form in this entry is the non-*j* present third singular **líkkōθi*, not the remodeled infinitive. Campbell states the class-II present endings as *lufas, -aþ (< -ōsi, -ōþi)*, and Ringe and Taylor likewise treat the class-II 2sg and 3sg as forms with stable *-a-* (Campbell 1959; Ringe and Taylor 2014).

Old English evidence Bosworth-Toller lemmatizes the verb as *liccian*, and the same consonantal frame appears in Campbell and Brunner's grammatical citations of *liccian* (Bosworth and Toller 1898; Campbell 1959; Sievers 1965). The Old English headword is therefore clear even though the entry here is not about the citation form.

The form treated here is the present third singular *liccaþ*. It is a selected paradigm form beside the lemma *liccian* and the related imperative *licca*, not a separately lemmatized headword.

Development to Old English **líkkōθi* first loses final *-i*, giving **líkkōθ*. Late shortening of unstressed **ō* then yields **líkkaθ*, written *liccaþ* (Campbell 1959; Ringe and Taylor 2014). Because this ending never contains *-j-*, the form does not pass through an i-umlauted *-eþ* stage; the regular class-II outcome is *-aþ*.

Paradigm comparison The comparison below is manual.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**líkkōjaną*	manual probe output *liccian*	liccian	ordinary dictionary headword of the verb, but not the selected finite cell
imperative singular companion	**líkkô*	manual probe output *licca*	licca	useful family control, but not the target of this entry
selected present third singular	**líkkōθi*	manual probe output *liccaþ*	liccaþ	exact match between the chosen input and the selected target

show (iptv.2sg) — OE sċēawa

Derivation: citation reconstruction **skawōną*; selected input **skáwô* > *sċēawa* (late analogy).

Derivation trace Proto input: **skáwô*

Earlier Germanic changes	Old English changes	
West Germanic	OE Aw Long Diphthong	**skḗawô*
[no change]	OE Sk Palatalization	**ʃḗawô*
Northwest Germanic	OE Unstressed Long Vowel Shortening	**ʃḗawa*
[no change]		

Outcome: *sċēawa*

Reconstruction and comparative evidence Orel reconstructs the verb as **skawōjanan* and cites Old English *sceáwian* beside Old Frisian *skawia*, Old Saxon *skawōn*, and Old High German *scouwōn* (Orel 2003). The selected input in this entry is not that infinitive but the imperative singular **skáwô*, a finite class-II cell that preserves bare **-ō* rather than the remodeled **-ōja-* formation (Ringe and Taylor 2014).

That distinction matters because the imperative singular provides the direct comparison for the Old English form treated here. The lexical history still belongs to the *sceáwian* verb, but the selected cell isolates the finite *-a* outcome more clearly than the citation form does.

Old English evidence Bright lists *scēawian* and explicitly gives the imperative singular *scēawa* under that headword (Bright 1971). The form treated here is therefore an attested finite paradigm form, not a reconstructed convenience form.

The spelling used in this entry is normalized *sċēawa*, while Bright's glossary gives source spelling *scēawa*. The ordinary Old English headword remains *scēawian*; *sċēawa* is the imperative singular chosen beside it.

Development to Old English From **skáwô*, Old English **aw* before vocalic material yields **skḗawô*, initial *sk* palatalizes, and late shortening of trimoric **-ô* gives **ʃḗawa*, written *sċēawa* (Campbell 1959). The result is therefore the expected finite singular form of the *scēawian* family rather than an analogical replacement of the headword.

Paradigm comparison The comparison below is manual.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**skáwōjaną*	manual probe output *sċēawian*	scēawian	ordinary dictionary headword of the verb, but not the selected finite cell
selected imperative singular	**skáwô*	manual probe output *sċēawa*	scēawa / normalized sċēawa	exact match between the chosen input and the selected target
present third singular companion	**skáwōθi*	manual probe output *sċēawaþ*	sċēawaþ	useful family control, but not the target of this entry

show (3sg) — OE sċēawaþ

Derivation: citation reconstruction **skawōną*; selected input **skáwōθi* > *sċēawaþ* (late analogy).

Derivation trace Proto input: **skáwōθi*

Earlier Germanic changes		**Old English changes**	
West Germanic		OE Aw Long Diphthong	**skḗawōθ*
PWGmc Early I Apocope	**skáwōθ*	OE Sk Palatalization	**ʃḗawōθ*
Northwest Germanic		OE Late O Shortening	**ʃḗawaθ*
[no change]			

Outcome: *sċēawaþ*

Reconstruction and comparative evidence Orel reconstructs the verb as **skawōjanan* and cites Old English *sceáwian* beside Old Frisian *skawia*, Old Saxon *skawōn*, and Old High German *scouwōn* (Orel 2003). The selected input in this entry is the present third singular **skáwōθi*, a finite class-II cell rather than the remodeled infinitive (Ringe and Taylor 2014).

Campbell states the class-II present endings as *lufas, -aþ (< -ōsi, -ōþi)*, and Ringe and Taylor likewise treat the class-II 2sg and 3sg as forms with stable *-a-* (Campbell 1959; Ringe and Taylor 2014). The relevant comparison is therefore the 3sg cell itself, not an i-umlauted alternative.

Old English evidence Bright lists the simplex headword *scēawian* and the imperative *scēawa*, and under *geond-scēawian* also records a third singular *-sceawað* (Bright 1971). The evidence thus establishes the *scēaw-* / *-awað* finite-cell pattern directly.

The form written here as *sċēawaþ* is the normalized simplex comparison form for that weak class-II pattern. It is therefore not a dictionary headword but a selected finite form aligned with the attested *scēaw-* evidence and the directly cited *-sceawað* ending pattern.

Development to Old English **skáwōθi* first loses final *-i*, giving **skáwōθ*. Old English **aw* before vocalic material yields **skǣ́wōθ*, initial *sk* palatalizes, and late shortening of unstressed **ō* gives **ʃǣ́awaθ*, written *sċēawaþ* (Campbell 1959). Campbell's chronology matters here: late-shortened *ō* yields *a* too late for Anglo-Frisian fronting, and the class-II 3sg ending therefore gives *-aþ* rather than *-eþ* (Campbell 1959; Ringe and Taylor 2014). Because the ending never contains *-j-*, no i-umlaut applies.

Paradigm comparison The comparison below is manual.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**skáwōjaną*	manual probe output *sċēawian*	scēawian	ordinary dictionary headword of the verb, but not the selected finite cell
imperative singular companion	**skáwô*	manual probe output *sċēawa*	scēawa	useful family control, but not the target of this entry
selected present third singular	**skáwōθi*	manual probe output *sċēawaþ*	normalized sċēawaþ; source-side pattern *-sceawað*	exact match for the selected cell

Part V. Reconstructed Old English comparators

These entries use an explicitly reconstructed Old English-stage comparator for the branch being modelled. The relevant comparison is therefore later than the Proto-Germanic citation form but still belongs to the lexical derivation layer.

knob — OE cnobba

Derivation: citation reconstruction **knúppaz*; selected input **knúbbô* > *cnobba* (reconstructed Old English comparator).

Derivation trace Proto input: **knúbbô*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Long Vowel Shortening	**knóbba*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**knóbbô*		

Outcome: *cnobba*

Reconstruction and comparative evidence The wider knob-family is not uniform. Kroonen's discussion of the Germanic n-stems points to related voiced and voiceless branches within this group (Kroonen 2011). The citation reconstruction **knúppaz* represents the broader cognate-set headword, while the selected input **knúbbô* represents the voiced weak-noun branch treated here.

That distinction matters because the Old English record is uneven. The better attested OE material belongs to the voiceless branch, but the present entry represents the reconstructed OE form that would continue the voiced branch behind later English **knob**.

Old English evidence Bosworth-Toller and Clark Hall preserve Old English evidence of the *cnopp / cnoppa* type (Bosworth and Toller 1898; Clark Hall 1960). Those forms are genuine Old English evidence, but they belong to the voiceless branch of the family.

The target *cnobba* is different in status. It is a **reconstructed Old English form**, not a directly attested one. The point of using it here is to give the voiced branch an explicit OE-stage representation instead of allowing the attested *cnoppa* branch to stand in for a different prehistory. The choice of *cnobba* is therefore a modeling and comparative decision rather than a settled point of Old English philology.

Development to Old English From the selected weak-noun input **knúbbô*, the regular Old English outcome is *cnobba*, with Proto-Germanic *kn-* represented in Old English as *cn-* and with the expected weak-noun ending.

The entry therefore does not claim that *cnobba* is attested. Its claim is different: if the voiced weak-noun branch is the one to be represented, then *cnobba* is the regular Old English form corresponding to that branch.

Reconstruction status The comparison below is manual. It keeps apart the reconstructed target and the better-attested neighboring forms.

Form	Status	Relevance to this entry
**knúbbô -> cnobba*	reconstructed OE form; trace-supported	selected target

Form	Status	Relevance to this entry
cnopp / *cnoppa*	attested OE branch	important control form, but belongs to the voiceless branch
cnæp	attested OE form from another family	not part of the present lexeme line

This remains the most review-sensitive item in the present pilot batch, because the choice between reconstructed *cnobba* and attested *cnoppa* is still a comparator-policy question rather than a settled point of OE attestation.

reek — OE rēac

Derivation: citation reconstruction _*ráukiz_; selected input _*ráukaz_ > *rēac* (reconstructed Old English comparator).

Derivation trace Proto input: _*ráukaz_

Earlier Germanic changes		Old English changes	
West Germanic		OE Au Fronting	_*ráeuka_
[no change]		OE Diphthong Leveling	_*rēaka_
		PWGmc Final Bare A Loss	_*rēak_
Northwest Germanic			
PGmc Final Z Deletion	_*ráuka_		

Outcome: *rēac*

Reconstruction and comparative evidence The wider noun family is represented by _*ráukiz_ / _*rauki-_, with Old English *rēc* as the attested noun reflex in the comparative dictionaries (Kroonen 2013, 446; Orel 2003, 338). The selected input _*ráukaz_ is therefore not the lexeme-level headword, but the form used here for the Old English derivation.

Old English evidence The attested noun is *rēc*, not *rēac*. Clark Hall records *rēc* as the noun and also preserves related forms such as *rēcels*; Kroonen likewise gives OE *rēc* under the noun family (Clark Hall 1960, 255; Kroonen 2013, 446). Clark Hall and Seebold also record verbal *rēac* as the preterite of *rēocan*, but that verbal form is separate from the noun treated here (Clark Hall 1960, 254; Seebold 1970, 380).

The selected target *rēac* is therefore a reconstructed West Saxon noun form, not a directly attested manuscript headword.

Development to Old English From _*ráukaz_, the regular West Saxon development gives *rēac*. The attested noun *rēc* belongs to the same lexical family, but reflects a later smoothed surface form rather than the regular noun target represented here.

Form note The distinction here is between an attested noun headword *rēc* and a reconstructed regular West Saxon target *rēac*. The latter is treated as the modelling target, while the former remains philological background.

strew — OE strīeġan

Derivation: _*stráwjaną_ > *strīeġan* (reconstructed Old English comparator).

Derivation trace Proto input: **stráwjaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Awj Glide Formation	**stráujaną*
[no change]	OE Au Fronting	**stráeujaną*
Northwest Germanic	OE Diphthong Leveling	**strēajaną*
[no change]	OE Heavy Syllable Nasal Apocope	**strēajan*
	OE Secondary Nasalization	**strēająn*
	OE I Umlaut	**strīejąn*
	OE Weak Tail Reduction	**strīejan*
	OE J Strengthening After Front Diphthong	**strīeʒan*

Outcome: *strīeġan*

Reconstruction and comparative evidence Kroonen cites the inherited weak verb as **straujan-* and gives Old English *streowian* as its dictionary continuation (Kroonen 2013, 483). Ringe and Taylor make the split within Old English explicit: the inherited class-I verb is continued by Anglian *strēgan*, while West Saxon *streowian* is a remodelled class-II verb (Ringe and Taylor 2014, sect. 6.1 n. 27).

The aw-series comparison is important here. Luick groups **strauwjan* with the same set as **hauwja-* and **kauwjan*, yielding Anglian *strēzan* beside West Saxon forms of the *hīez, ciezan* type (Luick 1914--1940, sect. 98). Fulk likewise allows an early West Saxon **striegan* directly from Proto-Germanic **straujana* (Fulk 2018, sect. 4.10 n. 1).

Old English evidence The attested inherited Old English form is *strēgan* in Anglian. The attested West Saxon citation forms are *strewian*, *streowian*, and *strēawian*, which belong to the remodelled class-II branch (Ringe and Taylor 2014, sect. 6.1 n. 27; Campbell 1959, sect. 753.7).

The target *strīeġan* is therefore a **reconstructed Old English form**, not an attested manuscript lemma. It is the reconstructed West Saxon reflex of the inherited class-I verb, chosen to keep the inherited branch distinct from the better-attested remodelled West Saxon lemma.

Development to Old English From **stráwjaną*, the inherited West Saxon development passes through **straujaną*, fronting and leveling to a **strēajan-* stage, i-umlaut to **strīejan*, and retention or strengthening of the glide after the front diphthong, written here as *ġ*. The resulting form is *strīeġan*.

This differs from Anglian *strēgan*, where smoothing removes the diphthongal sequence, and from West Saxon *strewian* / *streowian* / *strēawian*, where the verb has already been remodelled into class II (Fulk 2018, sect. 4.10 n. 1; Campbell 1959, sect. 753.7).

Reconstruction status The comparison below is manual. It keeps apart the attested inherited branch, the attested remodelled branch, and the reconstructed West Saxon comparator.

Form or branch	Status	Relevance to this entry
strēgan	attested Anglian inherited class-I form	proves that the inherited verb survived into Old English
strīeġan	reconstructed West Saxon inherited class-I form; trace-supported	selected target
strewian / *streowian* / *strēawian*	attested remodelled West Saxon class-II forms	genuine OE evidence, but not the inherited branch modeled here

Part VI. Known but unmodelled remodellings

These entries preserve cases where the historical remodelling is broadly understood, but the current deterministic transducer does not model that later reshaping directly.

fire — OE fȳre

Derivation: **fūri* yields regular *fȳr*; the selected target is *fȳre* (known but unmodelled remodelling).

Derivation trace Proto input: **fūri*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	**fȳri*
[no change]		OE High Vowel Apocope	**fȳr*
Northwest Germanic			
[no change]			

Transducer outcome: *fȳr*

Selected target: *fȳre*

Reconstruction and comparative evidence Kroonen places the lexeme in a heteroclitic family **fōr* ~ **fun-* and explains the front-mutated West Germanic forms from an oblique form of the **fu(w)eri* type (Kroonen 2013). The selected input **fūri* therefore does not function as an arbitrary substitute for the headword: it represents the specific inherited cell that supplies the *i* needed for i-umlaut.

That distinction matters because the Old English target combines a regular inherited form *fȳr* with an attested analogical surface form *fȳre*.

Old English evidence Bosworth-Toller records *fyr* as the noun 'fire' and also preserves oblique *fyre* in the Old English record (Bosworth and Toller 1898, 288). The first is the regular inherited outcome of the phonological development from the selected input; the second shows the later restoration of a final *-e* within the paradigm.

The entry therefore concerns the relation between a regular inherited oblique input and an attested Old English surface form that has undergone later morphological remodeling.

Development to Old English From **fūri*, i-umlaut changes *ū* to *ȳ*, and subsequent loss of the final high vowel after a heavy syllable yields *fȳr* (Ringe and Taylor 2014; Hogg 1992; Campbell 1959). The inherited phonology is complete at that point.

fȳre is later than that inherited output. Its final *-e* belongs to analogical restoration in the Old English paradigm rather than to the original Proto-Germanic ending. The form therefore remains *known_unmodelled*: the deterministic phonology is regular, but the attested surface form includes later morphological rebuilding.

Paradigm comparison The comparison below is manual. It distinguishes the lexeme-level etymological background from the inherited cell that actually produces the front-mutated form and from the later analogical surface result.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
lexeme-level heteroclitic headword	**fōr* ~ **fun-*	comparative background only	fire family	explains the wider lexeme, but not the selected oblique input
inherited oblique cell	*fūri	compact-trace output: *fȳr*	fȳr	regular inherited output from the selected input
later analogical surface form	—	attested *fȳre* with restored *-e*	fȳre	genuine OE target, but not the direct phonological output

tap — OE tæppa

Derivation: **táppô* yields regular *tappa*; the selected target is *tæppa* (known but unmodelled remodelling).

Derivation trace Proto input: **táppô*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**tæppô*
[no change]	OE A Restoration	**tappô*
Northwest Germanic	OE Unstressed Long Vowel Shortening	**tappa*
[no change]		

Transducer outcome: *tappa*

Selected target: *tæppa*

Reconstruction and comparative evidence Orel gives the noun under **tappòn* and already connects it with Old English *tæppa* (Orel 2003). The selected input is therefore the inherited noun itself; the entry does not depend on a different lexeme-level proto or a different inherited noun cell.

Old English evidence The Old English noun family is well attested. Orel gives *tæppa*, and Clark Hall records *tæppa* together with derivatives *tæppere* and *tæppestre* (Orel 2003; Clark Hall 1960, 305). The target is therefore a real Old English noun form, not a reconstructed convenience spelling.

Development to Old English From **táppô*, the regular inherited noun path gives *tappa*. The attested target *tæppa* therefore stands outside that regular phonological development.

The mismatch is historically intelligible, but it is not solved here by a new inherited input. A related j-verb pathway would give *teppan*, not the noun target *tæppa*. The entry accordingly remains *known_unmodelled*.

Form comparison

Form type	Input or form	OE output or comparison	Result
regular inherited noun path	*táppô	compact-trace output: *tappa*	regular output, but not the target
attested OE target	—	*tæppa*	genuine target form, but analogically remodelled in the present classification
related j-verb background	*táppjaną	*teppan*	related formation, but not the noun target

Part VII. Unexplained or deliberately unmodelled exceptions

These entries preserve a mismatch between the regular transducer output and the selected Old English target. They are retained as documented lexical exceptions rather than treated as evidence for further sound-change repair.

buck — OE bucc

Derivation: **búkkaz* yields regular *bocc*; the selected target is *bucc* (unexplained exception).

Derivation trace Proto input: **búkkaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**bókk*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**bókkaz*		
PGmc Final Z Deletion	**bókka*		

Transducer outcome: *bocc*

Selected target: *bucc*

Reconstruction and comparative evidence Kroonen and Orel both reconstruct the word with a geminate stop, **bukkaz*, and both also preserve parallel n-stem material behind Old English *bucca* (Kroonen 2013; Orel 2003). The selected input therefore remains identical with the lexeme label: no alternative inherited cell accounts for the form.

Old English evidence Old English preserves a mixed lexical picture. Campbell cites *bucca* in the exception set for this phonological environment, while Clark Hall and Bosworth-Toller show that Old English has both *bucca* and *bucc* (Campbell 1959; Clark Hall 1960; Bosworth and Toller 1898, 122). The a-stem citation form *bucc* is the target treated here, with *bucca* kept as genuine philological background from the same lexical family.

Development to Old English From **búkkaz*, the regular inherited path gives *bocc*. That is the form expected under the ordinary lowering pattern in this environment. *bucc* therefore remains outside the deterministic phonology.

No accepted inherited cell repairs the mismatch. A high-vowel alternative would introduce i-umlaut and produce a *byċċ*-type form rather than the target. *bucc* is therefore best treated as a documented exception, not as a regular paradigm-cell survival.

Form comparison

Form type	Input or form	OE output or comparison	Result
regular inherited noun path	*búkkaz	compact-trace output: *bocc*	regular output, but not the target
attested OE target	—	*bucc*	genuine target form, but unexplained in the present classification
parallel OE lexical background	—	*bucca*	related n-stem form, not the present target

fowl — OE fugol

Derivation: **fúglaz* yields regular *fogol*; the selected target is *fugol* (unexplained exception).

Derivation trace Proto input: **fúglaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**fógl*
[no change]		OE Epenthetic Vowel	**fógol*
Northwest Germanic			
NWGmc U Lowering	**fóglaz*		
PGmc Final Z Deletion	**fógla*		

Transducer outcome: *fogol*

Selected target: *fugol*

Reconstruction and comparative evidence The noun is the ordinary Germanic a-stem **fúglaz*, continued by forms such as Old Norse *fugl* and Old High German *fogal* (Kroonen 2013, 197; Orel 2003, 155). There is no stem-class or paradigm-cell dispute behind this entry. The comparative headword and the selected input are the same.

The difficulty lies instead in the stressed root vowel. Under the regular West Germanic and Old English development, that *u* should lower before the following non-high vowel, yielding an *o*-vocalism (Ringe and Taylor 2014, 42–43; Campbell 1959, 43).

Old English evidence Old English dictionaries record the noun as *fugol*, with variant spelling *fugel* (Bosworth and Toller 1898, 282; Clark Hall 1960, 138). The target is therefore an attested ordinary Old English noun, not a reconstructed or selectively chosen paradigm form.

The attested word already contains the crucial problem. Its medial *-o-* is the ordinary parasite vowel of Old English cluster phonology, but the root *fu-* retains *u* where the regular history predicts *fo-* (Campbell 1959, 150).

Development to Old English From **fúglaz*, the regular cascade yields *fogol*: the root vowel lowers before the following non-high vowel (Ringe and Taylor 2014, 42–43; Campbell 1959, 43), final *z* is lost, and the cluster is resolved by the usual medial vowel (Ringe and Taylor 2014, 345; Campbell 1959, 150). That is the expected inherited outcome.

The attested Old English noun is *fugol*, not *fogol*. Luick and later handbooks treat this preservation of *u* as a small inherited residue, not as a categorical sound law (Luick 1914--1940, 148; Ringe and Taylor 2014, 47). The item therefore remains a genuine lexical exception rather than the output of a recoverable regular mechanism.

Expected and attested forms The comparison below is manual. It distinguishes the regular prediction from the attested Old English noun.

Form	Status	Relevance to this entry
fogol	computed regular output from **fúglaz*	establishes the expected inherited outcome
fugol	attested Old English form	selected target; preserves unexplained root *u*
fugel	attested variant spelling	secondary spelling variant of the attested noun

The unresolved point lies only in the root vowel. The medial *-o-* is regular, but no accepted lautgesetzlich pathway has been found from **fúglaz* to attested *fugol*.

rust — OE rust

Derivation: **rústō* yields regular *rost*; the selected target is *rust* (unexplained exception).

Derivation trace Proto input: **rústō*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**róst*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**róstō*		
NWGmc Final Long O Raising	**róstu*		

Transducer outcome: *rost*

Selected target: *rust*

Reconstruction and comparative evidence The comparative dictionaries do not support a single citation reconstruction uniformly. Orel cites **rustaz sb.m./f.* with Old English *rust* and Old Saxon and Old High German *rost* (Orel 2003). The form **rústō* therefore stands here as a competing citation reconstruction rather than as the best-supported inherited headword.

That disagreement does not remove the central problem. Whether one starts from **rústō* or from source-supported **rustaz*, the regular citation-form history points toward *rost*, not toward the attested Old English noun.

Old English evidence The Old English noun is attested, not reconstructed. Clark Hall gives *rūst m.* (Clark Hall 1960), and Bosworth-Toller records *rúst (? and rust)* (Bosworth and Toller 1898, 677). The form is normalized here as *rust* from that attested record.

Those dictionary entries also matter morphologically. They support a masculine noun, which aligns better with Orel's **rustaz* than with the competing **rústō* preserved in the header.

Development to Old English Under the regular lowering of stressed *u* before a following non-high vowel, the citation-form input gives *rost*, not *rust* (Campbell 1959; Ringe and Taylor 2014). The same regular result follows from the comparative citation-form reconstruction **rustaz*.

A high-vowel comparator such as instrumental-type **rústu* would yield *rust* regularly, but that does not explain the attested citation form of the noun. No accepted regular pathway from the citation form to attested *rust* has been established.

Expected and attested forms The comparison below is manual. It separates the regular inherited outcomes from the attested Old English noun.

Form / interpretation	Status	Relevance to this entry
**rústō* -> *rost*	computed regular output from one competing citation reconstruction	shows that this citation reconstruction does not reach attested *rust*
**rustaz* -> *rost*	expected regular output from the source-supported citation reconstruction	shows that correcting the stem class does not solve the vowel problem
**rústu* -> *rust*	regular high-vowel comparator	useful negative control, but not a defensible citation-form solution
rust	attested Old English noun, normalized from *rūst* / *rúst* / *rust*	selected target; the citation-form development remains unexplained

wolf — OE wulf

Derivation: **wúlfaz* yields regular *wolf*; the selected target is *wulf* (unexplained exception).

Derivation trace Proto input: **wúlfaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**wólf*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**wólfaz*		
PGmc Final Z Deletion	**wólfa*		

Transducer outcome: *wolf*

Selected target: *wulf*

Reconstruction and comparative evidence The inherited noun is an a-stem: Kroonen gives **wulfa-*, and the regular Old English development of that citation form is the same one reflected by Old High German *wolf* (Kroonen 2013; Ringe and Taylor 2014). Campbell accordingly names Old English *wulf* as an exception to the regular lowering of stressed *u* before a following non-high vowel (Campbell 1959).

The older literature often notices that the exceptional words cluster near labials. Bülbring lists *full*, *wulle*, and *wulf* together, but he also says that the ordinary rule still gives *o* in comparable forms such as *folc* and *bolt* (Bülbring 1902). Luick rejects a categorical labial blocker on the same grounds and prefers a lexical or analogical account instead (Luick 1914--1940, 148).

Old English evidence Old English *wulf* is an attested noun, and the handbook tradition treats it as the ordinary lexical form while simultaneously recognizing its exceptional vowel (Campbell 1959; Sievers 1965). The oblique form *wulfe* also belongs to the record, but Sievers-Brunner warns that this type continues *wulfi* only with i-umlaut later abandoned across the paradigm (Sievers 1965).

That warning matters because the surviving oblique forms do not supply a clean regular route back to bare *wulf*. They belong to the same lexeme, but they do not remove the explanatory problem presented by the citation form.

Development to Old English Under the regular lowering of stressed *u*, the citation-form input gives *wolf*, not *wulf* (Campbell 1959; Ringe and Taylor 2014). The compact trace shows exactly that path: **wúlfaz* > **wólfaz* > **wólfa* > *wolf*.

A high-vowel oblique input would behave differently. The following high vowel would block the lowering of *u*, but the same environment would trigger i-umlaut, so the regular result would be *wylf* or *wylfe*, not bare *wulf* (Sievers 1965). The attested noun therefore remains unexplained at the citation-form level.

Expected and attested forms The comparison below is manual. It separates the regular inherited outcomes from the attested Old English noun.

Form / interpretation	Status	Relevance to this entry
**wúlfaz -> wolf*	computed regular output from the citation form	shows the regular development expected from the inherited a-stem
OHG wolf	comparative regular cognate	confirms that the *o*-vocalism is the ordinary outcome

Form / interpretation	Status	Relevance to this entry
**wúlfi / *wúlfis -> wylf / wylfe*	expected high-vowel control forms	shows why oblique high-vowel cells do not solve the noun's vowel history
wulf	attested Old English noun	selected target; the preservation of *u* remains unexplained

wool — OE wull

Derivation: **wúllō* yields regular *woll*; the selected target is *wull* (unexplained exception).

Derivation trace Proto input: **wúllō*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**wóll*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**wóllō*		
NWGmc Final Long O Raising	**wóllu*		

Transducer outcome: *woll*

Selected target: *wull*

Reconstruction and comparative evidence The inherited form is a feminine ō-stem **wúllō*. In the ordinary phonological history of West Germanic, stressed *u* lowers before a following non-high vowel, so the regular Old English outcome is an *o*-form. Campbell's discussion of the parallel adjective *full*, with OHG *foll* as the regular comparator, shows that the handbooks treat this as a genuine exception cluster rather than as a place where the rule itself is doubtful (Campbell 1959, sect. 115).

Bülbring likewise lists *wulle* among the traditional *u*-preserving exceptions (Bülbring 1902, sect. 116). The comparative evidence therefore establishes two things at once: the regular result should be *woll*, and Old English still has a lexical exception of the *wull / wulle* type.

Old English evidence The Old English target is given here as *wull*, a normalized lexeme form. Handbook discussion often cites *wulle*, the feminine weak form of the noun (Bülbring 1902, sect. 116). Both point to the same lexical item and to the same exceptional preservation of root *u*.

The OE evidence therefore does not remove the problem. It confirms that the language has a *u*-form where the regular phonology would have produced *o*.

Development to Old English From **wúllō*, the regular sequence is lowering of stressed *u* before a non-high vowel, followed by the ordinary later reductions of the ending. The regular outcome is therefore *woll*.

That regular derivation is not the attested Old English form. Luick rejects a simple phonological rule that would protect this word alone, and Ringe and Taylor state the larger problem plainly: we do not really know why **u* failed to lower in forms of this sort (Luick 1914--1940, 148; Ringe and Taylor 2014, sect. 2.3.1).

What remains unexplained The comparison below is manual. It separates the regular trace result from the attested lexical exception.

Form	Status	Relevance to this entry
**wúllō* -> *woll*	regular trace output	shows what the deterministic sound laws produce
wull / *wulle*	attested OE exception	selected target and philological fact to be recorded
high-vowel escape from another paradigm cell	unsupported for this noun	rejected because the feminine ō-stem paradigm supplies no suitable escape cell

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